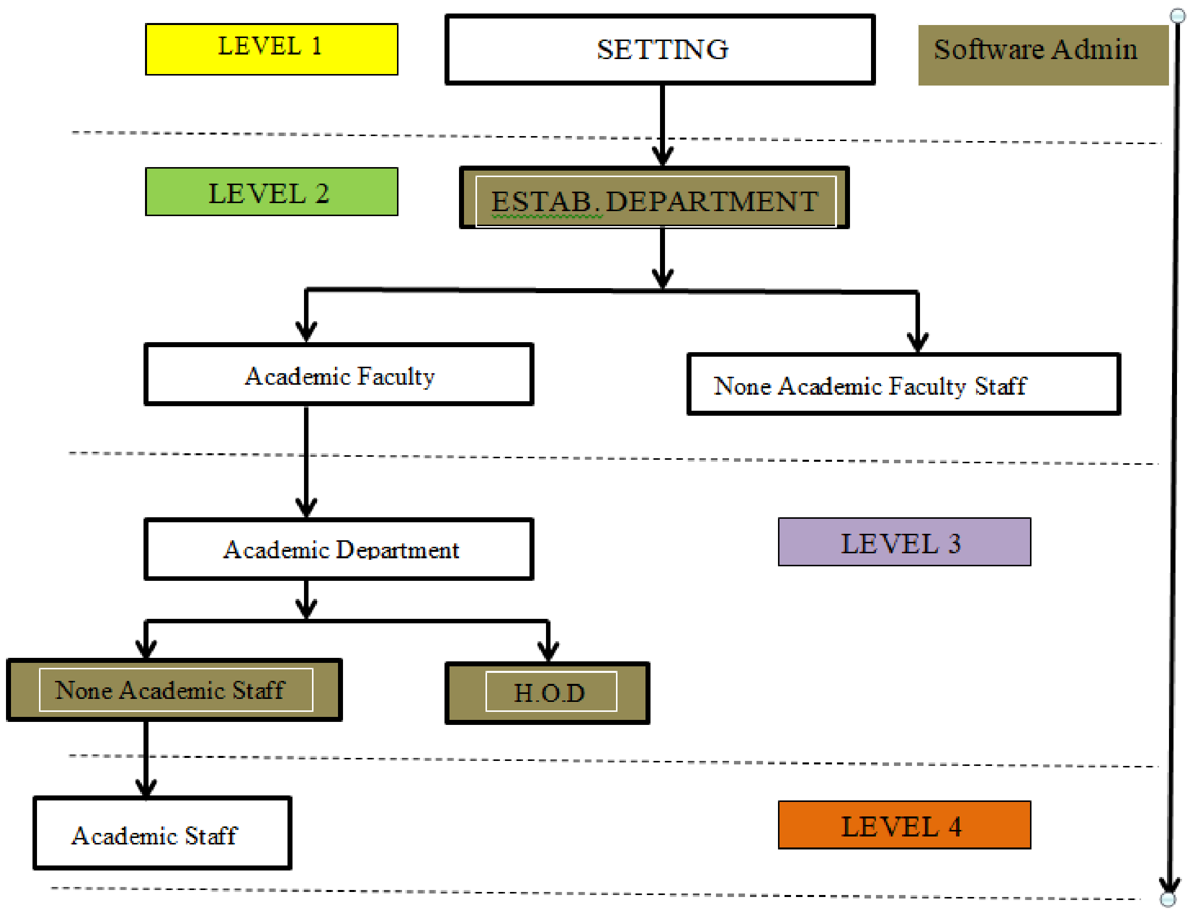
TITLE OF SOFTWARE

Determination of Work Force with it's factors in an Organization & Institution

SETTING (1)



SETTING: The Software Administrator belong to the level 1 group who does the setting of the software by establishing all departments level-by-level. He has the total control of the software like **creating identity for non-academic and academic staff** for the software to be able to recognize the departmental academic staff from departmental none-academic staff. He or she in collaboration with the ESTAB/PERSONNEL Department accepts the final settings set for the computations after the trial templates have been tried for understanding. In addition to his/her control features he/she has the ability of setting the software Institutional name, Logos (including the national coat of arm and institutions logo which will hence forth be seen on the software and on all documents-award certificates, Books of records and Hall of Fame record)

ESTAB/PERSONNEL: They have access right to all functions key (By being able to activate and de-activate functions keys available) that academic and none academic staff needs to access activities from the software

The following are the required details needed for each category setting (both Academic and Non academic staff)

1. Full Name of Officer (Block Letters) Surname first

*Dr./Mr./Mrs./Miss:…………………………..…………………………………………..…………………
……………. Surname Forenames

*(Delete whichever is not applicable)

2. Faculty/College: ……………….. Department:
……………….……………………………………………

3. (A) Personal Particulars

(i) Date of Birth (dd/mm/yy):	
(ii) Date of first appointment:	
(iii) Post/grade of first appointment:	
(iv) Date of Confirmation:	
(v) Present Post:	

(vi) Date Appointed to Present Post:	
(vii) Current Grade Level and Step:	
(viii) Functional GSM:	
(xi) E-mail Address:	
(x) Current Home Address:	

(B) Academic or/& Professional Qualifications held (CERTIFICATES **MUST** BE ATTACHED)	Year Obtained
(i)	
(ii)	
(iii)	
(iv)	
(v)	

ACADEMIC & NON-ACADEMIC FACULTY STAFF: They have right only to the function KEYS that regulates the departments under their faculty as given to them by the ESTAB/PERSONNEL department and also to printing relevant forms for their data entry and also for the none-academic staff to SATISFACTORLY approve data entered by none-academic department staff (More of their function will be given in detail later here).

NON-ACADEMIC STAFF/Department: The only have access to printing relevant forms for data entry and also to SATISFACTORLY approve data entered by academic staff when that needs verifications when such forms or original documents have been verified by them.

H.O.D/Department: Has access to **APROVE** all entries when satisfied with the satisfactory before the ESTAB/PERSONNEL department proceeds to carry out further analysis on data enter by various department.

ACADEMIC & NON-ACADEMIC DEPART. STAFF: These only have access to print relevant data entry forms for the purpose of collecting data related to them or their department and also the can view and print their individual results from the system after the ESTAB/PERSONNEL department are done with the system analysis for various model result computation.

DATA INPUT STRUCTURE (2)

From the Admin. Software setting, Academic and none-academic staff ID and details should be able to be captured by the Estab./Personnel department. i.e. it is either the setting should be done by choosing

FRESH STARTUP

The needed setting for Academic Year should have been chosen. This setting should be done by the Estab./Personnel

☐ semester ☐ Yearly

Note:

- Computation of K*(optimal), Stress value/level, Academic/None-academic staff number, Supervisory/managerial staff number can only be analyze semester-by-semester with the model in the software.
- Though data for staff Appraisal are collected semester-by-semester, Staff Appraisal can only be computed once a year (i.e. covering both semester in a given academic calendar year) by the ESTAB./PERSONNEL department.

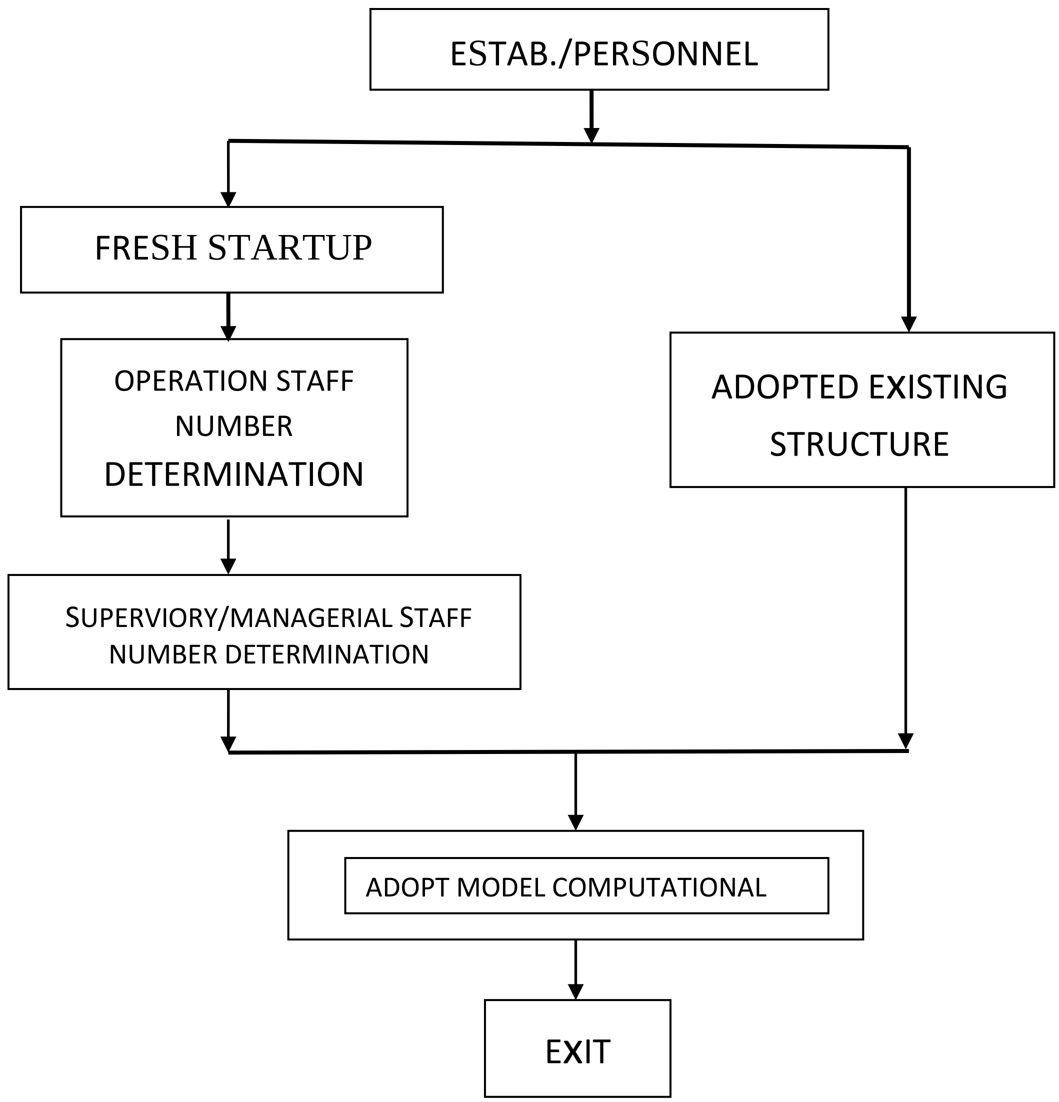


Figure ; ESTAB./PERSONNEL setup chart for data entry.

That is, it is either the setting should be for a [**FRESH STARTUP]** or [**ADOPTED EXISTING STRUCTURE]**. With [**FRESH STARTUP]** we will be able to find the required number of OPERATION staff (Under None-Academic) and SUPERVISORY/MANAGERIAL staff number (None-Academic).

For [**ADOPTED EXISTING STRUCTURE]** we import both None-Academic and Academic staff ID details for the institution. After these have been set, then each FACULTY and DEPARTMENTAL None Academic staff can be able to input the actual staff ID details at the beginning of a session base on semester-by-semester as required by the task needed to be perform.

Note: The software should be able to check/validate the staff ID entry at the beginning of the semester done by the departmental/Faculty None-academic staff against the staff imported details for the [**ADOPTED EXISTING STRUCTURE].** Why for the [**FRESH STARTUP]** structure, after the required number of OPERATION staff and SUPERVISORY/MANAGERIAL staff have been

determined, entry of staff ID's employed into the institution should be done first by the ESTAB./PERSONNEL Department to fit into the determine or obtained number (This is solely for the None academic staff).
The [**FRESH STARTUP]** only helps to determine the require number of none-academic staff after which the other procedure as obtained for [**ADOPTED EXISTING STRUCTURE]** also applies.
Base on what is chosen if semester, then allow departmental lecturers the privilege to set stress category frequencies. See **FORM 5** and **FORM 6.** Also allow them to be able to feel the feelings/frequencies **FORM 7**

Note: FORM 5 should not be printable by academic staff, departmental none academic depart. staff, H.O.Ds/DEANs, academic and none academic faculty staff as well as by Estab./Personnel Depart staff. i.e. the form should be none-printable at all level. **FORM 6 and FORM 7** should be made printable if so desired for data input.

Also, the academic staff nomenclature should have been captured at the beginning of an academic calendar i.e. by courses for each academic semester tutored by each lecturer involved.
see figure 1, figure 2, figure 3 and figure 4 for samples of such setting procedures for departments. Once this have been set at the Departmental/Faculty level by the None academic staff attached to the Department/Faculty, the ESTAB./PERSONNEL department should be able to access these information and then **[VERIFY/APPROVE]** for each department semester by semester after each of these activity has been done.

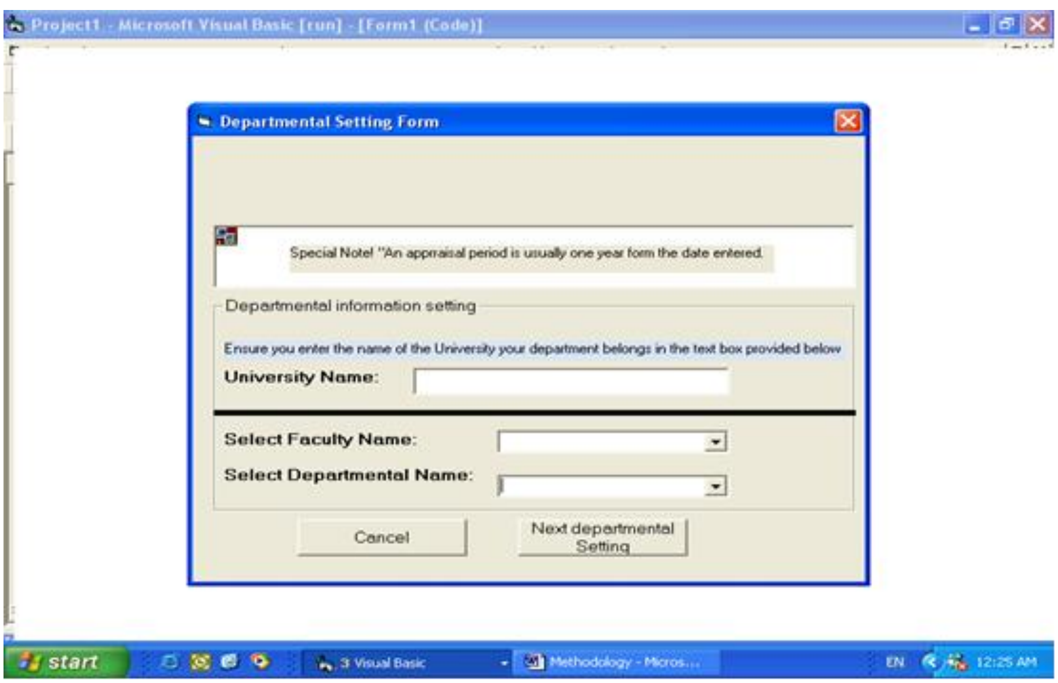


Figure 1 (FORM 1) Note: In place of University Name use Institution Name in your design.

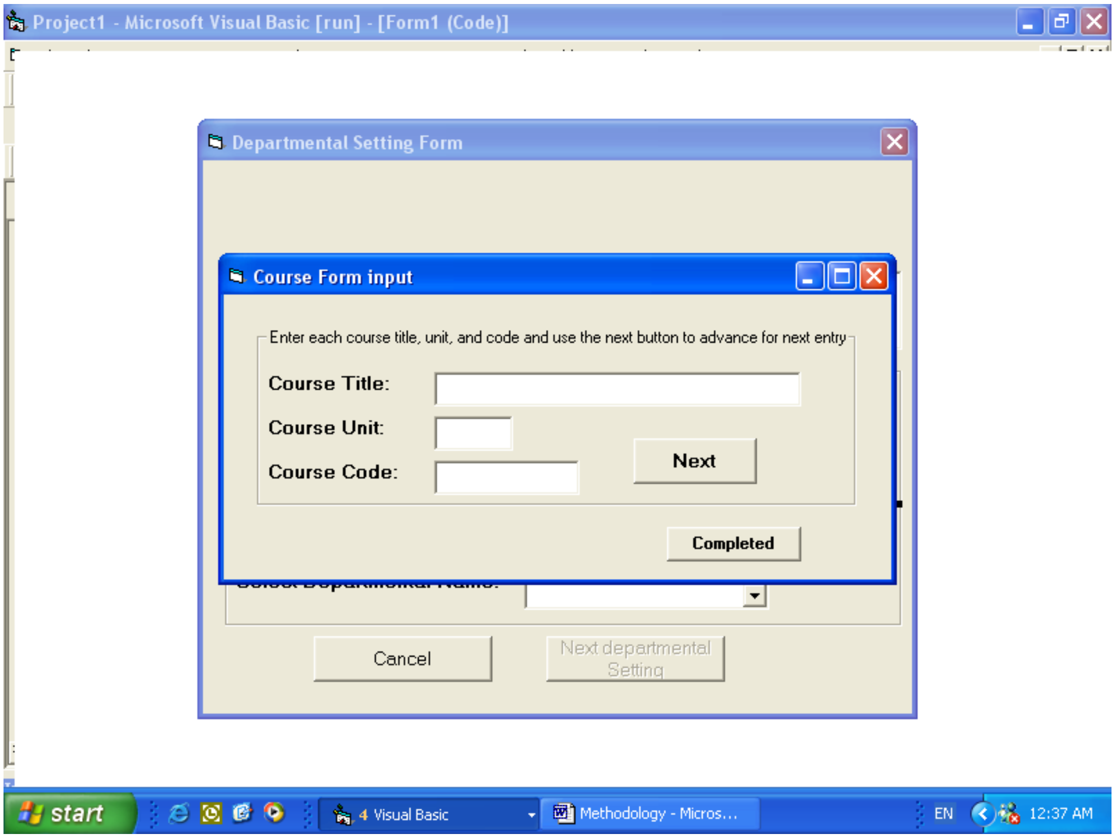


Figure 2 (FORM 2)

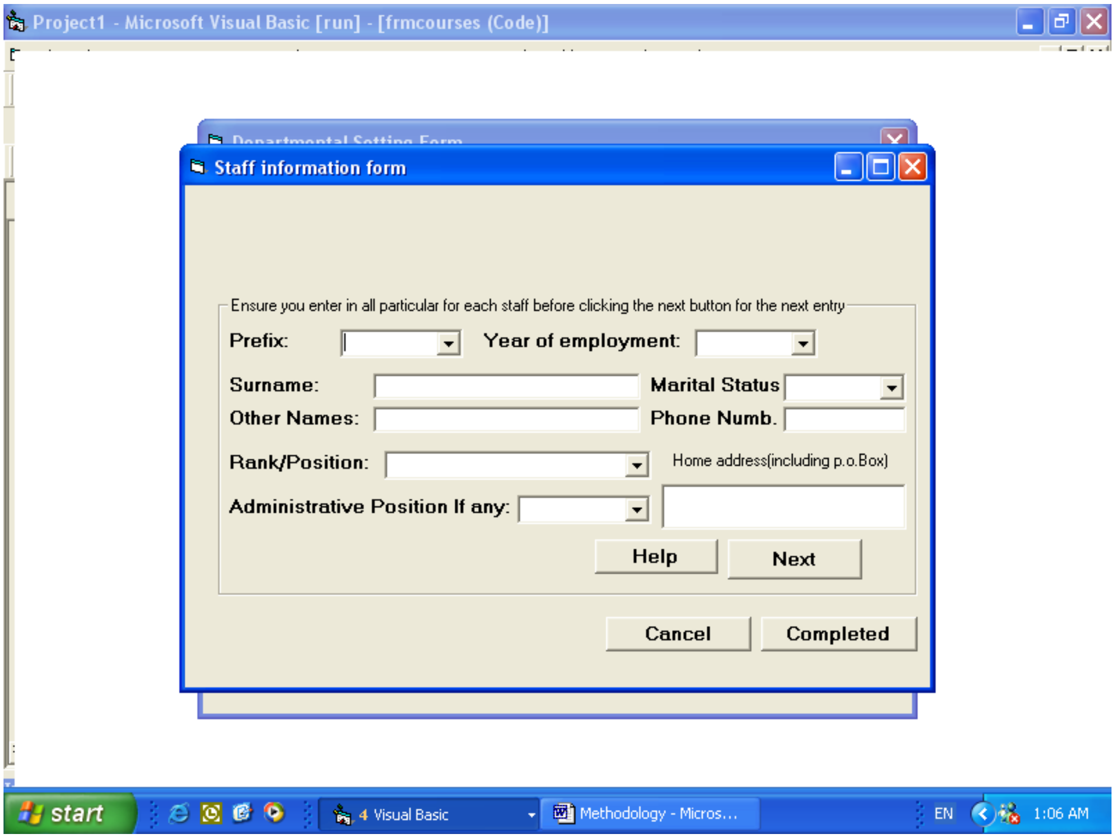


Figure 3 (FORM 3)

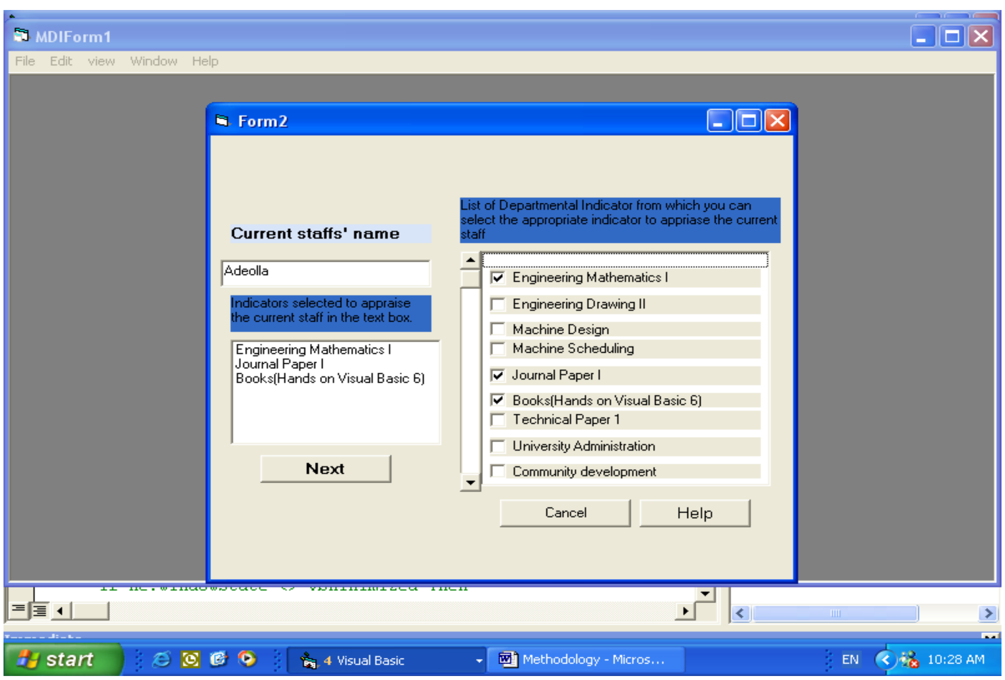
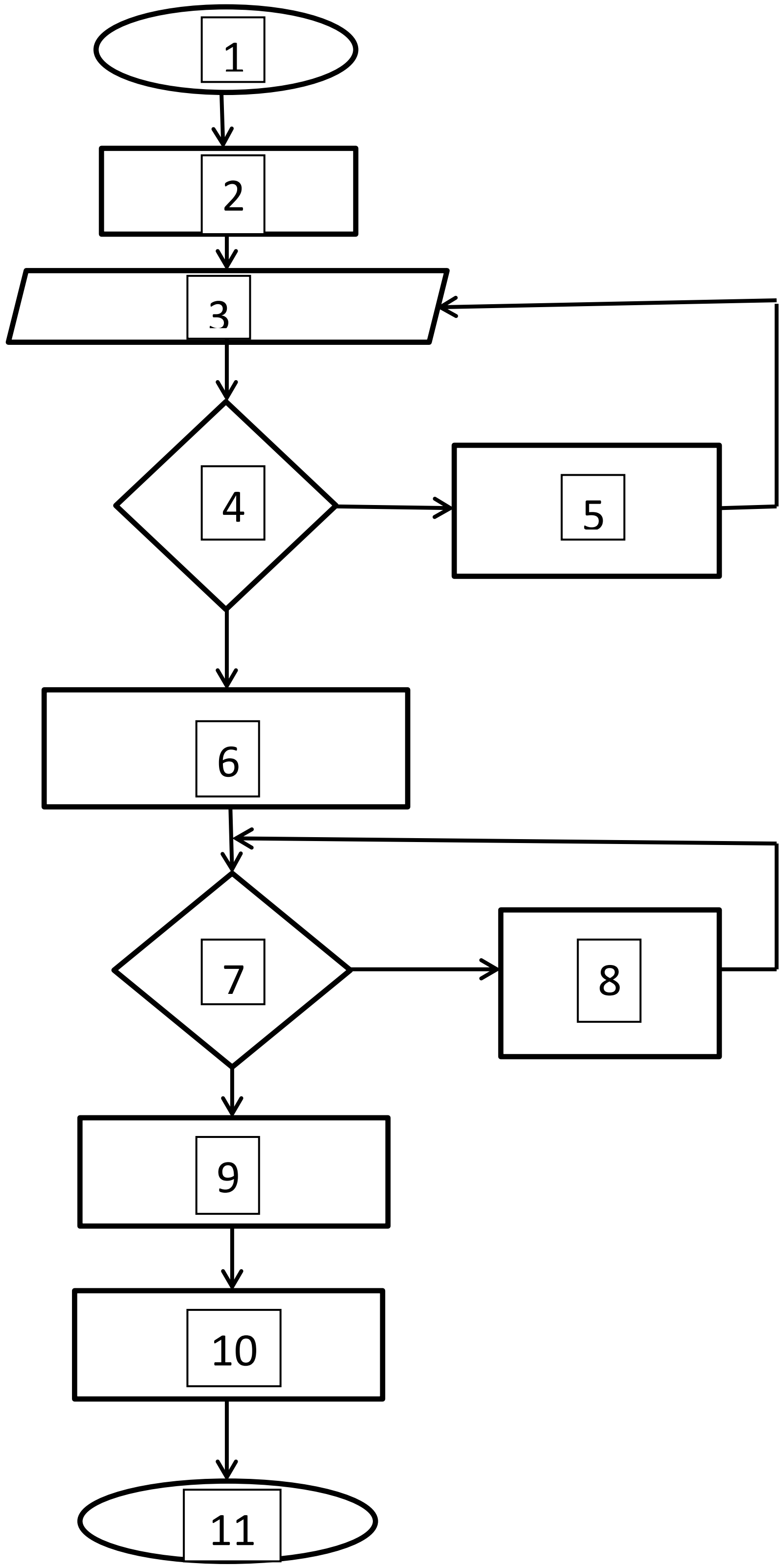


Figure 4 (FORM 4)



1) START
2) SETTINGS
3) DATA INPUT
4) ARE DATA INPUTS FIT
5) ELIMINATE DATA NOT FIT
6) PERFORM K*optimal, STRESS for Academic staff
7) IS STUDENT NUMBER INTAKE KNOWN
8) CONSULT ESTAB./PERSONNEL
9) DETERMINE NUMBERs of LECTURER, Predict future, Plots
10) DETERMINE Number of NONE-ACADEMIC STAFF, PREDICT, Plots.
11) END

Figure ; Flow chart for software

The following activities/features are what can then follow

Note: Before carrying out any analysis on data inputted into the system, first eliminate data that serves as outliers. See page 32

1) DATA INPUT (This could be yearly base or semester base according to the activity type)
SEMESTER-BASED DATA INPUT:

a) STRESS value/level by Department/Faculty

FORM 5 (SETTING THE STRESS THEME CATEGORY MAX. VALUES) (None printable by all levels except your own faculty via department platform) but accessible by departmental academic staff only. This can be set for once input or three/four times input depending on the ESTAB./PERSONNEL choice. i.e. this control is only done by the ESTAB./PERSONNEL Department. Four times input means input can be collected for three/four different academic year then the results of these for years academic calendar will be averaged and adopted for subsequent use however for each of the four academic calendar each result obtained for each year is used for that year but after the four year, **FORM 5** will be deactivated and will no more be used for the setting of **FORM 6** category parameter setting. We strictly adopt the average result of the four years input as our setting for **FORM 6**. If the once input option is adopted by the ESTAB./PERSONNEL department, then just the first year of use setting will strictly be use for setting the category parameter of **FORM 6** after which the **FORM 5** will be deactivated. However, we have a default setting for FORM 2 which will only be use for TRIAL test of the software and for learning purpose.

SETTING THE STRESS CATEGORY/STAFF
(1)

ORGANIZATIONAL										
	Less Likely	Summative Response Scales							Most Likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Time											$\frac{x}{10} \times \frac{116}{1}$
Paper Work											$\frac{x}{10} \times \frac{99}{1}$
Lack of Materials											$\frac{x}{10} \times \frac{34}{1}$
Extra Duties											$\frac{x}{10} \times \frac{32}{1}$
Physical Plant											$\frac{x}{10} \times \frac{19}{1}$
Meetings											$\frac{x}{10} \times \frac{15}{1}$
Class Size											$\frac{x}{10} \times \frac{15}{1}$
Poor Scheduling											$\frac{x}{10} \times \frac{13}{1}$
Interruptions											$\frac{x}{10} \times \frac{13}{1}$
Travels											$\frac{x}{10} \times \frac{12}{1}$

Conflicting Demand											$\frac{x}{10} \times \frac{11}{1}$
Athletics											$\frac{x}{10} \times \frac{4}{1}$

(2)

STUDENT											
	Less Likely	Summative Response Scales								Most Likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Student Discipline											$\frac{x}{10} \times \frac{97}{1}$
Student Apathy											$\frac{x}{10} \times \frac{38}{1}$
Low Student Achievement											$\frac{x}{10} \times \frac{37}{1}$
Student Absences											$\frac{x}{10} \times \frac{3}{1}$

(3)

ADMINISTRATIVE											
	Less Likely	Summative Response Scales								Most Likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Unclear Expectations											$\frac{x}{10} \times \frac{27}{1}$
Lack of Knowledge or Expertise											$\frac{x}{10} \times \frac{25}{1}$
Lack of Support (Backing, Recognition)											$\frac{x}{10} \times \frac{24}{1}$
Inconsistency											$\frac{x}{10} \times \frac{17}{1}$
Unreasonable Expectations											$\frac{x}{10} \times \frac{14}{1}$
Poor Evaluation Procedures											$\frac{x}{10} \times \frac{12}{1}$
Indecisiveness											$\frac{x}{10} \times \frac{10}{1}$
Lack of Opportunities for Input											$\frac{x}{10} \times \frac{10}{1}$
Failure to Provide											$\frac{x}{10} \times \frac{7}{1}$

Resources											
Lack of Follow-Through											$\frac{x}{10} \times \frac{6}{1}$
Harassment											$\frac{x}{10} \times \frac{5}{1}$
Favoritism											$\frac{x}{10} \times \frac{5}{1}$
Miscellaneous											$\frac{x}{10} \times \frac{4}{1}$

(4)

TEACHER											
	Less Likely	Summative Response Scales								Most Likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Conflict or Lack of Cooperation											$\frac{x}{10} \times \frac{59}{1}$
Incompetence or Irresponsibility											$\frac{x}{10} \times \frac{21}{1}$
Negative Attitude											$\frac{x}{10} \times \frac{7}{1}$
Lack of Communication											$\frac{x}{10} \times \frac{5}{1}$

(5)

PARENTS											
	Less Likely	Summative Response Scales								Most Likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Interference											$\frac{x}{10} \times \frac{24}{1}$
Nonsupport or Apathy											$\frac{x}{10} \times \frac{15}{1}$
Lack of Communication & Understanding											$\frac{x}{10} \times \frac{11}{1}$

(6)

OCCUPATIONAL											
	Less Likely	Summative Response Scales								Most Likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Lack of Professional Growth											$\frac{x}{10} \times \frac{12}{1}$
Low Salary											$\frac{x}{10} \times \frac{9}{1}$

Lack of Advancement											$\frac{x}{10} \times \frac{5}{1}$
Job Insecurity											$\frac{x}{10} \times \frac{4}{1}$

(7)

PERSONAL											
	Less Likely	Summative Response Scales								Most Likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Professional-Personal Conflict											$\frac{x}{10} \times \frac{12}{1}$
Conflict With Personal Values											$\frac{x}{10} \times \frac{10}{1}$
Miscellaneous											$\frac{x}{10} \times \frac{4}{1}$

(8)

ACADEMIC PROGRAM											
	Less Likely	Summative Response Scales								Most Likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Repetition											$\frac{x}{10} \times \frac{7}{1}$
Unrealistic Goals											$\frac{x}{10} \times \frac{7}{1}$
Low Standards											$\frac{x}{10} \times \frac{5}{1}$
Responsibility to Grade Students											$\frac{x}{10} \times \frac{4}{1}$

(9)

NEGATIVE PUBLIC ATTITUDE											
	Less Likely	Summative Response Scales								Most Likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Negative Public Attitude											$\frac{x}{10} \times \frac{9}{1}$

(10)

MISCELLANEOUS											
	Less Likely	Summative Response Scales								Most Likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Miscellaneous											$\frac{x}{10} \times \frac{27}{1}$

Figure 5 **(FORM 5)**

GROUPINGS
1, 2, 3 & 9 – for Students & Physical condition (Stress Factor)
39.0 + 17.8 + 16.9 + 1.0 = 74.7%

4, 5, 6 & 8 – for Time Pressure (Pressure Factor)
9.4 + 5.1 + 3.1 + 2.3 = 19.9%

7 – for Conflict
2.7 = 2.7%

Divide Miscellaneous (10) by a ratio of 3:2:1 to the three groups above where, 3 is for Students & Physical condition (Stress Factor), 2 is for Time Pressure and 1 is for conflict.

FORM 6 (DATA INPUT OF THE STRESS THEMES CATEGORY VALUE)
FORM 6 can be accessed by both academic and none-academic Department/Faculty staff, only after **FORM 5** have been used to set its Stress category Maximum values. Note: That means that the maximum value set on **FORM 6** for each staff varies from each others. This value shouldn't be alterable by any one once set. **FORM 6** can then be printed by each staff so that they can carefully study it and enter values which must not exceed the maximum vale they set. However, they can print a default of the **FORM 6** which we have in the system for a guide. Also as help tip, not all fields is mandatory to have values but where values are not necessary ZERO "0" must enter there in that field when inputting their scores in the software system.

STRESS THEMES
Frequencies of stress themes vis-à-vis stress categories

Note: It is not mandatory that all stress themes must have a score. Where score is not applicable record zero (0) there
(1)

STRESS THEMES										
Stress Category/Frequencies	Control of Time	Inference with Instruction	Overload		Under-load		General Performance	Threat to Self	Precipitate Change	
			Qualitative	Quantitative	Qualitative	Quantitative				
Organization (-)										
Student (-)										
Administrative (-)										

Teacher (-)										
Parent (-)										
Occupational (-)										
Personal (-)										
Academic Program (-)										
Negative Public Attitude (-)										
Miscellaneous (-)										
										Frequency of theme/ % of total frequency of all stress themes (3066)

Frequency of theme / % of total frequency of all stress themes

$$\frac{\sum_{i=1}^{n} stress\ cat/_{contri}}{\sum_{i=1}^{n} \frac{X_{ij}}{all}\ element\ in\ table} \times \frac{100}{1}$$

Figure 6 **(FORM 6)**

FORM 7 (DATA INPUT FOR THE FEELING/FREQUENCIES VALUE)

FORM 7 can also be printed by each staff so that they can carefully study it and enter values which must not exceed the maximum value they set with regard the STRESS CATEGORY/FREQUENCIES. However, they can print a default of the **FORM 7** which we have in the system for a guide. Also as help tip, not all fields is mandatory to have values but where values are not necessary they must enter ZERO "0" there in that field when inputting their scores in the software system.

IMPORTANT NOTE: For stress determination, only the H.O.Ds/DEANs (Deans can only **verify/approve/submit** staff data in the Deans office) needs to [**APPROVE/SUBMIT]** all entries hence He/She must be able to know the respective staff yet to enter his/her entries. The ESTAB./PERSONNEL Department must be able to know which departments/Faculties is yet to approve/submit their respective wards entries.

Stress Category/Freq	FEELINGS/FREQUENCIES				
	Anger Towards Others (-)	Depressive States (-)	Anxiety Anticipatory (-)	Physical Feelings (-)	Self-Blame (-)
Organization (-)					
Student (-)					
Administrative (-)					
Teacher (-)					
Parent (-)					
Occupational (-)					
Personal (-)					
Academic Program (-)					
Negative Public Attitude (-)					
Miscellaneous (-)					

$\frac{\sum_{i=1}^{n} X_{ij}/all\ elements}{\sum_{i=1}^{n} stress\ cat\ set/in\ table}$ = 1614/981 = 1.6453

This will give the indicated major feelings per stress category

Figure 7 **(FORM 7)**

ACADEMIC STAFF APPRAISAL PROCESS

b) APPRAISAL value/level by Department/Faculty for only Academic staff

This is a yearly activity i.e. The APPRAISAL Button can only be activated for use by the ESTAB./PERSONNEL department once a year. However, during each semester the following forms are needed to be filled as required. These forms are **FORM 8, FORM 9, FORM 10, FORM 11** and **FORM 12**. see sample of already designed data entry for these forms in figure 8, figure 9, figure 10, figure 11 and figure 12 below.

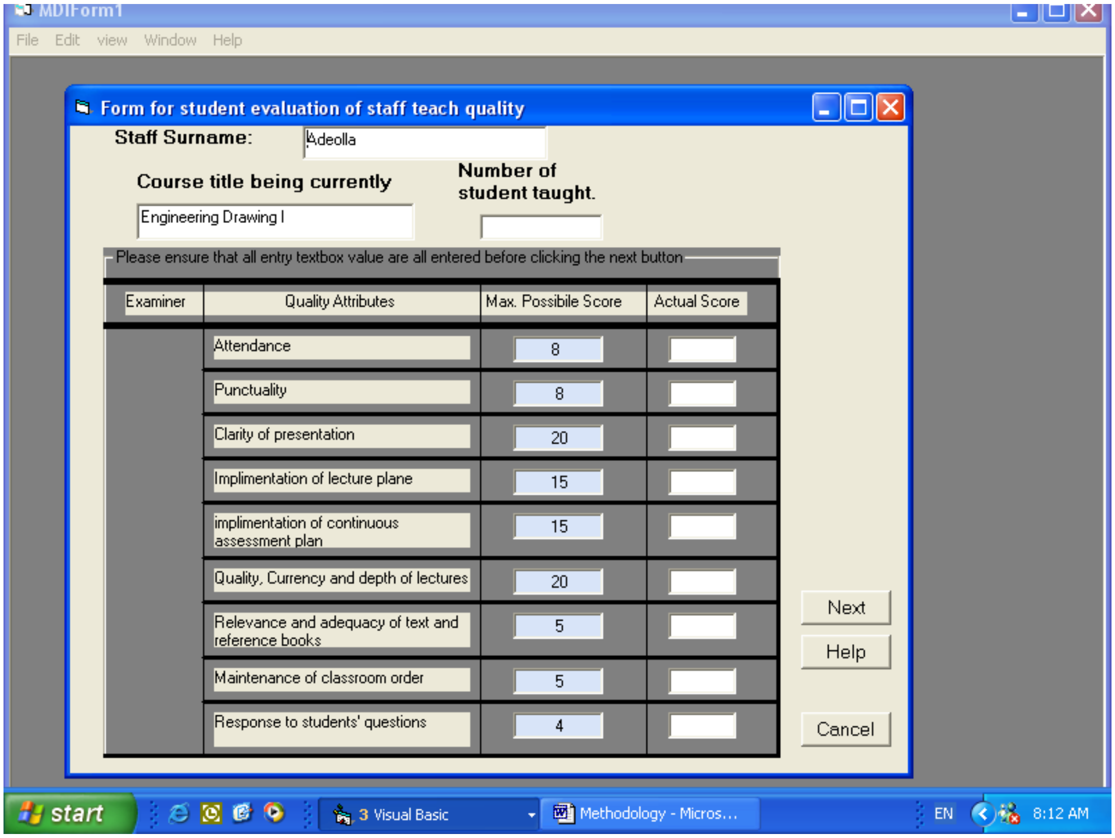


Figure 8 **(FORM 8)**

s/no	Assessed Criterion	Max Possible Score	Actual Score
1.	Attendance	8	
2.	Punctuality	8	
3.	Clarity of presentation	20	
4.	Implimentation of lecture plane	15	
5.	Implementation of continuous assessment plan	15	
6.	Quality, currency and depth of lectures	20	
7.	Relevance and adequacy of text and reference books	5	
8.	Maintenance of classroom order	5	
9.	Response to student’s questions	4	
10.	Aggregate (total) score	100	

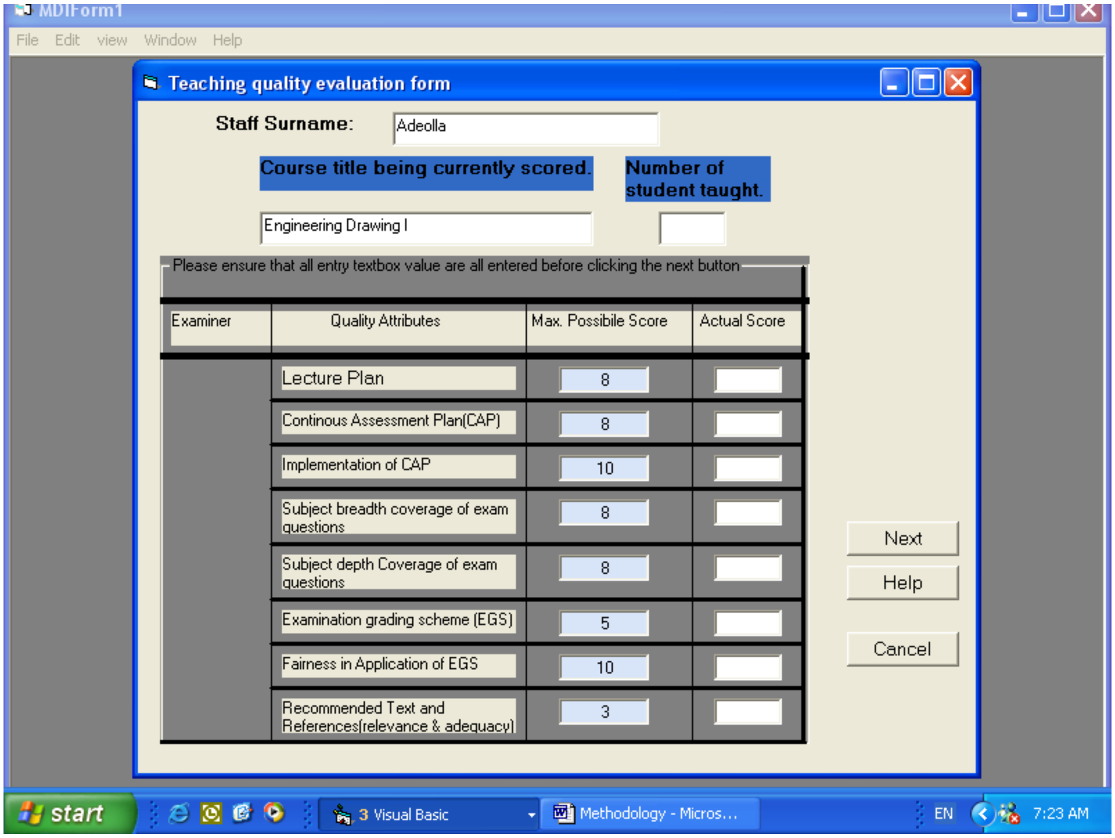


Figure 9 **(FORM 9)**

Examiner	**Quality Attributes**	**Max possible score**	**Actual Score**
External and or peer	Lecture plan	8	
“	Continous Assessment plan	8	
“	(CAP) implementation	10	
“	Subject Breath coverage of examination questions	8	
“	Subject Depth Coverage of examination question	8	
“	Examination grading Scheme	5	
“	Fourness in Application of EGS	10	
“	Recommended text and Reference (Relevance and Ad	3	
Class Students	Student Evaluation (implemer of lecture plan, CAP: Keepin lecture schedule: use of notes, Understanding of Subject, e.t.	40	

	Total		

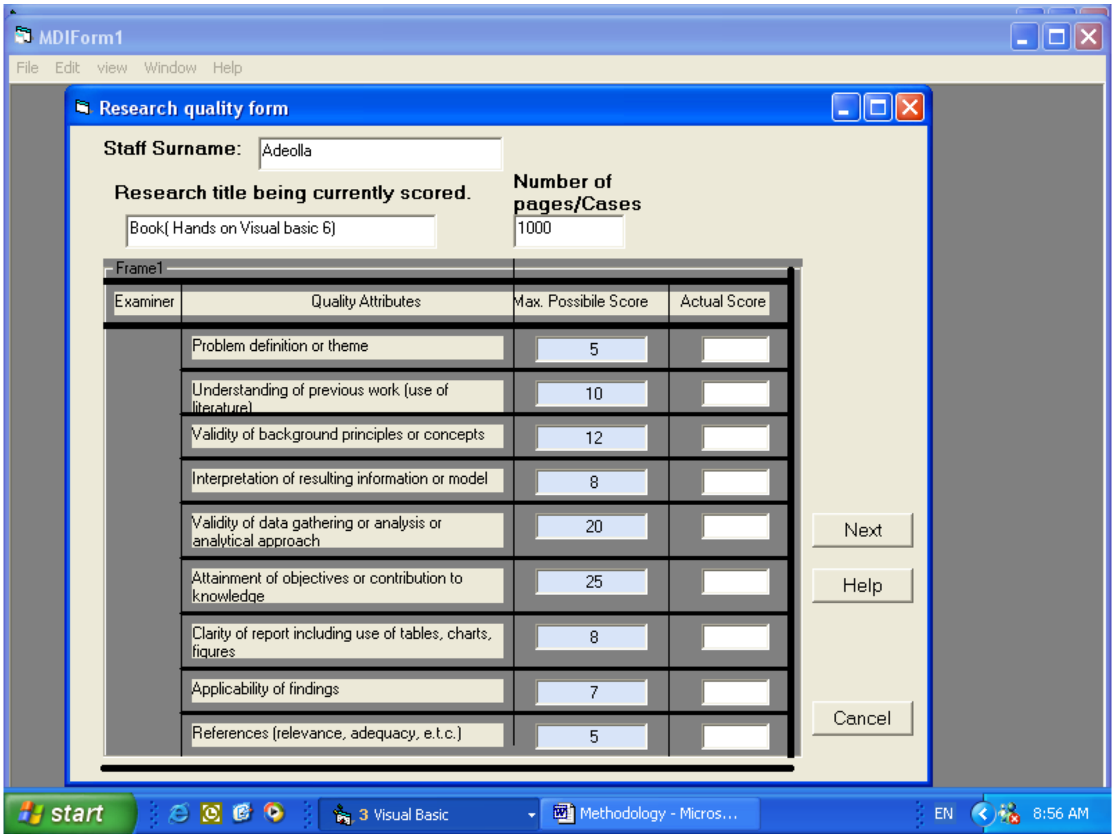


Figure 10 **(FORM 10)**

Assessor	Quality attributes of research	Max possible score	Actual Score
External and / or peer	Problem definition or scheme	5	
-	Understanding of previous work (use of lite	10	
-	Validity of background principles/ concepts	12	
-	Interpretation of resulting information/ model	8	
-	Validity of data gathering/ analysis or Analytical approach	20	
-	Attainment of objectives or contribution to knowledge	25	
-	Clarity of report including use of tables, Charts, figures	8	
-	Application of findings	7	

-	References (relevance, adequacy etc)	5	
	Total	100	

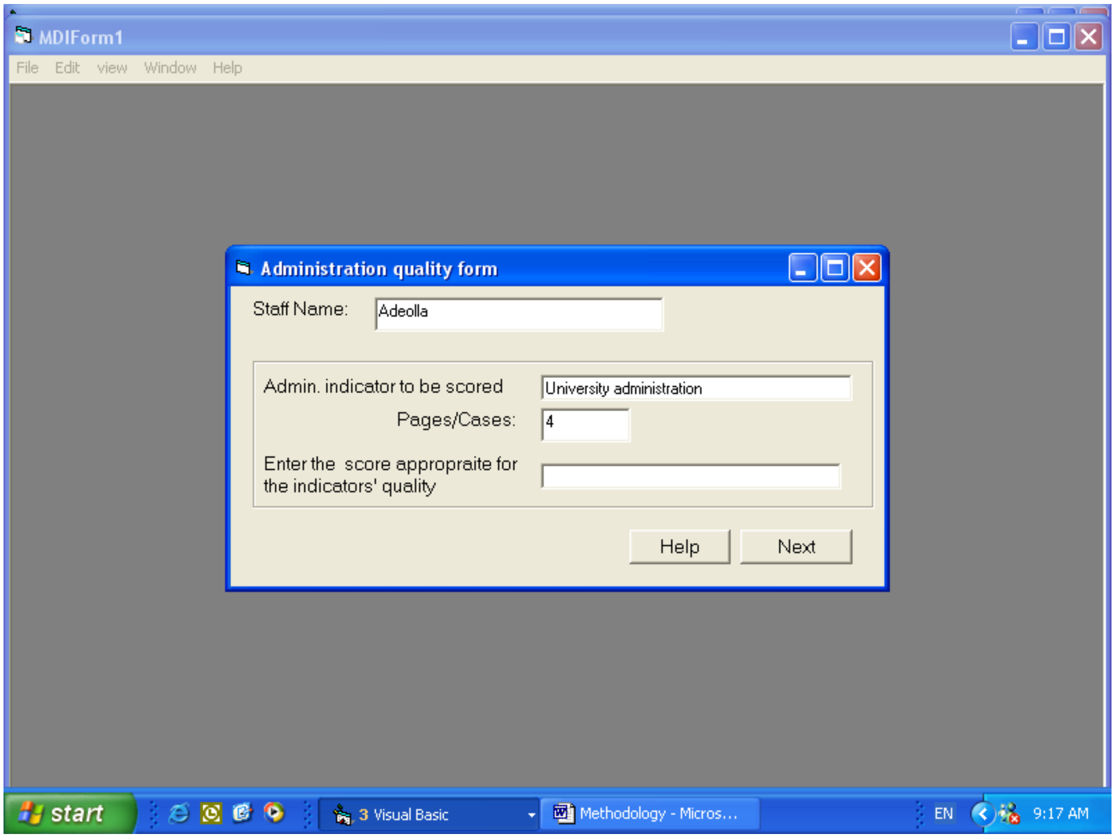


Figure 11 **(FORM 11)**

Note: For each achievement criteria performance measurement there should be a Box for Head of Department/Unit Max Score

Maximum Score

Appraisal

Maximum Score

HOD/Unit Max. Score

Yes

No

Appraisal Acceptance

HOD/UNIT

Justification____

Note: If Appraisal's Score is within plus/minus 5% of HOD/UNIT Max. Score then accept and record the average between both scores. i.e. Appraisal score attained and the HOD/Unit Max. Score

Note: If Appraisal's Score is below (minus 5% of HOD/UNIT Max. Score and the HOD/UNIT has justified his/her score without alteration) then accept and record only the HOD/UNIT Max. Score but do not store the score. Flag the appraisal score arrived at for further action by appraisal auditor.

Note: If Appraisal's Score is above (plus 5% of HOD/UNIT Max. Score and the HOD/UNIT has justified his/her score without alteration) then accept and record the average between both scores, and do not flag the score. i.e. Appraisal score attained and the HOD/Unit Max. Score

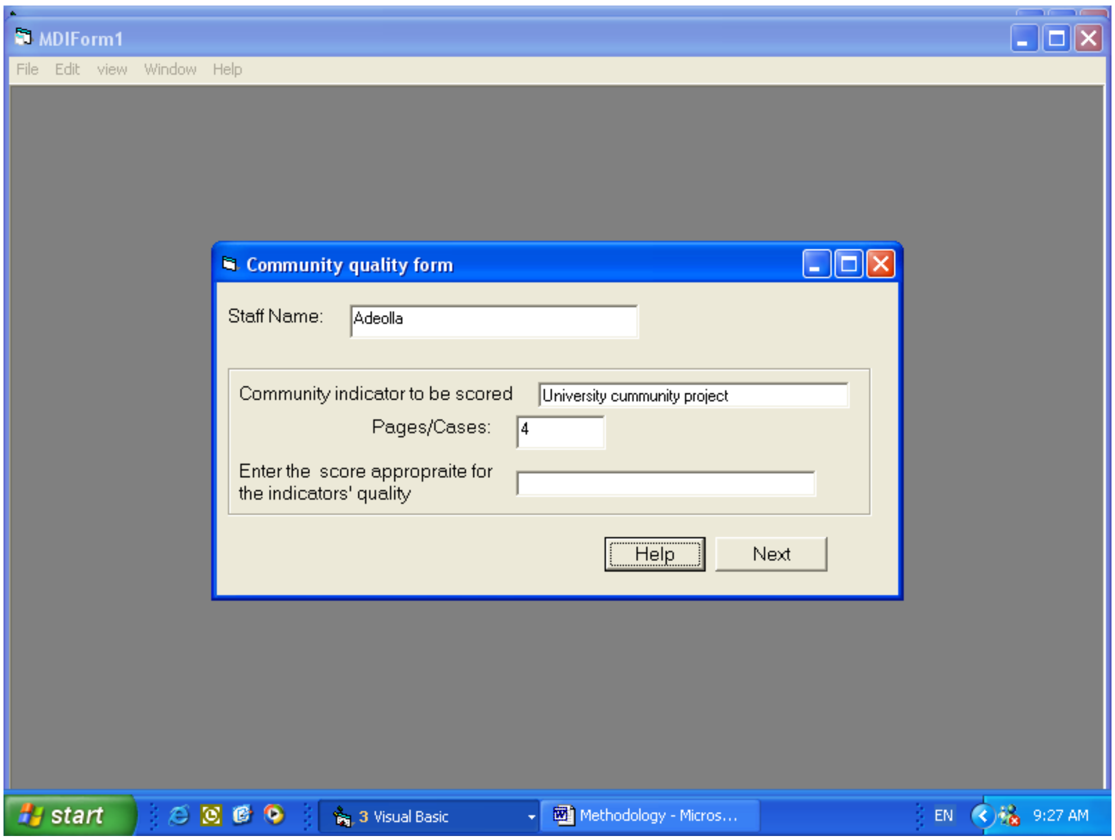


Figure 12 **(FORM 12)**

Note: An empty **FORM 8** to **FORM 9** can be printable and accessible by the none-academic staff attached to the department in-charge of such entries for both academic staff and none-academic staff before the hand filled forms can be used to fill its values into the software after they've been submitted to the department none-academic staff in-charge. i.e. **FORM 8** to **FORM 9** is only accessible by the none-academic department/faculty staff in-charge. However, **FORM 10** to **FORM 12** shouldn't be made printable but accessible online by each academic staff which is subject to adjustment by a superior **GROUP HEAD/H.O.D**.

IMPORTANT NOTE: A minimum of ten filled copies of **FORM 8** for each COURSE taught is needed to be submitted to the none-academic staff in-charge of the software along with the relevant filled copies of **FORM 9** (attached to each course) rightly scored for each course taught by their respective lecturer, and also with **FORM 10** rightly scored by each lecturer online for their research work with evidence **duly uploaded by each lecturer**. These however, will be [**VERIFIED]** w.r.t. **FORM 8**, and **FORM 9** by the none-academic staff in-charge of the software in the department. The H.O.D/DEAN still needs to [**APPROVE/SUBMIT]** all entries. Hence He/She must be able to know the respective staff yet to enter his/her entries. The ESTAB./PERSONNEL Department must be able to know which departments/Faculties is yet to approve/submit their respective wards entries.

Worth Assessment scheme

% Quality level	**Worth**
0 -30	0
31 – 33	1
34 – 36	2
37 – 39	3
40 – 42	4
43 – 45	5
46 – 48	6
49 – 51	7
52 – 54	8
55 – 57	9
58 – 60	10
61 – 63	11

64 – 66	12
67 – 69	13
70 – 72	14
73 – 75	15
76 – 78	16
79 – 81	17
82 – 84	18
85 – 87	19
88 – 100	20

An example of an annual targets for academic staff work output

s/no	**Work achievement parameters**	**Positions**	**Annual targets (expected value of output in position)**
1.	Teaching	Grade Asst/ Asst. Lecturer	70/150
		Lecturer II	165
		Lecturer I	192
		Senior Lecturer	221
		Professorial Cadre	252

2.	Research	Assistant Lecturer	25
		Lecturer II	70
		Lecturer I	116
		Senior Lecturer	168
		Professorial Cadre	273
3.	Administrator	Vice Chancellor	540
		DVC & Provost	492
		Dean	396
		Director/Ag. Head/Head	140
		Hall Warden	90
		other	?
4.	Inter. Community service	Assistant Lecturer	18
		Lecturer II	37
		Lecturer I	46
		Senior Lecturer	56
		Professorial Cadre	74
5.	Total (Teaching, Research & In Services)	Grad Asst/ Asst Lecturer	193

		Lecturer II	272
		Lecturer I	353
		Senior Lecturer	445
		Professorial Cadre	599

Ways of determining the quantity, quality and values of outputs was also covered in this model.

MODEL USED

The achievement parameter are categorically grouped into four. And these are:

a) Teaching

b) Research

c) Administration

d) Community service

Under teaching, the output specifications are :

i. Theoretical subjects taught
ii. Practical/clinical subjects taught
iii. Assessed scripts/projects
iv. Supervised projects, thesis, dissertations etc.
v. Others

The research outputs specifications includes:

i. Refereed journal papers
ii. Refereed proceeding papers
iii. Books

iv. Monograhps

v. Patents

vi. Designs

vii. Prototype

viii. Technical report

ix. Invited paper

x. Research report

xi. Conference papers/abstracts

xii. Others

Under Administration, outputs specifications are:

i. Formulated policy

ii. Developed program

iii. Policy work plan

iv. Execution of University program

v. Students/staff development program

vi. Others

For community services, outputs specifications are:

i. Certified documented contributed of value of the University.

ii. Others.

In order to determine the value of these output parameters, the quantity and the quality of each output first have to be obtained. Thus

Value of output = (quantity of output) x (quality of output)

Determining the quantity is with the use of standardized scheme. Such as that presently in use within the University system where for example forty-five (45) semester hours of practical/clinical demonstration is equivalent to one (1) unit of output or a fifteen (15) semester hours of theoretical teaching is equivalent to one (1) unit of output.

The overall output quantity for a course is therefore equal to:

BASIC UNITS + 0.02 * (Number of student)

To obtain the quality, various evaluation scheme form needs to be computed based on the achievement parameter grouping.

For teaching category, two forms is needed to be duly completed or scored and these are:

1. the form for student evaluation of academic staff teaching quality.
2. the teaching quality evaluation scheme form. See table 1 and 2 for a format of these forms.

For research category, the research paper quality evaluation scheme form needs to be filled. The administration and community service category group are quiet easy to score.

Having obtained the required quality of each output type, the corresponding worth is read from the worth table form.

It is actually the product of the corresponding worth and the quantity that determines the points. So:

Points = quantity * Worth of the quality.

To be able to rightfully grade each performance base on the achievement category, groups targets was set in correlation to the rank/position of the staffs needed to be appraised. See table 5 for an example of such target set.

With the computation of the Relative to Target Performance (RTP) we can rightfully grade performance.

$$RTP = \left(\frac{\text{Observed output} - \text{Target}}{\text{Target}} \right) 100$$

$$RTP \begin{cases} = +Ve, & \text{if achievement output is above target} \\ = 0, & \text{if achievement output equals target} \\ = -Ve, & \text{if achievement output is below target} \end{cases}$$

to further simplify performance grade, five performance level was adopted for use and these are:

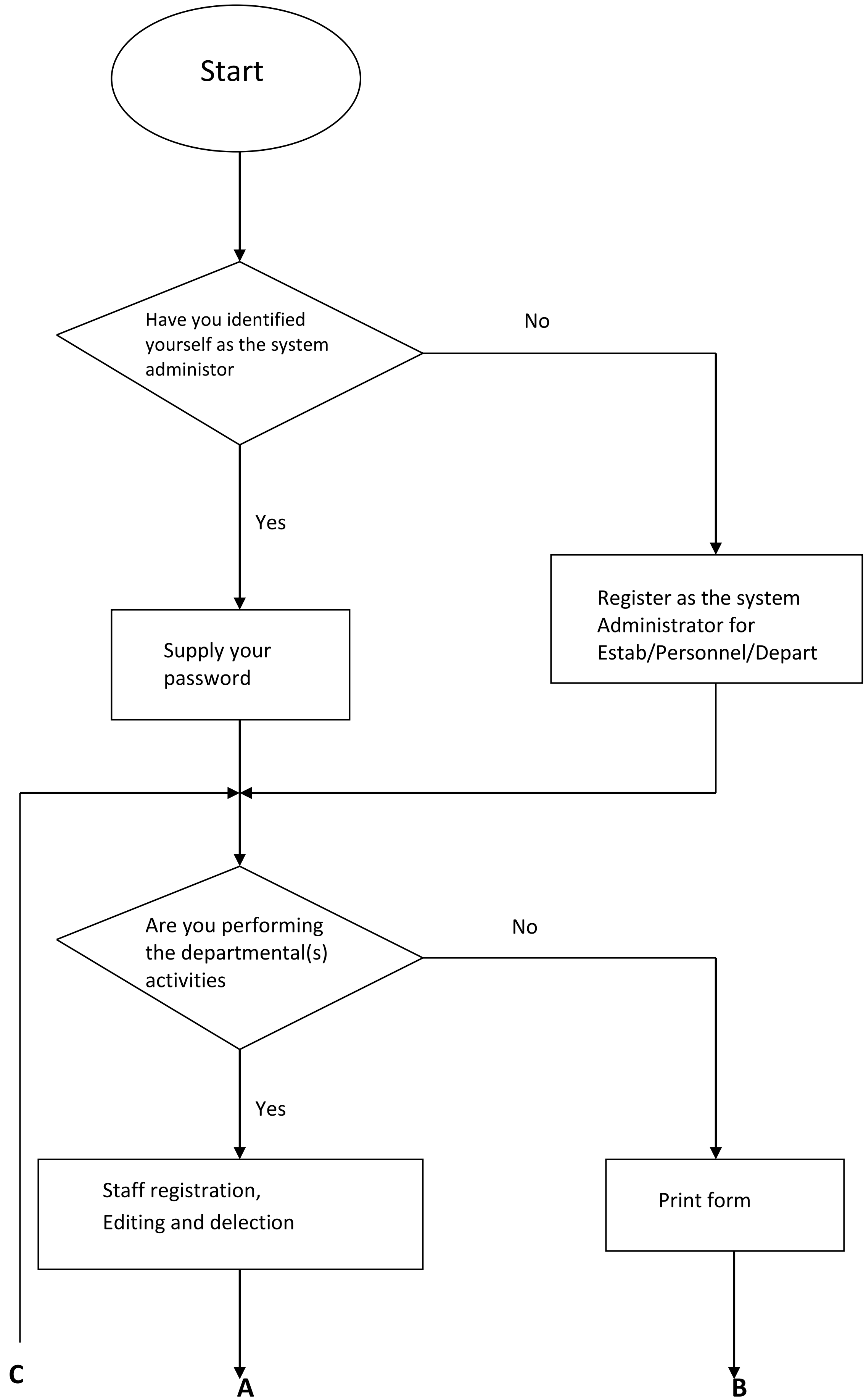
Good if his/her RTP is $\pm$ 5% of target achievement in any work parameter;

Fair if RTP is < 25% below the expected or target;

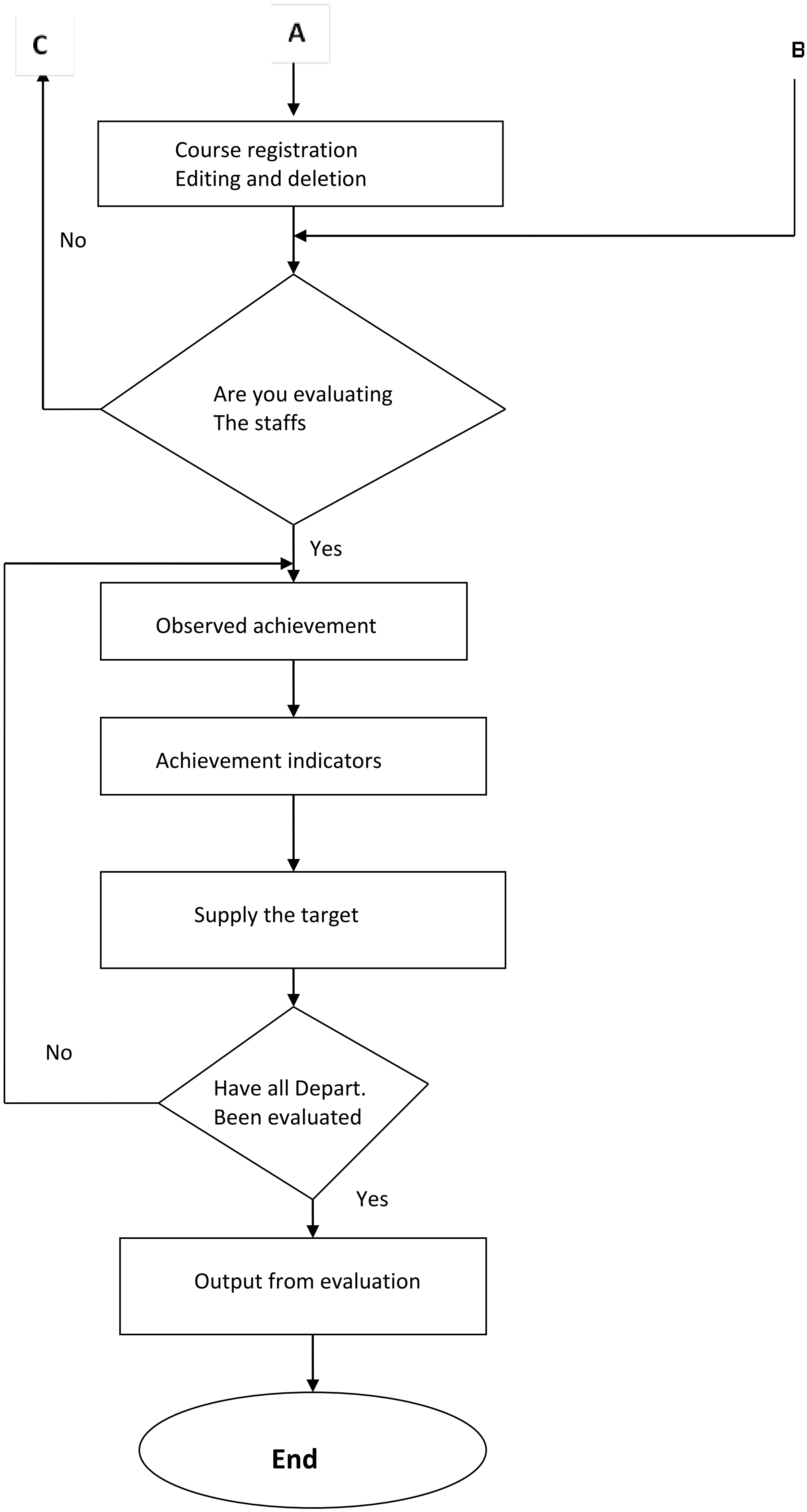
Poor if RTP is > 25% below the target;

Very Good if RTP is < 25% above the targeted;

Excellent if RTP is > 25% above the targeted.



Start
Have you identified yourself as the system administor
No
Yes
Supply your password
Register as the system Administrator for Estab/Personnel/Depart
Are you performing the departmental(s) activities
No
Yes
Staff registration, Editing and delection
Print form
C
A
B


C
A
B
Course registration
Editing and deletion
No
Are you evaluating
The staffs
Yes
Observed achievement
Achievement indicators
Supply the target
No
Have all Depart.
Been evaluated
Yes
Output from evaluation
End

Elimination all outliers/Data conditions to be met

CONDITIONAL IF STATEMENT (3)

This section is only accessible by the ESTAB./PERSONNEL department and can be perform when all associated H.O.Ds/DEANs have approved and submitted all entries under them. (However, should there be a bottleneck from any department, the ESTAB./PERSONNEL department have the right to deactivate such department/Deans office from partaking in the analysis to be performed. This feature can however be made flexible to later accept the defaulting department/deans office when their data is ready for use. This analysis can be remade and made to replace the existing one made within that academic year) When an analysis result performed has been [**APPROVED]** by the Head of the ESTAB./PERSONNEL officer in charge then the output result should be stored in the database and should be made un-alterable.

Note: None-academic staff details in any Faculties office should be processed under the staff in the ESTAB./PERSONNEL office

The follow should however follow when all data have confirm submitted for use

- Test for data fitness (We need to perform F-test) but first

1) Check for outliers Department-by-Department. Then eliminate all outliers found.
2) Avoid using (i.e. do not use) data sample size from any department whose data size is less than 15 (i.e. the data must be greater-than > 15)

Note: the dean is only the academic staff in the faculty office whose details and any other academic staff occupying a role/office under the faculty be captured by his department. For Institutions whose academic departmental staff are less than 15 the data entry should be done Faculty wise.

3) The remaining data after outliers have been remove should be subjected to random sampling Department-by-Department.
4) Calculate the SKEWNESS and KURTOSIS values for each department to check for NORMALITY .

If SKEWNESS and KURTOSIS values are zero or close to zero, it indicates NORMALITY of dataset. Accept for ranges $-1.0 \geq x \leq +1.0$ that is x for SKEWNESS and KURTOSIS must be greater than -1.0 which means (a value from -1 to 0) and x must be less than +1.0 which means (a value from 0 to +1.0). Anything outside these indicates NON-NORMALITY in distribution.

If condition 4 is satisfied then calculate the F-value using alpha = 0.05 but if condition 4 is not met the invoke a more stringent method either by using alpha level of $P < 0.001$ or adopting the KOLMOGOROV-SIMIRNOV and SHAPIW-WIK test or use the BROWN-FORSTY method for the NORMALITY test. Also, you can determine **Fmax** using HERTLEY test to <u>HOMOGENUITY</u> or the presence of HETEROGENERITY of variance in the dataset.

If Fmax > 3.0 it indicates HETEROGENEITY implying that a more strigent value of alpha level of 0.025 is to be applied for the determination of the F-value.

Note: The only time alpha can be change from 0.05 to any other value (such as 0.025 or even 0.01) is when the SKEWNESS, KURTOSIS and HETEROGENEITY of variance have shown that a more stringent method be used or applied for performing F-test on the adopted data to decide whether to accept or reject the data for computational purpose.

METHODS and FORMULAS OF NEEDED FOR SET 3

➢ How to determine outliers

Multiplying the interquartile range (IQR) by 1.5 will give us a way to determine whether a certain value is an outlier. If we subtract 1.5 x IQR from the first quartile, any data values that are less than this number are considered outliers.

- ➢ Steps to determine the interquartile range of a dataset

Steps:

Example

Rank	Value
1	6
2	7
3	15
4	36
5	39
6	41
7	41
8	43
9	43
10	47
11	49

1. **Step 1:** Put the numbers in order. ...

   i.e. (data values in ascending order)

2. Step 2: Find the median. ...

   i.e. Then you need to find the **rank of the median** to split the data set in two. If the number of data points is an **uneven value**, the rank of the median will be

   (n + 1) ÷ 2 = for example if the total number of data available is 11 then the rank will be (11 + 1) ÷ 2 = 6

   i.e the data occupying the sixth position is at the rank needed of the dataset to split the dataset into two equal half.

   But if the number of dataset is even in value then that is your **median (same rule applies in determining the lower and upper quartile)**

3. Step 3: Place parentheses around the numbers above and below the median. Not necessary statistically, but it makes Q1 and Q3 easier to spot. ...

   Then you need to split the lower half of the data in two again to **find the lower quartile**. The lower quartile will be the point of rank (5 + 1) ÷ 2 = 3. The result is Q1 = 15. The second half must also be split in two to find the value of the **upper quartile**. The rank of the upper quartile will be 6 + 3 = 9. So Q3 = 43.

4. Step 4: Find Q1 and Q3. ...
5. Step 5: Subtract Q1 from Q3 to find the interquartile range.

> Once you have the quartiles, you can easily measure the **spread**. The **interquartile range** (**Spread**) will be Q3 - Q1 (**upper quartile - lower quartile**), which gives 28 (43-15). The **semi-interquartile range** is 14 (28 ÷ 2) i.e (**Spread/2**) and the range is 43 (49-6) (Value in last position of dataset – Value in first position of dataset).

CONCLUSION: To check if there is any data which act as an outlier in the dataset

Determination of lower bound outliers

(IQR) by 1.5 - lower quartile

28 x 1.5 = 42 – 15 = 27

So therefore the first three values (6, 7, 15) should be discarded to perform our ANOVA (That is Analysis of Variance).

Determination of Upper bound outliers

(IQR) by 1.5 + Upper quartile

28 x 1.5 = 42 + 43 = 85

So therefore there is no value to be discarded in this regard (upper limits) to perform our ANOVA (That is Analysis of Variance).

DETAIL OF HOW TESTS IS TO BE PERFORMED ON THE DATASET (ASSUMPTION OF ERRORS VIOLATION TEST) BEFORE TAKING DECISION WHETHER TO PROCEED WITH THE ANALYSIS TO BE PERFORMED. (THE OPTION ARE TO USE **F-TEST** and **F* TEST** TO DECIDE)

A) ERROR INDEPENDENCE

3) Do a random sampling of the dataset for each group population (i.e. for each department)

4) Avoid the use of sampling size (dataset size from each department) that is less than 15 (i.e. use > 15)

Note: keep record of the department name whose dataset was not used. This will be displayed among the result.

Note: though number 3 & 4 above are useful for **avoiding the violation of the assumption of error independence**, The also play roles for avoiding the violation of Normality. Another important thing to note under assumption of error independence is that

5) SSE(R) ≥ SSE(F) i.e. SSE(R) can never be less than SSE(F). Where it is observed then the assumption of error independence has been violated.

B) NORMALITY OF ERROR

6) Plot the Skewness and Kurtosis of each dataset (as well as the combined dataset) so as to access the data Normality.

Note: Skewness and Kurtosis values that are zero or close to zero for any dataset indicates that that dataset is Normally distributed while values greater than or less than +1.0 and -1.0 respectively are considered indicative of a Normally distributed variable. Hence a more stringent approach must be applied. i.e. when determining the F-value the alpha level must be tougher (table of alpha of level 0.05 can't be used). A stringent alpha level of P < 0.001 should be used. When considering the T-distribution.

$$\text{Skewness} = \frac{n\sum_i^n (Y_i - \bar{Y})^3}{\sum_i^n (Y_i - \bar{Y})^3} \qquad \text{or}$$

$$\text{Skewness} = \frac{3(mean - median)}{Standard\ deviation}$$

Skewness Formula

$$\frac{n\sum_i^n (Y_i - \bar{Y})^4}{\sum_i^n (Y_i - \bar{Y})^4}$$

Kurtosis Formula

Where **Skewness** and **Kurtosis** are observed on a dataset which violates (i.e. -1.0 ≥ x ≤ +1.0) the use **Kolmogorov-Smirnov (K-S)** test or Shapiro-Wilk test which is due to their sensitivity. The Shapiro-Wilk test is far too powerful and won't be applied in this software. We can also use the **Brown-Forsythe test along with Kolmogorov-Smirnov test**.

$$F_0(\mathrm{X}) = K_n = \frac{(No.of\ observation \le \mathrm{X})}{Total\ number\ of\ observation}$$

Kolmogorov- Smirnov (K- S) Formula

K-S value of Maximum = 1.0 says the fit is good

K-S value of Maximum = 0.0 says the fit is not good

$$Mean = \bar{X} = \left(\sum_{i=1}^{N} X_i\right) \big/ N$$

Standard deviation formula of population

$$\sigma = \sqrt{\frac{(\mathrm{X}_1 - \bar{\mathrm{X}})^2 + (\mathrm{X}_2 - \bar{\mathrm{X}})^2 + --- (\mathrm{X}_n - \bar{\mathrm{X}})^2}{n}}$$

Standard deviation formula of the sample

$$\mathrm{S} = \sqrt{\frac{(\mathrm{X}_1-\bar{\mathrm{X}})^2+(\mathrm{X}_2-\bar{\mathrm{X}})^2+---(\mathrm{X}_n-\bar{\mathrm{X}})^2}{n-1}}$$

This express dispersion in the same units as mean

$$SS_{(Total)} = SS_{(BETWEEN)} + SS_{(ERROR)}$$

For a finite sample size N, the non-parametric standard deviation is

$$S = \left(\frac{N}{(N-1)}\right)(\sigma^2)^{\frac{1}{2}}$$

Formula for sum of square (SS)

In statistics $SS = \sum(X_i + \bar{X})^2$

$$SS_n = \frac{[n(n+1)(2n+1)]}{6}$$

Where n = 1, 2, 3, _ _ _ n

$SS_n = 1^2 + 2^2 + 3^2 \,\_\_\_\, n^2$

For mean sum of squares between the Groups $MS_B = SS_{(Between)}/M\text{-}1$

$F_{ratio} = MS_B/MS_E$

Analysis of Variance					
Source	DF	SS	MS	F	P
Regression	1=(n-1)-(n-2)	504.04(SSTR=SSR)	504.04	33.5899	0.000
Error	48=(n-2)	720.27(SSE(F)=SSE)	15.006		
Total	49	1224.32(SSE(R)=SSTO)	-		

In this example n=50 and it is a one-way sample test

$$F - Test = \frac{SSTR/(M-1)}{SSE/(n-M)} \; or \; \frac{SSR/M-1}{SSTO/n-M}$$

$$R^2 = \frac{SSR}{SSTO} = 1 - \frac{SSE}{SSTO}$$

For a residual data test against a full data test the following condition applies

$SSE(R) = \sum(y_i - \bar{y})^2 = SSTO$ note $df_R = n\text{-}1$

$SSE(F) = \sum(y_i - \hat{y})^2 = SSE$ note df_F = n-2

$$F^* = \left(\frac{SSE(R) - SSE(F)}{dF_R - dF_F}\right) / \left(\frac{SSE(F)}{dF_F}\right)$$

df_R = n-1 SSE(R)=SSTO

df_F = n-2 SSE(F) = SSE

$$F^* = \frac{\left(\frac{SSTO - SSE}{(n-1)-(n-2)}\right)}{\left(\frac{SSE}{(n-2)}\right)} = \frac{MSR}{MSE}$$

$$MSR = \frac{SS(R)}{DF}$$

$$MSE = \frac{SSE}{dF}$$

$$F = \frac{SS_{BG}/(g-1)}{SS_{wG}/(n-g)} = \frac{MS_{BG}}{MS_{WG}} = \frac{R^2/(g-1)}{(1-R^2)(n-g)}$$

$$\zeta^2 = \frac{SS_{BG}}{SS_{Total}} = R^2$$

The co-efficient of determination R^2 is

$$R^2 = \frac{SSR}{SSTO} = 1 - \frac{SSE}{SSTO}$$

Analysis of Variance Result				
Source	SS	Df	MS	F
Between Group	9656.49 (M=3)	3	3,218.83	5.869
Within Group	17,549.15(n-M)	32	548.41	
Total	27,205.64(n-1)	35	-	

In this example n=36 and it is a two-way sample test

n=N-1, N = Total sample number, M = Number of group in the test = g

The proportion of total sample variation Y that is explained by its linear relationship with X

Note:

$$0 \le R^2 \le 1$$

$$R^2 = 1 \rightarrow data\ perfectly\ linear$$

$$R^2 = 0 \rightarrow regression\ line\ horizontal\ (b_1 = 0)$$

The closer R^2 is to one, the greater the linear relationship between X and Y.

For $Y = f(x,p) = a + bx$ to determine the amount of work content. Only applicable to linear system

$$b = \frac{n\sum_{i=1}^{n} x_i y_i - (\sum_{i=1}^{n} x_i)(\sum_{i=1}^{n} y_i)}{n\sum_{i=1}^{n}(x_i)^2 - (\sum_{i=1}^{n} x_i)^2}$$

Where x = production or service and

a = intercept of the linear curve

b = gradient of the curve

$$a = \frac{(\sum_{i=1}^{n} y_i) - b\sum_{i=1}^{n} x_i}{n}$$

$Y = a + bx_i$

Kolmogorov- Smirnov (K- S) Formula

One Sample Test

No in each class	B.Sc	B.A	B.com	M.A	M.Com
	5	9	11	16	19

If 1 Students from each class I expected to join the drama club

Streams	Number of student interested in joining the club		$F_0(X)$	$F_T(X)$	$\lvert F_0(X) - F_T(X)\rvert$
	Observed	Theoretical			
B.Sc	5	12	5/60	12/60	7/60
B.A	9	12	14/60	24/60	10/60
B.com	11	12	25/60	36/60	11/60
M.A	16	12	41/60	48/60	7/60
M.Com	19	12	60/60	60/60	0/60
Total	n=60				

Test Statistic $|D|$

$$D = Maximize|F_0(X) - F_T(X)| = \frac{11}{60} = 0.183$$

From K-S table values for one sample test D at 5% significance level is $D_{0.05} = \frac{1.36}{\sqrt{n}}$ and n=60

$$D_{0.05} = \frac{1.36}{\sqrt{60}} = 0.175$$

Summary: Since the calculated value is 0.183 is greater than the critical value 0.175, hence we reject the null hypothesis and conclude that there is a difference among students of the different stream in their intention of joining the club.

Example (Kolmogorov- Smirnov)

C) HOMOGENEITY OF VARIANCE OR HOMOSCEDASTICITY (LEVENE'S TEST)

The violation of this assumption is called Heterogeneity of variance or Heteroscedasticity. To determine if this error exist we check for Fmax. Fmax is used to gauge the level of homoscedasticity. The Levene's test is an inferential statistic used to assess the equality of variances for a variable calculated for two or more groups. Hence, it tests the null hypothesis that the population variances are equal. If the resulting p-value of the test is less than some significance level (typically 0.05), the obtained differences in the sample variances are unlikely to have occurred based on random sampling from a population with equal variances.

$$F_{\max} = \frac{\mathbf{S}^2_{largest}}{\mathbf{S}^2_{\text{smallest}}}$$

Formular for F_{max}

If Fmax is $>$ 3 it indicates heterogeneity, then we need to use a more stringent alpha (α) of 0.025 for determining F-value.

NOTE: ITEM NUMBERS 3 TO 6 POINTS TOWARDS USING **F-TEST** FOR THE DECISION MAKING

7) The BROWN-FORSYTHE test is useful when the variance across the different groups are not equal. i.e. with the same sample sizes for each group F* = F but the denomination degrees of freedom will be different. It is a test for the equality (from an ordinary one-way analysis of variance on the absolute deviations) of group variances based on performing an Analysis of Variance

$$F^* = \frac{SS_B}{\sum_j \left(1 - \frac{n_j}{n}\right) S_j^2} \text{ or}$$

$$F^* = \frac{(N-g)\sum_{k=1}^{g} n_k (Z_k - \bar{Z})^2}{(g-1)\left\{\sum_{k=1}^{g}\sum_{i=1}^{n_k}(Z_{ki} - \bar{Z}_k)^2\right\}}$$

where:

$$Z_{ki} = \left|Y_{ki} - \tilde{Y}_k\right|$$

$$\bar{Z}_k = \frac{1}{n_k}\sum_{i=1}^{n_k} Z_{ki}$$

$$\bar{Z} = \frac{1}{N}\sum_{k=1}^{g}\sum_{i=1}^{n_k} Z_{ki}$$

$$\tilde{Y}_k = median\left(Y_{k1}, Y_{k2}, Y_{k3}, \dots, Y_{kn_k}\right)$$

If the assumptions are met, the distribution of this test statistic follows approximately the F distribution with degrees of freedom g – 1 and N – g.

Formula for BROWN-FORSYTHE test (F*)

Then F* ~ F(K -1, df) where the degree of freedom (also referred to as df*) are

$$df = \frac{1}{\sum_j\left(\frac{M_j^2}{n_j - 1}\right)}$$

$$M_j = \frac{\left(1 - \frac{n_j}{n}\right)S_j^2}{\sum_j\left(1 - \frac{n_j}{n}\right)S_j^2}$$

With the same sample size for each group, F* = F, but the denomination degrees will be different.

Note: When the variance are unequal for the data in each group (cell) of a class (e.g. Institution) F will be biased, especially when the cell size are unequal. In this case F* remains unbiased but valid.

Hence, for each Department (cell), check for their variance. If the variance are not equal, then using the F-test will be biased especially in the case when the cell size are unequal. So for such a case use F* to decide whether to reject or adopt the dataset for analysis.

IN SUMMARY CARRY OUT ALL THE FOLLOWING

1. Do a random sampling of the dataset for each group population

2. Avoid the use of sampling size (dataset size from each department) that is less than 15

3. Perform or calculate SSE(R)

4. Perform or calculate SSE(F)

Note: If SSE(R) ≥SSE(F) Where it is observed that SSE(F) is been greater than SSE(R) then the assumption of error independence has been violated.

5. Find the Skewness and Kurtosis value of the dataset.

6. Find if there is the violation of the assumption called Heterogeneity of variance or Heteroscedasticity by determining **Fmax**

7. For each Department (cell), check for their variance. If the variance are not equal, then using the F-test will be biased especially in the case when the cell size are unequal. So for such a case use F* to decide whether to reject or adopt the dataset for analysis.

Note: If values are within $-1.0 > x \leq +1.0$ then data is normally distributed we can proceed to checking the acceptance of the data for analysis using **F-Test** for alpha level of 0.05. If otherwise, make the alpha level more stringent for the F-value (such as using alpha level of 0.01 e.t.c) determination. If Fmax in item 6 under summary is > 3 it indicates heterogeneity, then we need to use a more stringent alpha level of 0.025 for determining F-value. Better still go to 8 below.

Note: When the variance are unequal for the data in each group (cell) of a class (Institution) F will be biased, especially when the cell size are unequal. In this case F* remains unbiased but valid

8. In place of a more stringent alpha level in using the F-Test method to decide, we can use the BROWN-FORSYTHE TEST to determine F*

ANALYSIS FOR STRESS DETERMINATION

Results and Discussion:

Using ANOVA (Analysis of variance) to analyze the workload for the faculty of Art.

The reason for this is to determine if the data collected from the faculty of Art actually spells out the workload experienced by the lecturer in the faculty.

$\bar{\bar{X}} = 1/n \Sigma \Sigma X = 1/74 (2837) = 38.34$

$SSTO = \Sigma \Sigma X^2 - n \bar{\bar{X}}^2$

$(2837)^2 - 74(38.34)^2 = 7939792.286$

$SSTR = \Sigma n_j \bar{X}_j^2 - n\bar{\bar{X}}^2$

$= [8(320) + 4(100) + 13(590) + 6(220) + 14(510) + 10(335) + 8(312) + 4(200)$

$+ 7(250)] - 74(38.34)^2 = (27486 - 108776.7144)$

$= -81290.7144$

$SSE = SSTO - SSTR$

$= 7939792.286 - [-81290.7144]$

$= 7939792.286 + 81290.7144$

$= 8021083$

$$F - test = \frac{SSTR/(m-1)}{SSE/(n-m)}$$

Note:

m = no of treatment (sample)

n = Total no of observations for all sample used

$$F - Test = \frac{-81290.7144/(9-1)}{8021083/(74-9)}$$

$= -0.08234$

F_{α} -----[8, 65]

Hence for α = 0.05 the rejection region is F > 2.89. since F = -0.08234 which is less than 2.89, then **accept the data for the model application.**

The research and Admin/Com. service analysis also carried out showed that the data collected from the faculty can be can applied to the model.

Critical values of F for the 0.05 significance level:

DF_1	1	2	3	4	5	6	7	8	9	10	15	20	30	40	60	120	∞
DF_2																	
1	161.5	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9	246.0	248.0	250.1	251.1	252.2	253.3	254.3
2	18.51	19.00	19.16	19.26	19.30	19.33	19.35	19.37	19.39	19.40	19.43	19.45	19.46	19.47	19.48	19.49	19.50
3	10.13	9.552	9.277	9.117	9.014	8.941	8.887	8.845	8.812	8.786	8.703	8.660	8.617	8.594	8.572	8.549	8.526
4	7.709	6.944	6.591	6.388	6.256	6.163	6.094	6.041	5.999	5.964	5.858	5.803	5.746	5.717	5.688	5.658	5.628
5	6.608	5.786	5.410	5.192	5.050	4.950	4.876	4.818	4.773	4.735	4.619	4.558	4.496	4.464	4.431	4.399	4.365
6	5.987	5.143	4.757	4.534	4.387	4.284	4.207	4.147	4.099	4.060	3.938	3.874	3.808	3.774	3.740	3.705	3.669
7	5.591	4.737	4.347	4.120	3.972	3.866	3.787	3.726	3.677	3.637	3.511	3.445	3.376	3.340	3.304	3.267	3.230

8	5.318	4.459	4.066	3.838	3.688	3.581	3.501	3.438	3.388	3.347	3.218	3.150	3.079	3.043	3.005	2.967	2.928
9	5.117	4.257	3.863	3.633	3.482	3.374	3.293	3.230	3.179	3.137	3.006	2.937	2.864	2.826	2.787	2.748	2.707
10	4.965	4.103	3.708	3.478	3.326	3.217	3.136	3.072	3.020	2.978	2.845	2.774	2.700	2.661	2.621	2.580	2.538
11	4.844	3.982	3.587	3.357	3.204	3.095	3.012	2.948	2.896	2.854	2.719	2.646	2.571	2.531	2.490	2.448	2.405
12	4.747	3.885	3.490	3.259	3.106	2.996	2.913	2.849	2.796	2.753	2.617	2.544	2.466	2.426	2.384	2.341	2.296
13	4.667	3.806	3.411	3.179	3.025	2.915	2.832	2.767	2.714	2.671	2.533	2.459	2.380	2.339	2.297	2.252	2.206
14	4.600	3.739	3.344	3.112	2.958	2.848	2.764	2.699	2.646	2.602	2.463	2.388	2.308	2.266	2.223	2.178	2.131
15	4.543	3.682	3.287	3.056	2.901	2.791	2.707	2.641	2.588	2.544	2.403	2.328	2.247	2.204	2.160	2.114	2.066
16	4.494	3.634	3.239	3.007	2.852	2.741	2.657	2.591	2.538	2.494	2.352	2.276	2.194	2.151	2.106	2.059	2.010
17	4.451	3.592	3.197	2.965	2.810	2.699	2.614	2.548	2.494	2.450	2.308	2.230	2.148	2.104	2.058	2.011	1.960
18	4.414	3.555	3.160	2.928	2.773	2.661	2.577	2.510	2.456	2.412	2.269	2.191	2.107	2.063	2.017	1.968	1,917
19	4.381	3.522	3.127	2.895	2.740	2.628	2.544	2.447	2.423	2.378	2.234	2.156	2.071	2.026	1.980	1.930	1.878
20	4.351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348	2.203	2.124	2.039	1.994	1.946	1.896	1.843
21	4.325	3.467	3.073	2.840	2.685	2.573	2.488	2.421	2.366	2.321	2.176	2.096	2.010	1.965	1.917	1.866	1.812
22	4.301	3.443	3.049	2.817	2.661	2.549	2.464	2.397	2.342	2.297	2.151	2.071	1.984	1.938	1.889	1.838	1.783
23	4.279	3.422	3.028	2.796	2.640	2.528	2.442	2.375	2.320	2.275	2.128	2.048	1.961	1.914	1.865	1.813	1.757
24	4.260	3.403	3.009	2.776	2.621	2.508	2.423	2.355	2.300	2.255	2.108	2.027	1.939	1.892	1.842	1.790	1.733
25	4.242	3.385	2.991	2.759	2.603	2.490	2.405	2.337	2.282	2.237	2.089	2.008	1.919	1.872	1.822	1.768	1.711
26	4.225	3.369	2.975	2.743	2.587	2.474	2.388	2.321	2.266	2.220	2.072	1.990	1.901	1.853	1.803	1.749	1.691
27	4.210	3.354	2.960	2.728	2.572	2.459	2.373	2.305	2.250	2.204	2.056	1.974	1.884	1.836	1.785	1.731	1.672
28	4.196	3.340	2.947	2.714	2.558	2.445	2.359	2.291	2.236	2.190	2.041	1.958	1.869	1.820	1.769	1.714	1.654
29	4.183	3.328	2.934	2.701	2.545	2.432	2.346	2.278	2.223	2.177	2.028	1.945	1.854	1.806	1.754	1.698	1.638
30	4.171	3.316	2.922	2.690	2.534	2.421	2.334	2.266	2.211	2.165	2.015	1.932	1.841	1.792	1.740	1.684	1.622
40	4.085	3.232	2.839	2.606	2.450	2.336	2.249	2.180	2.124	2.077	1.925	1.839	1.745	1.693	1.637	1.577	1.508
60	4.001	3.150	2.758	2.525	2.368	2.254	2.167	2.097	2.040	1.993	1.836	1.748	1.649	1.594	1.534	1.467	1.389
120	3.920	3.072	2.680	2.447	2.290	2.175	2.087	2.016	1.959	1.911	1.751	1.659	1.554	1.495	1.429	1.352	1.254
∞	3.842	2.996	2.605	2.372	2.214	2.099	2.010	1.938	1.880	1.831	1.666	1.571	1.459	1.394	1.318	1.221	1.000

Critical values of F for the 0.05 significance level:

DF_1	1	2	3	4	5	6	7	8	9	10
DF_2										
1	161.5	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9
2	18.51	19.00	19.16	19.26	19.30	19.33	19.35	19.37	19.39	19.40
3	10.13	9.552	9.277	9.117	9.014	8.941	8.887	8.845	8.812	8.786
4	7.709	6.944	6.591	6.388	6.256	6.163	6.094	6.041	5.999	5.964
5	6.608	5.786	5.410	5.192	5.050	4.950	4.876	4.818	4.773	4.735
6	5.987	5.143	4.757	4.534	4.387	4.284	4.207	4.147	4.099	4.060
7	5.591	4.737	4.347	4.120	3.972	3.866	3.787	3.726	3.677	3.637
8	5.318	4.459	4.066	3.838	3.688	3.581	3.501	3.438	3.388	3.347
9	5.117	4.257	3.863	3.633	3.482	3.374	3.293	3.230	3.179	3.137

10	4.965	4.103	3.708	3.478	3.326	3.217	3.136	3.072	3.020	2.978
11	4.844	3.982	3.587	3.357	3.204	3.095	3.012	2.948	2.896	2.854
12	4.747	3.885	3.490	3.259	3.106	2.996	2.913	2.849	2.796	2.753
13	4.667	3.806	3.411	3.179	3.025	2.915	2.832	2.767	2.714	2.671
14	4.600	3.739	3.344	3.112	2.958	2.848	2.764	2.699	2.646	2.602
15	4.543	3.682	3.287	3.056	2.901	2.791	2.707	2.641	2.588	2.544
16	4.494	3.634	3.239	3.007	2.852	2.741	2.657	2.591	2.538	2.494
17	4.451	3.592	3.197	2.965	2.810	2.699	2.614	2.548	2.494	2.450
18	4.414	3.555	3.160	2.928	2.773	2.661	2.577	2.510	2.456	2.412
19	4.381	3.522	3.127	2.895	2.740	2.628	2.544	2.447	2.423	2.378
20	4.351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348
21	4.325	3.467	3.073	2.840	2.685	2.573	2.488	2.421	2.366	2.321
22	4.301	3.443	3.049	2.817	2.661	2.549	2.464	2.397	2.342	2.297
23	4.279	3.422	3.028	2.796	2.640	2.528	2.442	2.375	2.320	2.275
24	4.260	3.403	3.009	2.776	2.621	2.508	2.423	2.355	2.300	2.255
25	4.242	3.385	2.991	2.759	2.603	2.490	2.405	2.337	2.282	2.237
26	4.225	3.369	2.975	2.743	2.587	2.474	2.388	2.321	2.266	2.220
27	4.210	3.354	2.960	2.728	2.572	2.459	2.373	2.305	2.250	2.204
28	4.196	3.340	2.947	2.714	2.558	2.445	2.359	2.291	2.236	2.190
29	4.183	3.328	2.934	2.701	2.545	2.432	2.346	2.278	2.223	2.177
30	4.171	3.316	2.922	2.690	2.534	2.421	2.334	2.266	2.211	2.165
31	4.16	3.31	2.91	2.68	2.52	2.41	2.32	2.26	2.20	2.15
32	4.15	3.30	2.90	2.67	2.51	2.40	2.31	2.24	2.19	2.14
33	4.14	3.29	2.89	2.66	2.50	2.39	2.30	2.24	2.18	2.13
34	4.13	3.28	2.88	2.65	2.49	2.38	2.29	2.23	2.17	2.12
35	4.12	3.27	2.87	2.64	2.49	2.37	2.29	2.22	2.16	2.11
36	4.11	3.26	2.87	2.63	2.48	2.36	2.28	2.21	2.15	2.11
37	4.11	3.25	2.86	2.63	2.47	2.36	2.27	2.20	2.15	2.10
38	4.10	3.25	2.85	2.62	2.46	2.35	2.26	2.19	2.14	2.09
39	4.09	3.24	2.85	2.61	2.46	2.34	2.26	2.19	2.13	2.08
40	4.085	3.232	2.839	2.606	2.450	2.336	2.249	2.180	2.124	2.077
41	4.08	3.23	2.83	2.60	2.44	2.33	2.24	2.17	2.12	2.07
42	4.07	3.22	2.83	2.59	2.44	2.32	2.24	2.17	2.11	2.07
43	4.07	3.21	2.82	2.59	2.43	2.32	2.23	2.16	2.11	2.06
44	4.06	3.21	2.82	2.58	2.43	2.31	2.23	2.16	2.10	2.05
45	4.06	3.20	2.81	2.58	2.42	2.31	2.22	2.15	2.10	2.05
46	4.05	3.20	2.81	2.57	2.42	2.30	2.22	2.15	2.09	2.04
47	4.05	3.20	2.80	2.57	2.41	2.30	2.21	2.14	2.09	2.04
48	4.04	3.19	2.80	2.57	2.41	2.30	2.21	2.14	2.08	2.04
49	4.04	3.19	2.79	2.56	2.40	2.29	2.20	2.13	2.08	2.03
50	4.03	3.18	2.79	2.56	2.40	2.29	2.20	2.13	2.07	2.03
51	4.03	3.18	2.79	2.55	2.40	2.28	2.20	2.13	2.07	2.02
52	4.03	3.18	2.78	2.55	2.39	2.28	2.19	2.12	2.07	2.02

53	4.02	3.17	2.78	2.55	2.39	2.28	2.19	2.12	2.06	2.02
54	4.02	3.17	2.78	2.54	2.39	2.27	2.19	2.12	2.06	2.01
55	4.02	3.17	2.77	2.54	2.38	2.27	2.18	2.11	2.06	2.01
56	4.01	3.16	2.77	2.54	2.38	2.27	2.18	2.11	2.05	2.01
57	4.01	3.16	2.77	2.53	2.38	2.26	2.18	2.11	2.05	2.00
58	4.01	3.16	2.76	2.53	2.37	2.26	2.17	2.10	2.05	2.00
59	4.00	3.15	2.76	2.53	2.37	2.26	2.17	2.10	2.04	2.00
60	4.001	3.150	2.758	2.525	2.368	2.254	2.167	2.097	2.040	1.993
61	4.00	3.15	2.76	2.52	2.37	2.25	2.16	2.09	2.04	1.99
62	4.00	3.15	2.75	2.52	2.36	2.25	2.16	2.09	2.04	1.99
63	3.99	3.14	2.75	2.52	2.36	2.25	2.16	2.09	2.03	1.99
64	3.99	3.14	2.75	2.52	2.36	2.25	2.16	2.09	2.03	1.98
65	3.99	3.14	2.75	2.51	2.36	2.24	2.15	2.08	2.03	1.98
66	3.99	3.14	2.74	2.51	2.35	2.24	2.15	2.08	2.03	1.98
67	3.98	3.13	2.74	2.51	2.35	2.24	2.15	2.08	2.02	1.98
68	3.98	3.13	2.74	2.51	2.35	2.24	2.15	2.08	2.02	1.97
69	3.98	3.13	2.74	2.51	2.35	2.24	2.15	2.08	2.02	1.97
70	3.98	3.13	2.74	2.50	2.35	2.23	2.14	2.07	2.02	1.97
71	3.98	3.13	2.73	2.50	2.34	2.23	2.14	2.07	2.02	1.97
72	3.97	3.12	2.73	2.50	2.34	2.23	2.14	2.07	2.01	1.97
73	3.97	3.12	2.73	2.50	2.34	2.23	2.14	2.07	2.01	1.96
74	3.97	3.12	2.73	2.50	2.34	2.22	2.14	2.07	2.01	1.96
75	3.97	3.12	2.73	2.49	2.34	2.22	2.13	2.06	2.01	1.96
76	3.97	3.12	2.73	2.49	2.34	2.22	2.13	2.06	2.01	1.96
77	3.97	3.12	2.72	2.49	2.33	2.22	2.13	2.06	2.00	1.96
78	3.96	3.11	2.72	2.49	2.33	2.22	2.13	2.06	2.00	1.95
79	3.96	3.11	2.72	2.49	2.33	2.22	2.13	2.06	2.00	1.95
80	3.96	3.11	2.72	2.49	2.33	2.21	2.13	2.06	2.00	1.95
81	3.96	3.11	2.72	2.48	2.33	2.21	2.13	2.06	2.00	1.95
82	3.96	3.11	2.72	2.48	2.33	2.21	2.12	2.05	2.00	1.95
83	3.96	3.11	2.72	2.48	2.32	2.21	2.12	2.05	2.00	1.95
84	3.96	3.11	2.71	2.48	2.32	2.21	2.12	2.05	1.99	1.95
85	3.95	3.10	2.71	2.48	2.32	2.21	2.12	2.05	1.99	1.94
86	3.95	3.10	2.71	2.48	2.32	2.21	2.12	2.05	1.99	1.94
87	3.95	3.10	2.71	2.48	2.32	2.21	2.12	2.05	1.99	1.94
88	3.95	3.10	2.71	2.48	2.32	2.20	2.12	2.05	1.99	1.94
89	3.95	3.10	2.71	2.47	2.32	2.20	2.11	2.04	1.99	1.94
90	3.95	3.10	2.71	2.47	2.32	2.20	2.11	2.04	1.99	1.94
91	3.95	3.10	2.71	2.47	2.32	2.20	2.11	2.04	1.98	1.94
92	3.95	3.10	2.70	2.47	2.31	2.20	2.11	2.04	1.98	1.94
93	3.94	3.09	2.70	2.47	2.31	2.20	2.11	2.04	1.98	1.93
94	3.94	3.09	2.70	2.47	2.31	2.20	2.11	2.04	1.98	1.93
95	3.94	3.09	2.70	2.47	2.31	2.20	2.11	2.04	1.98	1.93

96	3.94	3.09	2.70	2.47	2.31	2.20	2.11	2.04	1.98	1.93
97	3.94	3.09	2.70	2.47	2.31	2.19	2.11	2.04	1.98	1.93
98	3.94	3.09	2.70	2.47	2.31	2.19	2.10	2.03	1.98	1.93
99	3.94	3.09	2.70	2.46	2.31	2.19	2.10	2.03	1.98	1.93
100	3.94	3.09	2.70	2.46	2.31	2.19	2.10	2.03	1.98	1.93

CRITICAL VALUES OF F FOR THE 0.01 SIGNIFICANCE LEVEL:

DF_1	1	2	3	4	5	6	7	8	9	10
DF_2										
1	4052.19	4999.52	5403.34	5624.62	5763.65	5858.97	5928.33	5981.1	6022.5	6055.85
2	98.5	99	99.17	99.25	99.3	99.33	99.36	99.37	99.39	99.4
3	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35	27.23
4	21.2	18	16.69	15.98	15.52	15.21	14.98	14.8	14.66	14.55
5	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16	10.05
6	5.987	10.93	9.78	9.15	8.75	8.47	8.26	8.1	7.98	7.87
7	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62
8	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81
9	10.56	8.02	6.99	6.42	6.06	5.8	5.61	5.47	5.35	5.26
10	10.04	7.56	6.55	5.99	5.64	5.39	5.2	5.06	4.94	4.85
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.5	4.39	4.3
13	9.07	6.7	5.74	5.21	4.86	4.62	4.44	4.3	4.19	4.1
14	8.86	6.52	5.56	5.04	4.7	4.46	4.28	4.14	4.03	3.94
15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4	3.9	3.81
16	8.53	6.23	5.29	4.77	4.44	4.2	4.03	3.89	3.78	3.69
17	8.4	6.11	5.19	4.67	4.34	4.1	3.93	3.79	3.68	3.59
18	8.29	6.01	5.09	4.58	4.25	4.02	3.84	3.71	3.6	3.51
19	8.19	5.93	5.01	4.5	4.17	3.94	3.77	3.63	3.52	3.43
20	8.1	5.85	4.94	4.43	4.1	3.87	3.7	3.56	3.46	3.37
21	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51	3.4	3.31
22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35	3.26
23	7.88	5.66	4.77	4.26	3.94	3.71	3.54	3.41	3.3	3.21
24	7.82	5.61	4.72	4.22	3.9	3.67	3.5	3.36	3.26	3.17
25	7.77	5.57	4.68	4.18	3.86	3.63	3.46	3.32	3.22	3.13
26	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.18	3.09
27	7.68	5.49	4.6	4.11	3.79	3.56	3.39	3.26	3.15	3.06
28	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.23	3.12	3.03
29	7.6	5.42	4.54	4.05	3.73	3.5	3.33	3.2	3.09	3.01

30	7.56	5.39	4.51	4.02	3.7	3.47	3.31	3.17	3.07	2.98
31	7.53	5.36	4.48	3.99	3.68	3.45	3.28	3.15	3.04	2.96
32	7.5	5.34	4.46	3.97	3.65	3.43	3.26	3.13	3.02	2.93
33	7.47	5.31	4.44	3.95	3.63	3.41	3.24	3.11	3	2.91
34	7.44	5.29	4.42	3.93	3.61	3.39	3.22	3.09	2.98	2.89
35	7.42	5.27	4.4	3.91	3.59	3.37	3.2	3.07	2.96	2.88
36	7.4	5.25	4.38	3.89	3.57	3.35	3.18	3.05	2.95	2.86
37	7.37	5.23	4.36	3.87	3.56	3.33	3.17	3.04	2.93	2.84
38	7.35	5.21	4.34	3.86	3.54	3.32	3.15	3.02	2.92	2.83
39	7.33	5.19	4.33	3.84	3.53	3.31	3.14	3.01	2.9	2.81
40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89	2.8
41	7.3	5.16	4.3	3.82	3.5	3.28	3.11	2.98	2.88	2.79
42	7.28	5.15	4.29	3.8	3.49	3.27	3.1	2.97	2.86	2.78
43	7.26	5.14	4.27	3.79	3.48	3.25	3.09	2.96	2.85	2.76
44	7.25	5.12	4.26	3.78	3.47	3.24	3.08	2.95	2.84	2.75
45	7.23	5.11	4.25	3.77	3.45	3.23	3.07	2.94	2.83	2.74
46	7.22	5.1	4.24	3.76	3.44	3.22	3.06	2.93	2.82	2.73
47	7.21	5.09	4.23	3.75	3.43	3.21	3.05	2.92	2.81	2.72
48	7.19	5.08	4.22	3.74	3.43	3.2	3.04	2.91	2.8	2.72
49	7.18	5.07	4.21	3.73	3.42	3.2	3.03	2.9	2.79	2.71
50	7.17	5.06	4.2	3.72	3.41	3.19	3.02	2.89	2.79	2.7
51	7.16	5.05	4.19	3.71	3.4	3.18	3.01	2.88	2.78	2.69
52	7.15	5.04	4.18	3.7	3.39	3.17	3.01	2.87	2.77	2.68
53	7.14	5.03	4.17	3.7	3.38	3.16	3	2.87	2.76	2.68
54	7.13	5.02	4.17	3.69	3.38	3.16	2.99	2.86	2.76	2.67
55	7.12	5.01	4.16	3.68	3.37	3.15	2.98	2.85	2.75	2.66
56	7.11	5.01	4.15	3.67	3.36	3.14	2.98	2.85	2.74	2.66
57	7.1	5	4.15	3.67	3.36	3.14	2.97	2.84	2.74	2.65
58	7.09	4.99	4.14	3.66	3.35	3.13	2.97	2.84	2.73	2.64
59	7.09	4.98	4.13	3.66	3.35	3.12	2.96	2.83	2.72	2.64
60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63
61	7.07	4.97	4.12	3.64	3.33	3.11	2.95	2.82	2.71	2.63
62	7.06	4.97	4.11	3.64	3.33	3.11	2.94	2.81	2.71	2.62
63	7.06	4.96	4.11	3.63	3.32	3.1	2.94	2.81	2.7	2.62
64	7.05	4.95	4.1	3.63	3.32	3.1	2.93	2.8	2.7	2.61
65	7.04	4.95	4.1	3.62	3.31	3.09	2.93	2.8	2.69	2.61
66	7.04	4.94	4.09	3.62	3.31	3.09	2.92	2.79	2.69	2.6
67	7.03	4.94	4.09	3.61	3.3	3.08	2.92	2.79	2.68	2.6
68	7.02	4.93	4.08	3.61	3.3	3.08	2.91	2.79	2.68	2.59
69	7.02	4.93	4.08	3.6	3.3	3.08	2.91	2.78	2.68	2.59

70	7.01	4.92	4.07	3.6	3.29	3.07	2.91	2.78	2.67	2.59
71	7.01	4.92	4.07	3.6	3.29	3.07	2.9	2.77	2.67	2.58
72	7	4.91	4.07	3.59	3.28	3.06	2.9	2.77	2.66	2.58
73	7	4.91	4.06	3.59	3.28	3.06	2.9	2.77	2.66	2.57
74	6.99	4.9	4.06	3.58	3.28	3.06	2.89	2.76	2.66	2.57
75	6.99	4.9	4.05	3.58	3.27	3.05	2.89	2.76	2.65	2.57
76	6.98	4.9	4.05	3.58	3.27	3.05	2.88	2.76	2.65	2.56
77	6.98	4.89	4.05	3.57	3.27	3.05	2.88	2.75	2.65	2.56
78	6.97	4.89	4.04	3.57	3.26	3.04	2.88	2.75	2.64	2.56
79	6.97	4.88	4.04	3.57	3.26	3.04	2.87	2.75	2.64	2.55
80	6.96	4.88	4.04	3.56	3.26	3.04	2.87	2.74	2.64	2.55
81	6.96	4.88	4.03	3.56	3.25	3.03	2.87	2.74	2.63	2.55
82	6.95	4.87	4.03	3.56	3.25	3.03	2.87	2.74	2.63	2.55
83	6.95	4.87	4.03	3.55	3.25	3.03	2.86	2.73	2.63	2.54
84	6.95	4.87	4.02	3.55	3.24	3.03	2.86	2.73	2.63	2.54
85	6.94	4.86	4.02	3.55	3.24	3.02	2.86	2.73	2.62	2.54
86	6.94	4.86	4.02	3.55	3.24	3.02	2.85	2.73	2.62	2.53
87	6.94	4.86	4.02	3.54	3.24	3.02	2.85	2.72	2.62	2.53
88	6.93	4.86	4.01	3.54	3.23	3.01	2.85	2.72	2.62	2.53
89	6.93	4.85	4.01	3.54	3.23	3.01	2.85	2.72	2.61	2.53
90	6.93	4.85	4.01	3.54	3.23	3.01	2.85	2.72	2.61	2.52
91	6.92	4.85	4	3.53	3.23	3.01	2.84	2.71	2.61	2.52
92	6.92	4.84	4	3.53	3.22	3	2.84	2.71	2.61	2.52
93	6.92	4.84	4	3.53	3.22	3	2.84	2.71	2.6	2.52
94	6.91	4.84	4	3.53	3.22	3	2.84	2.71	2.6	2.52
95	6.91	4.84	4	3.52	3.22	3	2.83	2.7	2.6	2.51
96	6.91	4.83	3.99	3.52	3.21	3	2.83	2.7	2.6	2.51
97	6.9	4.83	3.99	3.52	3.21	2.99	2.83	2.7	2.6	2.51
98	6.9	4.83	3.99	3.52	3.21	2.99	2.83	2.7	2.59	2.51
99	6.9	4.83	3.99	3.52	3.21	2.99	2.83	2.7	2.59	2.51
100	6.9	4.82	3.98	3.51	3.21	2.99	2.82	2.69	2.59	2.5

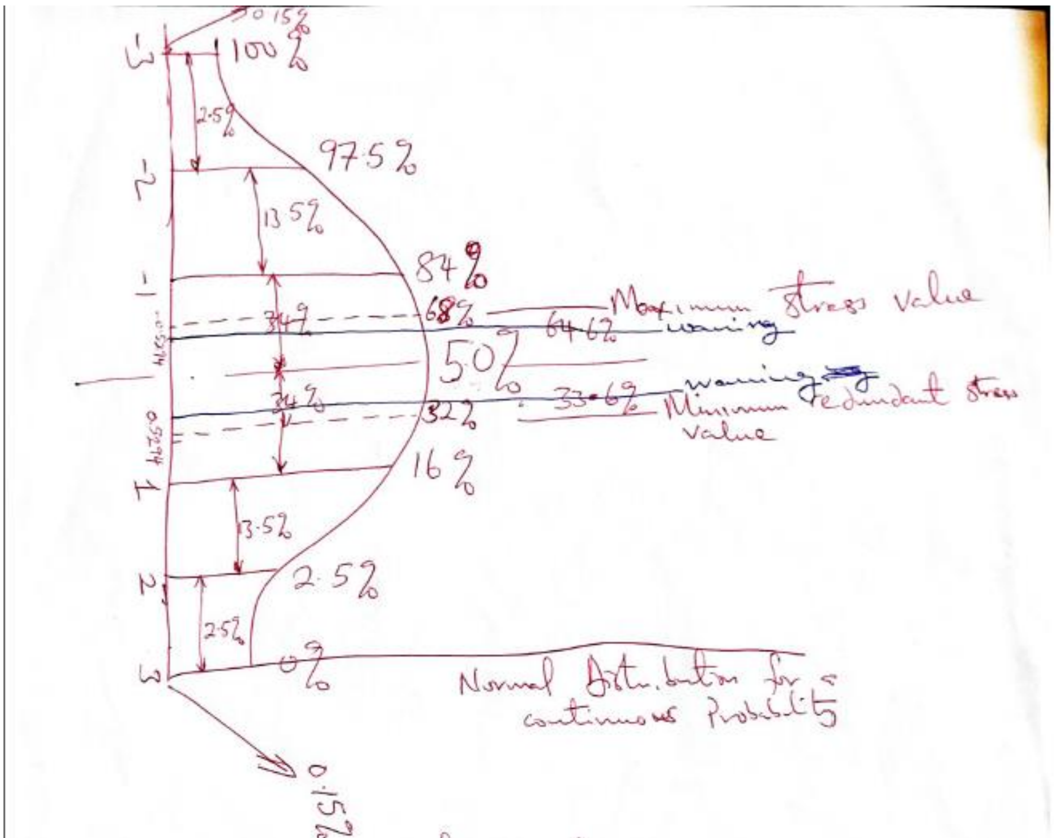
TO DETERMINE THE STRESS THE LECTURERS IN THE FACULTY OF ART IS GOING THROUGH.

In determining the teachers' stress three latent stress variables are used to described it. And these are:

- Students and (Physical) Conditions: Stress arising from interactions with students and from the teaching environment in general.
- Time Pressure: Stress arising from having to do too much in an insufficient time period and work intruding on home life.
- Conflict: Stress arising from staff tensions and role conflict (having to do things which are inconsistent with an individual's expectations).

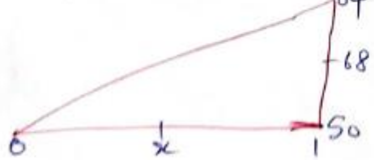
With the use of SPSS (*see appendix*) under three scale factors (Mean, S.D and number of items) the following was observed.

	Student and Physical condition	Time pressure	Conflict
Mean	76.3750	8.6375	3.3463
S.D.	69.7965	3.6291	2.1416
No. Items	80	80	80



For the maximum stress value
$84-16=68\%$
For the minimum redundant stress value
$16+16=32\%$

$$\frac{1-0}{1-x}=\frac{84-50}{84-68} \quad \therefore x = 0.5294$$



The following must always be checked for all faculties (for academic staff) and for Non academic staff

4. Leave Records

(A) Total number of days absent on sick leave during the period coved by this Report	**From**	**To**	**No. of days**
(i) Hospitalization			
(ii) Treatment Received Abroad (where applicable)			
(iii) Sick Leave			
Total			
(B) Maternity Leave			
(C) (i) Annual Leave			
(ii) Casual Leave			
Total number of days spent on Annual/Casual Leave			

ANALYSIS FOR DETERMINATION OF STUDENT-TEACHER RATIO (K*)

$$H = \frac{T_s + T_i + T_A + J}{T + (G - D)} \qquad \text{............................eq.21}$$

Where:

T: Total available teaching – learning man – hours for consultation and assessment of the variable activities.

$T_{s:}$ Effective students' consulting man – hours with respect to formal working hours Per-week.

T_i: Effective lecturers' consultation man – hours with respect to formal working hours Per-week.

J: Effective lecturers' consultation man – hour with respect to non-formal working hours Per-week.

T_A: Actual man – hours lecturer used for student work assessment per week.

A_1: Weekly hours scheduled for formal lectures per week, and

A_2: those scheduled for other K – independent activities notably research and community services during formal working hours per week.

D = total official hours available for all lecturers' formal activities in the week.

G = Total hours available in the week

K: Max number of students per staff

In terms of B and C as defined earlier;

$$T = KB + B + C = B(K+1) + C \qquad \text{............eq.22}$$

$$T_s = K(B - W) \qquad \text{............eq.23}$$

$$T_i = B(1 - P_0) \qquad \text{............eq.24}$$

$$T_A = t_4 K \qquad \text{............eq.25}$$

$$J = (1 - S_0)G - D \qquad \text{............eq.26}$$

B: The weekly hours scheduled for students to consult the lecturer.

C: The hours scheduled for assessing students' work within a week.

Note:

W = expected hours a student waits before receiving lecturer's attention

P_0 = Probability of having no student in the consultation system.

t_4 = Standard hours a lecturer needs to completely assess all work done by a single student in one week.

S_0 = Probability of having gone into more research work outside formal working hours per week. (J/(G-D))

where:

J: hours spent on research outside formal working hours/week

(G-D): Total hours outside formal working hours/week

Substituting all the above expression into eq.2, teaching – learning man – hour utilization function H will become.

$$H = \frac{K\left(B-W\right)+B\left(1-P_0\right)+t_4K+\left(1-S_0\right)G-D}{B\left(K+1\right)+C+\left(G-D\right)}$$ ……………eq.27

B and C is said to depend on how much time is left for non-formal classroom teaching activities.

$$F = D - A_1 = D - Y.\alpha$$ ..…………..eq.28

Thus:

λ = rate at which students consult the lecturer or number of students that comes to a teacher: (some may not necessarily be chanced to see him/her) per hour during consultation period.

μ = rate at which lecturer attends to student or number of students the teacher can attend to per hour during consultation.

Y = average weekly lecture hours per course.

α = number of courses to be lectured per week.

F = non – formal classroom lecture hours per week.

J = non-formal working hour time used for research per week.

S_0 = Proportion of time spent outside D (i.e. Outside the total official hours available for all lecturers'

formal activities in the week).

In determining an expression for B, A_2 and C we have it that

$$B = t_1F = t_1\left(D-Y.\alpha\right)$$ ………eq.29

$$A_2 = t_2F = t_2\left(D-Y.\alpha\right)$$ ……….eq.30

$$C = t_3F = t_3\left(D-Y.\alpha\right)$$ ……….eq.31

And $t_3 = \left[1-\left(t_1+t_2\right)\right]$ hence eq11 becomes

$$C = \left[1-\left(t_1+t_2\right)\right]\left(D-Y.\alpha\right) \qquad \text{……….eq.32}$$

Where $t_{1,}$ t_2 and t_3 are the proportions of F scheduled for B, A_2 and C activities; and is expressed as:

$$t_1+t_2+t_3=1 \qquad \text{(sum of proportions).}$$

Substituting eq29 and eq32 into eq27 will yield

$$H\left(t,K\right)=\frac{K\left[t_1\left(D-Y.\alpha\right)-W\right]+t_1\left(D-Y.\alpha\right)\left(1-P_0\right)+t_4K+\left(1-S_0\right)G-D}{\left(K+1\right)\left(D-Y.\alpha\right)t_1+\left[1-\left(t_1+t_2\right)\right]\left(D-Y.\alpha\right)+\left(G-D\right)} \qquad \text{…….eq.33}$$

W and P_0 are expressed as shown below.

$$W=\frac{\sum_{n=2}^{k}\left(n-1\right)C_n^k.n!\left[\frac{\lambda}{\mu}\right]^n P_0}{\mu\left(1-P_0\right)}+\frac{1}{\mu} \qquad \text{……...eq.34}$$

where:

$$P_0=\left[1+\sum_{n=2}^{k}C_n^k.n!\left[\frac{\lambda}{\mu}\right]^n\right]^{-1} \qquad \text{………eq.35}$$

Substituting W and P_0 into eq.33, H (t, K) will become

$$H\left(t,K\right)=\frac{K\left[t_1\left(D-Y.\alpha\right)-\left(\frac{\sum_{n=2}^{k}\left(n-1\right)C_n^k.n!\left[\frac{\lambda}{\mu}\right]^n P_0}{\mu\left(1-P_0\right)}+\frac{1}{\mu}\right)\right]+t_1\left(D-Y.\alpha\right)\left(1-\left[1+\sum_{n=2}^{k}C_n^k.n!\left[\frac{\lambda}{\mu}\right]^n\right]^{-1}\right)+t_4K+\left(1-S_0\right)G-D}{\left(K+1\right)\left(D-Y.\alpha\right)t_1+\left[1-\left(t_1+t_2\right)\right]\left(D-Y.\alpha\right)+\left(G-D\right)}$$

………………eq.36

Which will further give

$$H(t,K)=\frac{K\left[t_1(D-Y.\alpha)-\left(\frac{\sum_{n=2}^{k}(n-1)C_n^k.n!\left[\frac{\lambda}{\mu}\right]^n\left[1+\sum_{n=2}^{k}C_n^k.n!\left[\frac{\lambda}{\mu}\right]^n\right]^{-1}}{\mu\left(1-\left[1+\sum_{n=2}^{k}C_n^k.n!\left[\frac{\lambda}{\mu}\right]^n\right]^{-1}\right)}+\frac{1}{\mu}\right)\right]}{(K+1)(D-Y.\alpha)t_1+\left[1-(t_1+t_2)\right](D-Y.\alpha)+(G-D)}+\frac{t_1(D-Y.\alpha)\left(1-\left[1+\sum_{n=2}^{k}C_n^k.n!\left[\frac{\lambda}{\mu}\right]^n\right]^{-1}\right)}{(K+1)(D-Y.\alpha)t_1+\left[1-(t_1+t_2)\right](D-Y.\alpha)+(G-D)}$$

$$+\frac{t_4K+(1-S_0)G-D}{(K+1)(D-Y.\alpha)t_1+\left[1-(t_1+t_2)\right](D-Y.\alpha)+(G-D)}$$

…………….eq.37

where :

t_1 proportion of non – classroom teacher’s time for students’ consultation.

t_2 proportion of non – classroom teacher’s time meant for research and community service.

The numerator terms of eq. (27) are:

K(B – W) : expresses the average weekly formal classroom lecture hours per week in consultation with respect to student.

$B(1 – P_0) + t_4K$: expresses the average weekly non - formal classroom lecture hours per week in consultation with respect to lecturer.

$(1 – S_0)G - D$: expresses the average weekly private hours per week in consultation (Research) with respect to lecturer.

(G – D): included in the denominator is the total time left outside the formal classroom lecture hours per week in consultation with respect to lecturer.

Hence it is concluded that for any parameter set, t, the student/teacher ratio problem is

that of determining the value of K which Max H(t,K), stated as standard constrained optimization problem:

Max H(t, K) …………eq.38

Subject to

$t_4K \leq t_3 (D - Y\alpha)$ from assumption i) ………eq.39

$W \leq t_3 (D - Y\alpha)$ from assumption v) ………..eq.40

$\lambda < \mu$ from assumption vii …………eq.41

$J < (G - D)$ from assumption xi …………eq.42

$2 \leq K < \infty$

Hence the properties of H(t, k) and Optimal student – teacher ratio was outlined as follow

$0 \leq H(t, K) \leq 1$, obvious from its definition in equation

H(t, K) is discrete function since K, the variable, is discrete

H(t, K) is concave, having satisfied the concavity conditions of (Gottfried and Weissman; 1973)

$H(t, K^*) \geq H(t, K^* + 1), H(t, K^* - 1)$ follows properties (ii) and (iii); K* stands for true optimal student-teacher ratio.

Trial model for student staff ratio

$$H(t,K)=\frac{K\left[t_1(D-Y.\alpha)-W\right]+t_1(D-Y.\alpha)(1-P_0)+t_4K}{(K+1)(D-Y.\alpha)t_1+\left[1-(t_1+t_2)\right](D-Y.\alpha)} \qquad \text{………eq.13}$$

W and P_0 are expressed as shown below.

$$W=\frac{\sum_{n=2}^{k}(n-1)C_n^k.n!\left[\frac{\lambda}{\mu}\right]^n P_0}{\mu(1-P_0)}+\frac{1}{\mu} \qquad \text{……....eq.14}$$

where:

$$P_0=\left[1+\sum_{n=2}^{k}C_n^k.n!\left[\frac{\lambda}{\mu}\right]^n\right]^{-1} \qquad \text{………eq.15}$$

Substituting eq.14 and eq.15 into eq.13 H (t, K) will become

$$H(t,K)=\frac{K\left[t_1(D-Y.\alpha)-\left(\frac{\sum_{n=2}^{k}(n-1)C_n^k.n!\left[\frac{\lambda}{\mu}\right]^n P_0}{\mu(1-P_0)}+\frac{1}{\mu}\right)\right]+t_1(D-Y.\alpha)\left(1-\left[1+\sum_{n=2}^{k}C_n^k.n!\left[\frac{\lambda}{\mu}\right]^n\right]^{-1}\right)+t_4K}{(K+1)(D-Y.\alpha)t_1+\left[1-(t_1+t_2)\right](D-Y.\alpha)}$$

……..eq16

Which can still be expressed as

$$H(t,K)=\frac{K\left[t_1(D-Y.\alpha)-\left(\frac{\sum_{n=2}^{k}(n-1)C_n^k.n!\left[\frac{\lambda}{\mu}\right]^n\left[1+\sum_{n=2}^{k}C_n^k.n!\left[\frac{\lambda}{\mu}\right]^n\right]^{-1}}{\mu\left(1-\left[1+\sum_{n=2}^{k}C_n^k.n!\left[\frac{\lambda}{\mu}\right]^n\right]^{-1}\right)}+\frac{1}{\mu}\right)\right]}{(K+1)(D-Y.\alpha)t_1+\left[1-(t_1+t_2)\right](D-Y.\alpha)}+\frac{t_1(D-Y.\alpha)\left(1-\left[1+\sum_{n=2}^{k}C_n^k.n!\left[\frac{\lambda}{\mu}\right]^n\right]^{-1}\right)+t_4K}{(K+1)(D-Y.\alpha)t_1+\left[1-(t_1+t_2)\right](D-Y.\alpha)}$$

When the value of K* for when H(t,K) maximum/optimal and we know the student population for a particular school is known then to determine the number of teachers needed to service such number of students we

$$No\ of\ Academic\ staff = \frac{Student\ Population}{Optimal\ Student{-}teacher\ ratio}$$

Then if an academic staff mix is given to be (0.5, 0.3, 0.2) for number of lectures, senior lectures and professors respectively, then we will have for example if number of academic staff is obtained to be 182

- Number of lecturers = .5 x 182 = 91
- Number of senior lecturers = .3 x 182 = 55
- Number of professors = .2 x 182 = 36

Case study

In one university, the following was found to be the distribution of staff in operation position:

Registry	=	672
Maintenance & Works	=	248
Bursary	=	197
Library Services	=	88
Health Services	=	128
Audit	=	77
Security Services	=	84
Total (N_0)	=	1678

And on the study of the human dynamic working parameter $(\boldsymbol{\lambda}, \boldsymbol{\mu})$, the following were observed:

Supervisory level	:	$\boldsymbol{\mu = 15.75}$;	$(\boldsymbol{\lambda = 2.25})$
Management level 1	:	$\boldsymbol{\mu = 12.75}$;	$(\boldsymbol{\lambda = 3.0})$
Management level 2	:	$\boldsymbol{\mu = 3.75}$;	$(\boldsymbol{\lambda = 2.75})$
Top Management level	:	$\boldsymbol{\mu = 1.0}$;	$(\boldsymbol{\lambda = 0.75})$

and it is now required to determine the number of non-teaching staff in the university, excluding laboratory and workshop staff. We are required to first determine the number of supervisory and management staff in each department/unit.

K_1^* (2.25, 15.75) = 11

K_2^* (3.0, 12.75) = 8

K_3* (2.75, 3.75) = 4

K_4* (0.75, 1.0) = 3

So therefore,

Registry

No (registry) in operation position = 672

Number of supervisory staff (N_1*) = $\frac{No\ (registry)}{K_1^*(2.25,\ \ 15.75)}$

= $672/_{11} = 61$

Registry

No management staff level 1 (registry) = $\frac{N_2^*{=}61}{K_2^*(3.0,\ \ 12.75)}$

= $61/_{8} = 8$

Registry

No management staff level 2 (registry) = $\frac{N_3^*{=}8}{K_3^*(2.75,\ \ 3.75)}$

= $8/_{4} = 2$

Registry

Top management staff level (registry) = $\frac{N_4^*{=}2}{K_4^*(0.75,\ \ 1.0)}$

= $2/_{2} = 1$

The total number of staff in registry = N_0 + N_1* + N_2* + N_3* + N_4*

= 672 + 61 + 8 + 2 + 1 = 744

If for any cogent reason, the school decides to have a cut or deduction of staff requirements by 50 percent. It will then be determined by:

operation position = 50% of 672 = 336

supervisory staff position = 50% of 61 = 31

Management staff position = 50% of (8 + 2 + 1) = 6

The entire procedure will have to be done for all other departments.

DETERMINATION OF NUMBER OF NON-ACADEMIC STAFF

ESTIMATING THE AMOUNT OF WORK CONTENT

(First Method)

1) **Observed Time**: time a task is being observed over and recorded by a time study officer.

2) Normal Rating: refers to the value 100 on a **performance rating.**

………………………….75 80 85 90 95 (**100**) 105 110 115 120 125……………………

3) Performance Rate: the expression value of the pace of human work.

A) Plain Estimating: Involving time study Includes Analytical estimating: This involves breaking down the main task into element and then applying the above rules on them.

S/No	Task(i)	Standard Man-Hours	Annual Frequency	Total Standard Man-hours per year (TSMH(i))
				Summation J

4) Basic Time: the time work content of a task or duty here relaxation time is not included.

Basic time(hour) = observed time x (observed performance Rating/Normal rating)

5) **Standard Time(hour)** = Basic time + Relaxation time allowance + contingency time allowance

Table 5a Workload Classification

S/No	Class of Work	Workload (KG)
1	Negligible Light	≤ 0.4535
2	Very Light	0.4536 – 2.7215
3	Light	2.7216 – 9.0718
4	Medium	9.0719 – 18.1436
5	Heavy	18.1437 – 27.2155
6	Very Heavy	27.2156 – 50.8023
7	Unduely Heavy	> 50.8023

Table 5. page 86 Relaxation allowance guide

FACTORS	TYPICAL EXAMPLES	ALLOW-

	To convert lb (multiply by 0.454 kilogram) to KG			ANCES
A. Energy Output (affecting muscular recovery)		Equivalent to handling	Men	Women
			Per cent	Per cent
Negligible…………	Light bench work-seated	No load	0 - 6	0 – 6
Very Light………..	Light bench work-standing	0 – 5lb	6 – $7\frac{1}{2}$	6 - $7\frac{1}{2}$
Light………………	Light shoveling	5 – 20lb	$7\frac{1}{2}$ -12	$7\frac{1}{2}$ – 16
Medium………….	Hack sawing or filling	20 – 40lb	12 – 19	16 – 30
Heavy…………….	Swinging heavy hammer 7 – 28lb	40 – 60lb	19 – 30	-
Very Heavy………	Loading weights	60 – 112lb	30 – 50	-
Exceptional………	Loading heavy sacks	Above 112lb	Requires special considerations	Requires special considerations
B. Posture			Per cent	
1. Sit	Normal sedentary work		0 -1	
2. Stand (both feet)	Whenever body is erect and support on feet only		1 - $2\frac{1}{2}$	
3. Stand (one foot)	Standing on one leg (using a foot control)		$2\frac{1}{2}$ – 4	
4. Lying down	On side, face or back		$2\frac{1}{2}$ - 4	
5. Crouch	When body is bent, but support on feet or knees		4 - 10	
C. Motions			Per cent	
1. Normal	Free swing of hammer		0	
2. Limited	Limited swing of hammer		0 – 5	
3. Awkward	Carrying heavy load in one hand		0 – 5	
4. Confined (limbs only)	Working with arms above head		5 – 10	
5. Confined (whole body)	Working at thin coal seam		10 – 15	
D. Visual Fatigue			Lighting Good	Lighting Poor/Variable
1. Intermittent eye attention	Reading meters or gauges		0%	1%
2. Nearly continuous	Precision machine work		2%	2%

eye attention			
3. Continuous eye attention – varying focus	Inspecting moving or stationary cloth for faults	2%	5%
4. Continuous eye attention – fixed focus	Inspecting minute and/or moving objects	4%	8%
E. Personal needs		$2\frac{1}{2}$%	$4\frac{1}{2}$%
F. Thermal conditions	Temperature	Humidity: Normal	Humidity: Excessive
1. Freezing	Below 30^0 F	Over 10%	Over 12%
2. Low	32^0 – 55^0F	0 – 10%	12 – 5%
3. Normal	55^0 - 75^0F	0%	5%
4. High	75^0 - 100^0F	0 – 40%	5 – 100%
5. Excessive	Above 100^0F	Over 40%	Over 100%
G. Atmospheric Conditions			Per cent
1. Good…………….	Well ventilated rooms or fresh air		0
2. Fair……………….	Badly ventilated air, presence of non-toxic but foetid odours or non-injurious fumes		0 – 5
3. Poor………………	Presence of toxic dusts or heavy concentration of non-toxic dusts involving use of breathing filters		5 – 10
4. Bad………………	Presence of toxic fumes or dusts involving use of respiratory		10 – 20
H. Other influences of Environment			Per cent
1. Clean, healthy, dry and bright surroundings, low noise level			0
2. Where work cycle is continuously repetitive and between 5 and 10 seconds			0 – 1
3. Where work cycle is continuously repetitive and less than 5 seconds			1 - 3
4. Where there is a complete absence of company Day – men			1
Day – women, Night – men			2
5. Excessive noise e.g. riveting (allowance related to continuity of noise)			0 – 5
6. Where effect of such disturbing influences might be detrimental to quality of output			0 – 5
7. Vibration of floors or machines, e.g. pneumatic drilling (allowance related to continuity of vibration)			5 – 10
8. Extreme conditions, e.g. dirt, noise, etc			5 – 15

Note:

1. This table is intended as a guide in assessing the relaxation allowances for individual elements
2. Although allowances are usually additive for the individual elements, assessment for elements incurring high allowances may require some reduction when followed by elements incurring less.

Source: R. M. CURRIE (1959)

(Second method)

This involves three different procedures

B) Factored Estimating: This method allows for higher accuracy in estimating

Includes Analytical estimating: This involves breaking down the main task into element and then applying the above rules on them.

This method include/involves:

i, Determining a corrective factor $\mathrm{X}_i = \frac{O_i - E_i}{O_i}$ where

O_i= the time observed through time study on task i

E_i= the time estimated for task i

X_i = corrective factor for task i

$$\mathrm{X} = \frac{\mathrm{X}_1 + \mathrm{X}_2 + ----+\mathrm{X}_i + ----+\mathrm{X}_n}{N}$$

N = number of sub estimated task in a major task

Therefore, corrected Estimate = (Peron’s estimate) x (1 + X)

Example

Task 1: writes cleaners pay voucher (0.60 hrs)

Task 2: writes permanent junior staff voucher (1.3 hrs)

Task 3: writes senior staff voucher (1. 2 hrs)

Task 4: writes car maintenance pay voucher (0.3 hrs)

Corrected factor are

Task 1: $O_1 = 0.6$, $E_1 = 0.75$

$$X_1 = \frac{0.6-0.75}{0.6} = -0.25$$

Task 2: $O_2 = 1.30$, $E_2 = 1.60$

$$X_2 = \frac{1.30-1.60}{1.30} = -0.23$$

Task 3: $O_3 = 1.20$, $E_3 = 1.30$

$$X_3 = -0.08$$

Task 4: $O_3 = 0.03$, $E_3 = 0.04$

$$X_4 = -0.33$$

Corrected Factor $\mathbf{X} = \frac{X_1+X_2+X_3+X_4}{4} = -0.22$

Consider for instance an estimate of his/her on "write voucher for casual" been given a 1.8 hrs

Corrected Estimate = person's estimate × (1 + **X**)

= 1.80(1 + (-0.22))

= 1.404 hrs

The Basic time = corrected estimate × (observed Rating/Normal Rating)

= 1.404 × (125/100) = 1.755 if 125 is his/her actual performance rating.

At 9% allowance

His/her **standard time** = 1.755 + (0.09 × 1.755) = 1.9130 hrs.

(THIRD METHOD CALLED WORK SAMPLING APPROACH)

Standard Man-hour = [standard Hour of a task] × [number of worker involved]

Total standard Man-hour of a Task = [Annual frequency of task occurrence] × [standard Man-hour]

To determine the proportion of time an observed staff is found busy on the pay job, we need Table (fig 6.1) on page 124 in the textbook as shown below

ESTABLISHMENT:				DATE:	
DEPARTMENT:				SECTION:	
ANALYST:			AUTHORISED BY:		
START TIME:		END TIME:		NO. OF OBSERVATIONS:	
Position	Work Status	Observations	Performance Rating	Total	%
	Busy				
	Not Busy				
	Busy				
	Not Busy				
	Busy				
	Not Busy				
	Busy				
	Not Busy				
	Busy				
	Not Busy				
	Busy				
	Not Busy				
	Busy				
	Not Busy				
	Busy				
	Not Busy				
	Busy				
	Not Busy				

Figure q: Work Sampling Data Collection Form

Work sampling approach for determining available work

PROCEDURE

1. Identify all operation positions of the establishment from her records. Visit the respective positions and map out one or more routes for visiting all in a trip. If the area is too large, more than one analyst may be used. In this case, the whole establishment may be subdivided into zones to allow each analyst cover a zone.

2. Design the data collection form. One familiar to figure q may be adequate. Notice that the size of the form depends on the number of positions to be studied.

3. Take preliminary number of observations to determine the study parameter. P the proportion of time an observed staff is found busy on the paid job. Care should be taken to ensure that only assigned activities are performed during the busy mode. One busy doing unrelated activities should be marked for the not-busy mode.

For multiple positions, the analyst may randomly pick only few for this preliminary stage. The number of observations at this point may be between twenty to forty. P is then determined with the following expression:

P = (Number of time staff were found busy / Total number of observation for all observed staff)

4. Now compute number of observations for the study given the desired accuracy or level of confidence. Thus let

N = number of observations

A = desired relative accuracy

K = number of standard deviations away from the mean

$$AP = K\sqrt{\frac{P(1-P)}{N}}$$

$$N = \frac{K^2(1-P)}{A^2P}$$

Hence, if observed that with K = 2, A = 5%, often a confidence limit of 95% is adequate. The value of N from the above formula

$$N = \frac{2^2}{0.05^2}\left(\frac{1-P}{P}\right)$$

5. Make schedule of study

In order to capture data at low, moderate and peak hours, days, weeks and months, the study may be spread over many days and months. For large number of observations, the study may be spread over the twelve months. In this case, the days to be studied may be picked as follow:

Let Z_x be the date the study will be carried out; where

Z = the month identity; and

x = the day of observation

x is picked randomly with the following formula:

$x_i = 1.0 + 27R_i$ for the month of February.

$x_i = 1.0 + 29R_i$ For April, June, September and November

$x_i = 1.0 + 30R_i$ For January, March, May, July, August, October and December

R_i is a random number lying between 0 and 1.0 which can be taken from Tables or scientific calculators. For a particular month, computed x_i corresponding to a working day is retained. If public holiday or weekend, it is discarded and the process repeated until the number of desired days are determined.

For the hours of study, the schedule will be made for each day. The random number scheduling formula may again be adopted as follows:

$$Y_i = Y_{i-1} + [A + (B - A)R_i]$$

where:

Y_i = scheduled observation time

Y_0 = clock time when work commences in the establishment

A = minimum duration the analyst considers sufficient for observation and rest per cycle (minutes)

B = maximum possible duration (minute)

For n observations per day of W working minutes, B can be determined from the empirical formula.

$$\frac{A+B}{2} = \frac{W}{n}$$

So for 8 – hour working day,

$$B = \frac{960-nA}{n}$$

For an example an observer or analyst estimated the minimum time to go around all observed positions, record and rest to be 30 minutes, for ten observations per day, then

A = 30, and n = 10

$$B = \frac{960-10\times 30}{10} = 66\,minutes$$

Considering 8am as work resumption time, using the expression given above, so therefore

$$Y_i = 8.00 + [30 + (66-30)R_i]$$

Then choosing ten random number which lies between 0 and 1

For R_i chosen to be {0.662, 0.227, 0.230, 0.300, 0.904, 0.809, 0.626, 0.406, 0.345, 0.939, 0.603}

Arranging them in an ascending order

$$Y_i = \{8.53,\ \ 9.31,\ \ 10.09,\ \ 10.49,\ \ 11.51,\ \ 12.50,\ \ 1.42,\ \ 2.63,\ \ 3.08,\ \ 4.46\}$$

When a schedule is to be changed for next day, different set of random numbers is used. Note that towards the end of the schedule, one or two points may fall outside the working hours. At such instances, they are discarded, while those within the period are generated as replacers. On the other hand, if the schedule has room for more observation times, they may be added.

6. Carrying out the study

The analyst first fills the recording form indicating the date, department or section. The start time and names of positions to be observed are indicated. The study may now commence keeping strictly to the schedule. At each position, the following are noted and then recorded as soon as he/she moves a little away:

- Busy or not busy mode; and
- Performance rating of the individual (figure q)

At the end of each study day, the number of busy mode for each position is counted; total number of observations and end of study time recorded. Also, determined is the average performance rating for each position.

7. Data Analysis

The following useful quantities may be computed using the collected data:

- Utilization Factor for each position i, (U_i)
- Estimated Annual Man-hours, (EAM_i)
- Estimated Basic Man-hours, (EBM_i)
- Estimated Standard Man-hours (ESM_i)

The following expressions may be used to compute these quantities:

$$U_i = \frac{Number\ of\ observed\ busy\ mode\ for\ position\ i}{Total\ number\ of\ observations}$$

$$EAM_i = U_i \times [Available\ hours\ of\ work\ in\ the\ year]$$

$$EBM_i = EAM_i \times \frac{Performance\ Rating\ in\ position\ i}{100}$$

$$ESM_i = EBM_i + \frac{EBM_i \times PA_i}{100}$$

where PA_i is percentage allowance for type of job done in position i. Its values may be obtained from table 5. With all operation positions studied, the total available standard man-hours (TAM) for operations in an establishment is given by:

$$TAM = \sum_{i=1}^{M} ESM_i$$

Position	Proportion of Time Busy ()	Estimated Annual Standard Man-hours
	%(i) divide by 100	(TSMH(i))=ESMi
Total		Summation Z

To determine number of staff position = (Grand total standard man-hours per annum= Summation Z)/(available hours per annum) × (use factor)

Note: Use factor is the proportion of time expected for a staff to actually be on the job he/she is paid for. It doesn't include time for conversation, lunch, toilet, sleeping, going out e.t.c. But accommodate sick-leave, official meeting, establishment social, annual leave, official travels e.t.c.

DETERMINING OF **PERSONNEL UTILIZATION** OF DECISION CENTRE AT LEVEL i CENTRE j (H_{ij})

$$H_{ij} = 1 - \frac{L_{ij}\mathrm{W}_{ij}}{A_{ij}(K_{ij}+1)} - \frac{P_{ij}}{K_{ij}+1}$$

where

H_{ij} is personnel utilization of decision centre at level i

L_{ij} is average number of cases in for supervisor's or manager's attention in a day

W_{ij} is average time (hours) a case waits and receives attention from the superior

P_{ij} the proportion of timed the superior has no case to attend to

A_{ij} amount of time (hours) scheduled for work in a day and

K_{ij} the span of control at decision centre j of the ith level

Model Assumptions

1. Every employee has at least, one job to perform in the organization

2. The organization has terminal activities distinct from the supervisory and pure decision activities

3. The conditions mentioned in proposition 4. 2 hold such that every employee is highly motivated

4. The conditions cited in proposition 4.1 hold (one-boss structure)

5. Standard workload (suitable for the position) and not maximum possible workload I assigned to every staff.

6. The business organization is the type specified in proposition 8.1

7. The workload of a boss (superior) at decision centre (i, j) is proportional to his/her span of control, K_{ij}

8. Request, response of directives, situational reporting, clarification, authorization, counseling are regular feature of superior-subordinate relationships.

9. Arrival of cases for and departure from the boss are stochastic events.

10. First come, first served consultation discipline is observed, the superior attends to one subordinate's at a time.

11. Superior is experienced enough to handle a decision centre. Otherwise, there will be a large heap of cases at every moment.

12. A boss in charge of a decision position oversees only the specific number of immediate subordinates (K_{ij}) assigned.

13. Data for parameter estimation are collected from the interactional, stochastic and dynamic system, when it has passed from the transient to a steady state.

14. The time a case leaves it location and travel to the superior's desk is negligible.

This condition satisfies one channel queuing model with restricted queue length and arriving population. Using Kendal-Lee notation, (Taha, 1976), the queuing system is (M|M|1); (FCFS|K_{ij}|K_{ij})

Where;

M : Poisson arrival/departure distribution

1 : one server

FCFS : Queuing discipline, first come, first served

K_{ij} : Maximum number allowed in the system/calling source.

Under steady state condition

$L_{ij} = \bar{L}_{ij} + 1 - P_{ij}$

and

$$W_{ij} = \frac{\bar{L}_{ij}}{\mu_{ij}(1-P_{ij})} + \frac{1}{\mu_{ij}} = \left[\frac{\bar{L}_{ij}+1-P_{ij}}{1-P_{ij}}\right]\frac{1}{\mu_{ij}}$$

Where

$\bar{L}_{ij}$ = average number of cases which waited in a day to receive the attention of the boss

μ_{ij} = the rate at which the superior at decision center j, level i, attend to his/her subordinate.

W_{ij} = average time (hours) a case waits and receives attention from the superior

H_{ij} is personnel utilization of decision centre at level i

P_{ij} the proportion of timed the superior has no case to attend to

λ_{ij} = the rate at which the subordinate arrive to consult the superior; and

$$\mathrm{W}_{ij} \leq A_{ij}$$

$$H_{ij} = 1 - \frac{\frac{\left(\bar{L}_{ij} + 1 - P_{ij}\right)^2}{\mu_{ij}\left(1 - P_{ij}\right)}}{A_{ij}\left(K_{ij} + 1\right)} - \frac{P_{ij}}{K_{ij} + 1}$$

Also, from Taha (1976)

$$P_{ij} = \left[1 + \sum_{n=1}^{K_{ij}} C_n^{K_{ij}} n!\, \rho_{ij}^n\right]^{-1}$$

and

$$\bar{L}_{ij} = \frac{\sum_{n=2}^{K_{ij}} (n-1) C_n^{K_{ij}} n!\, \rho_{ij}^n}{1 + \sum_{n=1}^{K_{ij}} C_n^{K_{ij}} n!\, \rho_{ij}^n}$$

$$H_{ij} = 1 - \frac{\frac{\left(\frac{\sum_{n=2}^{K_{ij}} (n-1) C_n^{K_{ij}} n!\, \rho_{ij}^n}{1 + \sum_{n=1}^{K_{ij}} C_n^{K_{ij}} n!\, \rho_{ij}^n} + 1 - \left[1 + \sum_{n=1}^{K_{ij}} C_n^{K_{ij}} n!\, \rho_{ij}^n\right]^{-1}\right)^2}{\mu_{ij}\left(1 - \left[1 + \sum_{n=1}^{K_{ij}} C_n^{K_{ij}} n!\, \rho_{ij}^n\right]^{-1}\right)}}{A_{ij}\left(K_{ij} + 1\right)} - \frac{\left[1 + \sum_{n=1}^{K_{ij}} C_n^{K_{ij}} n!\, \rho_{ij}^n\right]^{-1}}{K_{ij} + 1}$$

This will become

$$H_{ij}\left(K_{ij}, A_{ij}, \mu_{ij}, \lambda_{ij}\right) = 1 - \frac{1}{A_{ij}\left(K_{ij}+1\right)}\left[\frac{\frac{\sum_{n=2}^{K_{ij}} (n-1) C_n^{K_{ij}} n!\left(\frac{\lambda_{ij}}{\mu_{ij}}\right)^n}{1 + \sum_{n=1}^{K_{ij}} C_n^{K_{ij}} n!\left(\frac{\lambda_{ij}}{\mu_{ij}}\right)^n} + 1 - \left[1 + \sum_{n=1}^{K_{ij}} C_n^{K_{ij}} n!\left(\frac{\lambda_{ij}}{\mu_{ij}}\right)^n\right]^{-1}}{\mu_{ij}\left(1 - \left[1 + \sum_{n=1}^{K_{ij}} C_n^{K_{ij}} n!\left(\frac{\lambda_{ij}}{\mu_{ij}}\right)^n\right]^{-1}\right)}\right]^2 - \frac{\left[1 + \sum_{n=1}^{K_{ij}} C_n^{K_{ij}} n!\left(\frac{\lambda_{ij}}{\mu_{ij}}\right)^n\right]^{-1}}{K_{ij}+1}$$

where

$$\rho_{ij} = \frac{\lambda_{ij}}{\mu_{ij}};\ \lambda_{ij} < \mu_{ij}$$

A_{ij} = the value of working hour assigned to the boss and subordinate who report to decision position (i, j)

Let $Q_1 = K_{ij}$ in a fair organization and the optimal value at the supervisory level be Q_1*. The associated optimal number of supervisor is given by the expression:

$N_1^* = N_0/Q_1^*$

M number of decision levels

N_i number of decision positions at level i; and

K_{ij} as earlier defined (span of control)

Observe that the number of variables in include K_{ij} the span of control, M and Ni, it may also be noted here that

N_0 the number of operation positions which perform terminal activities (number of lowest level cadre of staff)

N_1 number of supervisory positions

N_2 purely decision or management positions at management level 2;

N_m 1, is the topmost or chief executive position

Typical Sample Form used for Determining λ_{ij} *and* μ_{ij}

$$N_i = \begin{cases} 1, if\ computed\ value\ is\ only\ a\ decimal \\ nearest\ higher\ whole\ number\ if\ decimal\ component\ > 0.5; \\ nearest\ lower\ whole\ number\ if\ decimal\ component\ < 0.5; and \\ the\ resulting\ whole\ number, otherwise. \end{cases}$$

Work Study Data Entry Form

Total Number of Cases (TNC) = 389 Number of Cases Attended To (TCC) = 324 Total Hours of Study (TTS) = 200				
Case	Arrival Time	Attending to Case		
		Start Time (T_{in})	Completion Time (T_{out})	Time Span $T_{ij\alpha} = T_{out} - T_{in}$
1	7.40	7.40	7.58	18 min
2	8.23	8.25	8.47	22 min
3	8.32	8.47	10.07	80 min
4	8.33	-	-	-
5	9.01	-	-	-
6	9.34	10.07	10.51	44 min
7	10.42	10.51	11.22	31 min
8	12.09	12.09	12.12	3 min
9	2.21	2.21	2.27	6 min
10	8.35	8.35	8.37	2 min

11 12 13 14 15 16 17 18 19 20 21	9.10 9.13 9.20 12.06 2.08 2.22 3.14 3.24 7.41 7.53 8.41	9.10 - 10.16 12.06 2.08 2.36 3.14 - 7.41 8.26 9.17	10.15 - 10.24 1.01 2.10 3.02 3.27 - 7.46 9.17 10.05	65 min - 8 min 55 min 2 min 26 min 13 min - 5 min 51 min 48 min
386 387 388 389	1.15 1.32 2.37 3.10	1.56 2.19 2.46 3.22	2.19 2.46 3.22 3.28	23 min 27 min 36 min 6 min
$\sum_{\alpha=1}^{324} t_{ij\alpha} =$				$^{3410}/_{60}\ hours$

Data collected for one supervisor in Dana (Nigeria) Limited

TNC_{ij} = total number of cases at position (i,j)

TTS_{ij} = total time (hours) for study at position (i,j)

From the above table

$$\mu_{ij} = \frac{total\ completed\ cases\ at\ position\ (i,j)}{time\ (hours)\ spent\ to\ treat\ all\ completed\ cases}$$

$$\mu_{ij} = \frac{TCC_{ij}}{\sum_{\alpha=1}^{324} t_{ij\alpha}}$$

$$\lambda_{ij} = \frac{TNC_{ij}}{TTS_{ij}} = \frac{389}{200} = 1.945\ cases/hour$$

TCC = 324 (i.e. total cases attended to)

$$\sum_{\alpha=1}^{324} t_{ij\alpha} = \frac{3410}{60}$$

$$\mu_{ij} = \frac{324}{^{3410}/_{60}} = 5.70087\ cases/hour$$

Obtaining similar study for 4 more supervisors in the same company gave a total of the below

$\lambda_{ij} = 1.945$ $\mu_{ij} = 5.700$

$\lambda_{ij} = 1.672$ $\mu_{ij} = 7.501$

$\lambda_{ij} = 2.045$ $\mu_{ij} = 6.243$

$\lambda_{ij} = 1.827$ $\mu_{ij} = 5.341$

$\lambda_{ij} = 1.746$ $\mu_{ij} = 8.132$

$$\overline{\lambda}_{ij} = \frac{\sum_{i=1}^{5} \lambda_{ij}}{5} = \frac{1.995 + 1.672 + 2.045 + 1.827 + 1.746}{5} = 1.847$$

$$\overline{\mu}_{ij} = \frac{\sum_{i=1}^{5} \mu_{ij}}{5} = \frac{5.700 + 7.501 + 6.243 + 5.341 + 8.132}{5} = 6.5834$$

STEPS FOR DETERMINING λ_{ij} AND μ_{ij}

1. Pick study period to cover low, moderate and peak level of activities. It may last from three to twelve month.

2. At each decision position j, level i, note the time study commenced.

3. Keep count of each arrival and note whether or not the case waited for and received attention. Let the total number of cases at position j of level i be TNC_{ij}. It may be recalled that a case may be a telephone call, a memo or letter, e-mail or fax message, personnel appearance for consultation. Etc. a case may come from either the immediate boss of the decision position of interest or its immediate subordinates.

4. Record the amount of time (hours) the personnel occupying position (i, j) actually spent attending to each case. Let the time for case α be $t_{ij\alpha}$

5. Keep count of completed cases within the study period at position (i, j). Let this number be TCC_{ij}

6. Record the total working time (hours) actually spent observing the activities at decision position of interest. Let this time be TTS_{ij}

7. Compute λ_{ij} and μ_{ij} as follows:

$$\lambda_{ij} = {total\ number\ of\ cases\ at\ position\ (i,j)} \Big/ {total\ time\ (hours) for\ \text{study at the position}}$$

that is,

$$\lambda_{ij} = \frac{TNC_{ij}}{TTS_{ij}}$$

and

$\mu_{ij} = \frac{TCC_{ij}}{\sum_{\alpha=1}^{TCC_{ij}} t_{ij\alpha}}$

$$\bar{\mu}_{ij} = \sum_{j=1}^{N_i} {}^{\mu_{ij}}/_{N_i}$$

$$\bar{\lambda}_{ij} = \sum_{j=1}^{N_i} {}^{\lambda_{ij}}/_{N_i}$$

The above are their respective means.

WASTED MAN-HOUR

$D_{ij} = \left[\frac{\left(\bar{L}_{ij}+1-P_{ij}\right)^2}{1-P_{ij}}\right]\frac{a_{ij}}{\mu_{ij}} + b_{ij}A_{ij}P_{ij}$UR COST

$$\bar{L}_{ij} = \frac{\sum_{n=2}^{K_{ij}}(n-1)C_n^{K_{ij}} n!\, \rho_{ij}^n}{1 + \sum C_n^{K_{ij}} n!\, \rho_{ij}^n}$$

Therefore

$$D_{ij} = \left[\frac{\left(\frac{\sum_{n=2}^{K_{ij}}(n-1)C_n^{K_{ij}} n!\, \rho_{ij}^n}{1 + \sum C_n^{K_{ij}} n!\, \rho_{ij}^n} + 1 - \left[1 + \sum_{n=1}^{K_{ij}} C_n^{K_{ij}} n!\, \rho_{ij}^n\right]^{-1}\right)^2}{1 - \left[1 + \sum_{n=1}^{K_{ij}} C_n^{K_{ij}} n!\, \rho_{ij}^n\right]^{-1}}\right]\frac{a_{ij}}{\mu_{ij}} + b_{ij}A_{ij}\left[1 + \sum_{n=1}^{K_{ij}} C_n^{K_{ij}} n!\, \rho_{ij}^n\right]^{-1}$$

Once we have been able to attain K*(λ, μ) for a department, then from your time study determine the number of staff for each task in an organization. Then from there which will be N_0*

Note: To determine the total number of staff in an organization, first we need to

1. find all the departments involved and

2. determine the standard man-hours for a normal staff to perform the activity using the 1st, 2nd or 3rd (third) method presented on page 54 forward

3. determine the corresponding use factor for each task of each department

Note: Use factor is the proportion of time expected for a staff to actually be on the job he/she is paid for. It doesn't include time for conversation, lunch, toilet, sleeping, going out e.t.c. But accommodate sick-leave, official meeting, establishment social, annual leave, official travels e.t.c.

Still considering for Dana (Nigeria) Limited, the observer observed that the firm works, on the average for 220 days per annum and 8 hours a day. Things considered to collate the data are

- Inventory of all the manual skills in Dana (Nigeria) Limited at the lowest levels
- The amount of annual work (man-hours) available for each skill
- The use factor per skill type
- The rate at which supervisors attend to their immediate subordinates. That is, determine $\mu_{ij}, \forall_{ij}$
- The rate at which supervisors report to their respective direct bosses (μ_{2j})
- The rate at which subordinates report to their immediate bosses (λ_{ij})
- Use the information to design an optimal organization structure to run Dana manufacturing establishment.

S/No	JOB	Standard man-hours	Use Factor
1	Cleaning	5367	.72
2	Mail Delivery and Record	3456	.95
3	Welding	8466	.75
4	Assembling and Finishing	7662	.85
5	Secretarial Duties	6895	.84
6	Machine	12664	.92
7	Packaging	5983	.78
8	Receiving Visitors	1456	.80
9	Telephone Operating	1364	.76
10	Warehousing	6995	.94
11	Equipment Repairing	2966	.86
12	Copy Machine Operating	2236	.85

Note: use factor is the proportion of time expected for a staff to actually be on the job for which he/she is paid. It can be determined experimentally for any work situation and can also be prescribed as a company policy

Number of staff $= \dfrac{Grand\ total\ standard\ manhours\ per\ annum}{(Available\ hours)\times(use\ factor)}$

To determine the number of cleaners $= \frac{5367}{8\times220\times.72} = 4.235 = 4$

Carrying out similar computations,

Number of secretarial staff = 5

Number of messengers = 2

Numbers of assembly personnel = 5

Number of welders = 6

Number of machinists	=	8
Number of packaging staff	=	4
Number of receptionist	=	1
Number of telephone operator	=	1
Number of store-men	=	4
Number of equipment service men	=	2
Number of copy machine operators	=	2
Total number of operation positions (N_0)=		**44**

Where a fair organization structure is assumed the following was generated for the different managerial levels

A = 8.0

$\boldsymbol{\bar{\lambda}_1 = 1.25}$, $\boldsymbol{\bar{\mu}_1 = 5.50}$

$\boldsymbol{\bar{\lambda}_2 = 0.75}$, $\boldsymbol{\bar{\mu}_2 = 1.50}$

$\boldsymbol{\bar{\lambda}_3 = 0.50}$, $\boldsymbol{\bar{\mu}_3 = 1.0}$

Using:

Maximise $H(K_{ij}, N_i, M, \phi_c) = \sum_{i=1}^{M} \sum_{i=1}^{N_i} H_{ij}(K_{ij}, \theta_c) / \sum_{i=1}^{M} N_i$ general formular as expressed above in page 63

Often, parameters $(\boldsymbol{\lambda}, \boldsymbol{\mu})$ are generated for three levels which are

- Supervisory
- Intermediate
- Top

So when you keep dividing the operation number arrived at with the **K*** value obtained for each level, we proceed until we arrive at N* = 1. This might involve generating other levels parameters

K_1* (1.25, 5.50) = 7

K_2* (0.75, 1.50) = 3

K_3* (0.50, 1.0) = 3

N_1* = N_0/K_1* if N_0 is determined to be 44 and K_1* is 7 for K(K_1, λ_1, μ_1)

Then N_1* = 44/7 = 6

$N_2^* = N_1^*/K_2^* = 6/3 = 2$

$N_3^* = N_2^*/K_3^* = 2/3 = 1$

So therefore,

M* = 3 (number of management levels)

$H^* = 6 \times H_1(K_1^*, \lambda_1, \mu_1) + 2 \times H_2(K_2^*, \lambda_2, \mu_2) + 1 \times H_3(K_3^*, \lambda_3, \mu_3) / 6 + 2 + 1$

H* = 0.9365 which must not be greater than 1

Then the size of the organization will be

$S = N_0 + N_1^* + N_2^* + N_3^*$

= 44 + 6 + 2 + 1 = 53

To determine the breath or average number of the management position per level (Z)

$$Z = \frac{\sum_{i=1}^{M} N_i}{M} = \frac{6+2+1}{3} = 3$$

M = 3,

$Z = (N_1^* + N_2^* + N_3^*) / M = 3$

The structure:

Therefore; the organization is of the form $\epsilon = (S; M, Z)$

$$\epsilon = (53; 3,3)$$

$$\psi = \begin{bmatrix} 3, & 1 \\ 2, & 2 \\ 1, & 6 \\ 0, & 53 \end{bmatrix}$$

Where the different categories of operations are group into supervisory section, like e.g.

Section 1: Welding (6)

Section 2: Assembling and Equipment Maintenance (7)

Section 3: Machining (8)

Section 4: Secretarial Duties (7)

Section 5: Packaging and Warehousing (8)

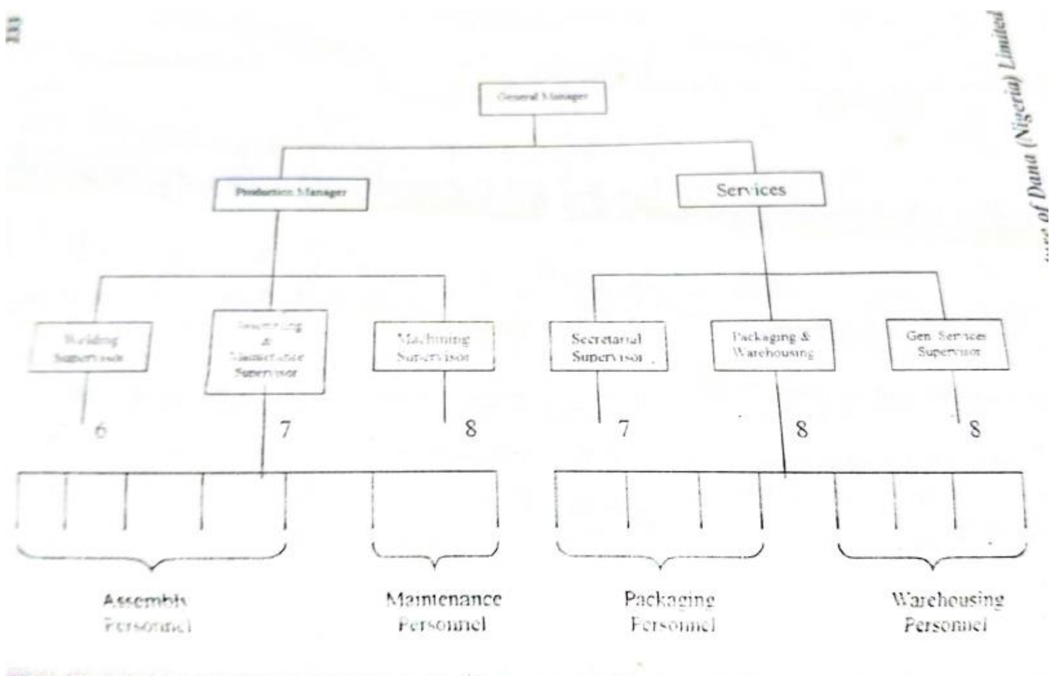
Section 6: General Service (Cleaners, Telephone Operator, Receptionist, Copy Machine Operator) (8)

By grouping of supervisors into departments

(i) Production Management Departments (Section 1, 2 and 3)

(ii) Services Department (Section 4, 5 and 6)

Organisation structure



Computation of personnel utilization after the grouping into sections and departments

$$H = \frac{1H_{11}(6,8,1.25,5.5) + 1H_{12}(7,8,1.25,5.5) + 1H_{13}(8,8,1.25,5.5)}{1+2+3+2+1} + \frac{2H_{21}(3,8,0.75,1.5) + 1H_{31}(2,8,0.5,1.0)}{1+2+3+2+1}$$

$$H = \frac{1 \times 0.9599 + 2 \times 0.9621 + 3 \times 0.956 + 2 \times 0.8941 + 1 \times 0.8222}{9}$$

$$H = 0.9292$$

To determine the percentage redundancy of the junior staff of the real structure the following formula is applied

$$Real\ Percentage\ Redundancy\ (PR\%) = \left(\frac{Real\ number\ of\ junior\ staff - Ideal\ number\ of\ junior\ staff}{ideal\ number\ of\ junior\ staff}\right) \times 100$$

To determine the personnel redundancy of an organization if

Example

$(K_1^*, \lambda_1, \mu_1)$ = (9, 6.75, 9.5)

PR_1 = 1.17% i.e. Minimum personnel redundancy

N_1 = 11

$(K_2^*, \lambda_2, \mu_2)$ = (4, 9.76, 12.49)

PR_2 = 2.20%

N_2 = 3

$(K_3^*, \lambda_3, \mu_3)$ = (3, 3.74, 4.50)

PR_3 = 5.02%

N_3 = 1

M* = 3

Therefore personnel redundancy

$$\overline{PR} = \frac{(11)(1.17) + (3)(2.20) + (1)(5.02)}{11 + 3 + 1} = 1.63\%$$

If we want to determine the percentage redundancy of the junior staff of the real structure, the following formula is applied.

$$Real\ Percentage\ redundancy = \frac{Real\ number\ of\ junior\ staff - Ideal\ number\ of\ junior\ staff}{ideal\ number\ of\ junior\ staff}$$

So, if we are given

- Real number of junior staff = 67
- Ideal number of junior staff = 62
- Real percentage

$$Real\ Percentage\ Redundancy = \frac{67 - 62}{62} \times 100 = 8.06\%$$

Determining Cost Savings/Losses

The unit cost averages are given for only two personnel levels: lowest and management cadres. Thus,

Average lowest cadre pay rate (C_1) = N10,620 per month

Average management pay rate (C_2) = N52,200 per month

Number of personnel in

- Lowest level existing = 147
- Management cadre existing = 76
- Rationalized lowest cadre = 101
- Rationalized management cadre = 15

Therefore, D_1 = 147 – 101 = 46

D_2 = 76 -15 = 61

Total Cost Savings (TCS) = C_1D_1 + C_2D_2

TCS = $(10,620 \times 46 + 52,200 \times 61) \times 12$ per annum

= N44,072,640 per annum

PROCEDURE FOR PROJECTING/PREDICTING FUTURE PERSONNEL REQUIREMENTS

Step 1: Determine, either through known plan or forecast, the type and volume of additional business an organization will have in the future period of interest. This may be from corporate plan for expansion, diversification or through known growth rate.

Step 2: Identify all possible organizational terminal activities required to accomplish the business and classify into the various skills.

Step 3: Determine or forecast the annual volume or frequency of each activity.

Step 4: Applying the procedures in chapter five, determine the amount of standard man-hours per type of skill required for the additional business. For steady growth conditions, **regression models** may be used as found in page 37.

Step 5: Compute the number of operation positions per skill and the total number (N_0).

Step 6: Determine the human dynamic interactional parameters $(\boldsymbol{\lambda}, \boldsymbol{\mu})$ as outline before now. For a steady growth or expansion, parameter values earlier determined with the firm's data may be used. For a case of diversification, parameters estimated with data observed in most similar operating environment may be used for an initial solution. Improvements may be made in subsequent years.

Step 7: With $(\boldsymbol{\lambda}, \boldsymbol{\mu})$ known, determine the number of supervisory, intermediate and top management positions as discussed earlier.

Below is a case study where the above procedure can be used to make projection. It may be noted that this is a case of steady business growth. The organizational structure design parameters and span of control for this firm is summarized below.

Tinka Plc Parameter Value and Span of Control

Departments/Units	Level i	Parameter Value			
		A	$\overline{\lambda}_1$	$\overline{\mu}_1$	Optimal Span of Control
Production	1	8	1.749	14.501	12
	2	8	1.304	5.512	7
	3	8	1.02	3.546	5
Warehousing	1	8	1.495	15.507	14
	2	8	0.985	3.978	6
	3	8	0.757	2.010	4
Maintenance	1	8	2.001	9.499	8
	2	8	1.252	4.013	5
	3	8	0.750	1.02	3
Units Combined	1	8	1.750	16.251	13
	2	8	1.523	3.512	4
	3	8	1.523	3.512	4

The use of linear regression can be used to predict the span of control. For projecting the amount of available manual work for future periods as obtained, we have

Production: Yp = 505584.29 + 48764.18x

Warehousing: Yw = 70945.29 + 8556.89x

Secretarial Support: Ys = 20675.14 + 1031.607x

Maintenance: Ym = 126212.57 + 2537.25x

Accounting & Finance: Ya = 17500.06 + 802.02x

Human Resources: Yh = 14488.425 + 842.29x

Welfare: Ywd = 9390.486 + 316.486x

Transport: Yt = 9828.286 + 454.857x

where:

x = 1, 2, 3, ……. 10, 11, 12, …..

is the code for years 1991, 1992, 1993, ……. 2000, 2001, 2002, …… While Y is the standard man-hour available in the respective departments/unit. Then apply the formular

$$\text{Number of staff} = \frac{Grand\ total\ standard\ manhours\ per\ annum}{(Available\ hours) \times (use\ factor)}$$

Where available hours is 300 working days per annum as is the norm in Tinka Plc and they operate two shifts each lasting for eight hours per day

therefore

available hours = 300 x 8 = 2400

NON ACADEMIC STAFF APPRASIAL

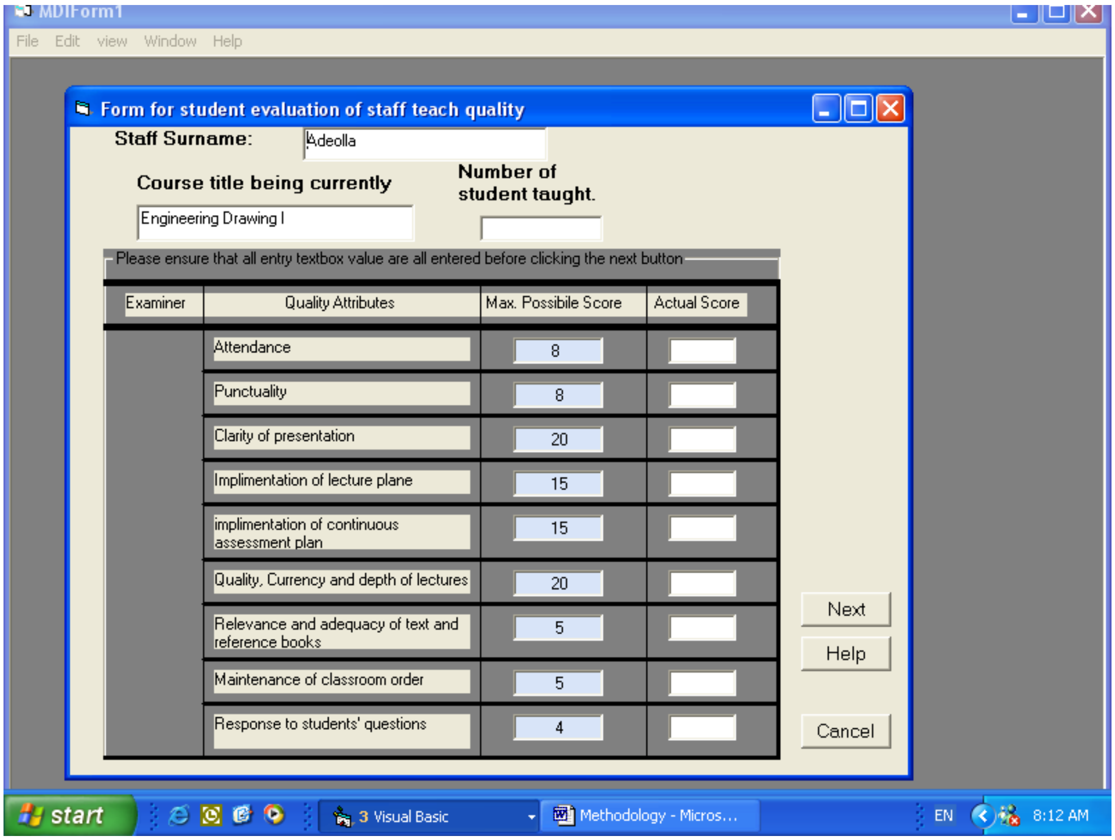


Figure 8 **(FORM 8) TO BE KNOW AS ACTIVITY FORM to be use to determine the entry score for Hardwork (quantity) in staff appraisal sample adopted from Academic staff Appraisal.**

s/no	**Assessed Criterion**	**Max Possible Score**	**Actual Score**
1.	Attendance	8	
2.	Punctuality	8	
3.	Clarity of presentation	20	
4.	Implementation of activity plane	15	

5.	Implementation of continuous assessment plan	15	
6.	Quality, currency and depth of activities done	20	
7.	Relevance and adequacy of text and Other aid items on the job	5	
8.	Maintenance of office order	5	
9.	Response to staffs questions from supervisors	4	
10.	Aggregate (total) score	100	

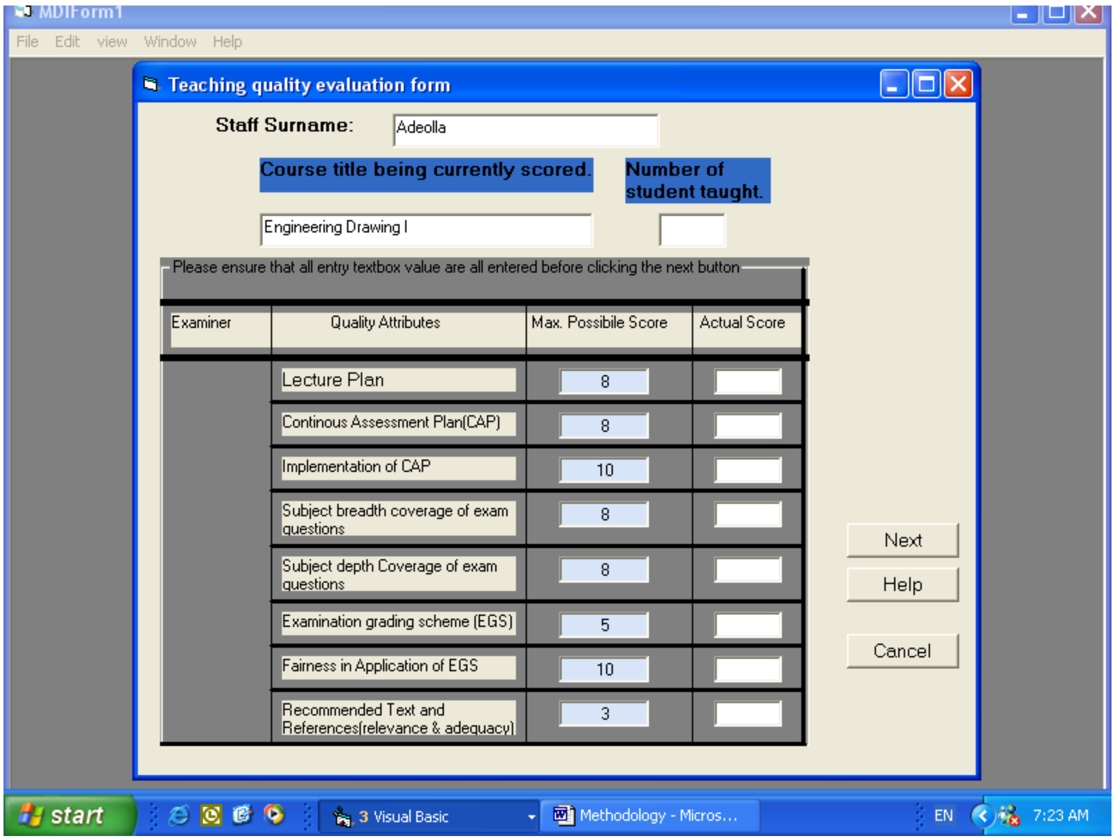


Figure 9 **(FORM 9) TO BE KNOWN AS TRAINING QUALITY EVALUATION FORM**

Examiner	**Quality Attributes**	**Max possible score**	**Actual Score**
External and or peer	Activities plan	8	
"	Continous Assessment plan	8	
"	(CAP) implementation	10	
"	Training coverage and	8	

	examination questions on trainings		
"	Subject Depth Coverage of examination question during traing	8	
"	Examination grading Scheme from training	5	
"	Fourness in Application of EGS	10	
"	Recommended text, and training materials with Reference (Relevance and Adequacy)	3	
CO-STAFF	staff Evaluation (implementation of Training plan, CAP: Keeping meetings schedule: use of notes, text: Understanding of Subject, e.t	40	
	Total	**100**	

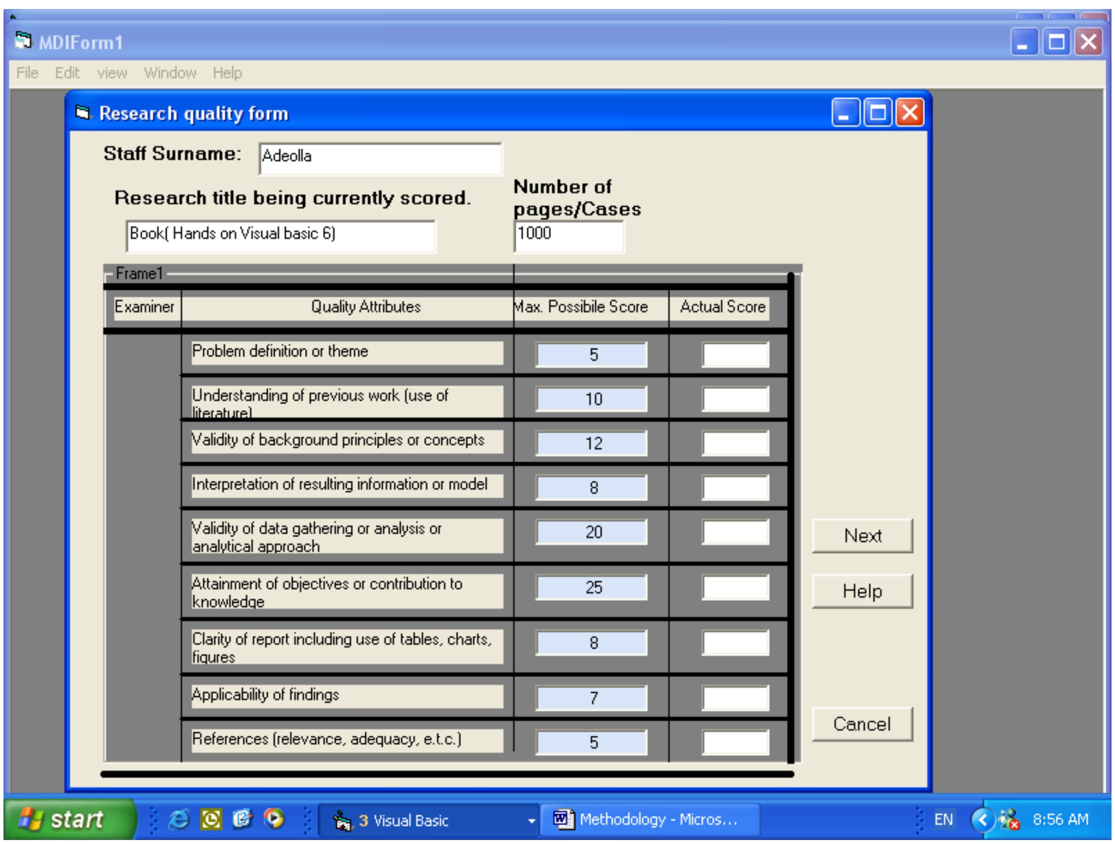


Figure 10 **(FORM 10) TO BE KNOWN AS FAULT SOLVING QUALITY ATTRIBUTES FORM to solve how Quality of work is to be computed during appraisal.**

Assessor	Quality attributes of research	Max possible score	Actual Score
External and / or peer	Problem definition or scheme	5	
-	Understanding of previous work	10	
-	Validity of background principles/ concepts	12	
-	Interpretation of resulting information/ model	8	
-	Validity of data gathering/ analysis or Analytical approach	20	
-	Attainment of objectives or contribution to knowledge	25	
-	Clarity of report including use of tables, Charts, figures	8	
-	Application of findings	7	
-	References (relevance, adequacy etc)	5	
	Total	100	

Note: For each achievement criteria performance measurement there should be a Box for Head of Department/Unit Max Score

Maximum Score

Appraisal

Maximum Score

HOD/Unit Max. Score

Yes

No

Appraisal Acceptance

HOD/UNIT

Justification____

Note: If Appraisal's Score is within plus/minus 10% of HOD/UNIT Max. Score then accept and record the average between both scores. i.e. Appraisal score attained and the HOD/Unit Max. Score

Note: If Appraisal's Score is below (minus 10% of HOD/UNIT Max. Score and the HOD/UNIT has justified his/her score without alteration) then accept and record only the HOD/UNIT Max. Score but do not store the score. Flag the appraisal score arrived at for further action by appraisal auditor.

Note: If Appraisal's Score is above (plus 10% of HOD/UNIT Max. Score and the HOD/UNIT has justified his/her score without alteration) then accept and record the average between both scores, and do not flag the score. i.e. Appraisal score attained and the HOD/Unit Max. Score

a) WORKS & PHYSICALPLANING, TECHNICAL STAFF, TECHNOLOGIST, QUANTITY SURVEYORS, MEDICAL PRACTITIONERS, CILPU, LABORATORY ASSISSTANCE, HEALTH CENTRE										
PERFORMANCE RELATED ASPECT										
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR COMPETENCE										
	Less likely	Summative Response Scales							Most likely	Formula
	1	2	3	4	5	6	7	8	10	
Hardwork (quantity)										$\frac{x}{10} \times \frac{55}{1}$
Quality of work										$\frac{x}{10} \times \frac{10}{1}$
Initiative										$\frac{x}{10} \times \frac{60}{1}$
Expertise										$\frac{x}{10} \times \frac{30}{1}$
Supervision										$\frac{x}{10} \times \frac{40}{1}$
Reporting										$\frac{x}{10} \times \frac{20}{1}$
Work Planning										$\frac{x}{10} \times \frac{30}{1}$
Creativity										$\frac{x}{10} \times \frac{60}{1}$
	TOTAL									
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR INTEGRITY										
	Less likely	Summative Response Scales							Most likely	Formula
	1	2	3	4	5	6	7	8	10	
Leadership										$\frac{x}{10} \times \frac{100}{1}$
Dedication										$\frac{x}{10} \times \frac{70}{1}$
Honesty										$\frac{x}{10} \times \frac{60}{1}$
Self-discipline										$\frac{x}{10} \times \frac{40}{1}$
Responsibility										$\frac{x}{10} \times \frac{40}{1}$

Reliability											$\frac{x}{10} \times \frac{40}{1}$
Punctuality (earliest arrival time to work often)											$\frac{x}{10} \times \frac{30}{1}$
Regularity or Absenteeism											$\frac{x}{10} \times \frac{30}{1}$
	TOTAL										
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR COMPATIBILITY											
	Less likely	Summative Response Scales								Most likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Team work											$\frac{x}{10} \times \frac{80}{1}$
Contributions to the immediate community											$\frac{x}{10} \times \frac{20}{1}$
Hospitality											$\frac{x}{10} \times \frac{20}{1}$
Special contributions to section/branch											$\frac{x}{10} \times \frac{20}{1}$
Relation to customer											$\frac{x}{10} \times \frac{10}{1}$
	TOTAL										
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR USE OF RESOURCES											
	Less Likely	Summative Response Scales								Most likely	
	1	2	3	4	5	6	7	8	9	10	
Use of resources											$\frac{x}{10} \times \frac{400}{1}$
	TOTAL										
b) STAFF SCHOOL, COLLEGE, INTERNATIONAL SCHOOL											
PERFORMANCE RELATED ASPECT											
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR COMPETENCE											
	Less likely	Summative Response Scales								Most likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Hardwork (quantity)											$\frac{x}{10} \times \frac{100}{1}$
Quality of work											$\frac{x}{10} \times \frac{100}{1}$
Initiative											$\frac{x}{10} \times \frac{60}{1}$
Expertise											$\frac{x}{10} \times \frac{30}{1}$
Supervision											$\frac{x}{10} \times \frac{40}{1}$

Reporting											$\frac{x}{10} \times \frac{20}{1}$
Work Planning											$\frac{x}{10} \times \frac{30}{1}$
Creativity											$\frac{x}{10} \times \frac{60}{1}$
	TOTAL										
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR INTEGRITY											
	Less likely	Summative Response Scales								Most likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Leadership											$\frac{x}{10} \times \frac{100}{1}$
Dedication											$\frac{x}{10} \times \frac{70}{1}$
Honesty											$\frac{x}{10} \times \frac{60}{1}$
Self-discipline											$\frac{x}{10} \times \frac{40}{1}$
Responsibility											$\frac{x}{10} \times \frac{40}{1}$
Reliability											$\frac{x}{10} \times \frac{40}{1}$
Punctuality (earliest arrival time to work often)											$\frac{x}{10} \times \frac{30}{1}$
Regularity or Absenteeism											$\frac{x}{10} \times \frac{30}{1}$
	TOTAL										
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR COMPATIBILITY											
	Less likely	Summative Response Scales								Most likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Team work											$\frac{x}{10} \times \frac{80}{1}$
Contributions to the immediate community											$\frac{x}{10} \times \frac{20}{1}$
Hospitality											$\frac{x}{10} \times \frac{20}{1}$
Special contributions to section/branch											$\frac{x}{10} \times \frac{20}{1}$
Relation to customer											$\frac{x}{10} \times \frac{10}{1}$
	TOTAL										
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR USE OF RESOURCES											
	Less Likely	Summative Response Scales								Most likely	

	1	2	3	4	5	6	7	8	9	10	
Use of resources											$\frac{x}{10} \times \frac{400}{1}$
	TOTAL										

c) COMPUTER CENTER, ACCOUNTING STAFF, EXECUTIVE OFFICERS ADMINISTRATIVE OFFICERS											
PERFORMANCE RELATED ASPECT											
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR COMPETENCE											
	Less likely	Summative Response Scales								Most likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Hardwork (quantity)											$\frac{x}{10} \times \frac{100}{1}$
Quality of work											$\frac{x}{10} \times \frac{100}{1}$
Initiative											$\frac{x}{10} \times \frac{60}{1}$
Expertise											$\frac{x}{10} \times \frac{30}{1}$
Supervision											$\frac{x}{10} \times \frac{40}{1}$
Reporting											$\frac{x}{10} \times \frac{20}{1}$
Work Planning											$\frac{x}{10} \times \frac{30}{1}$
Creativity											$\frac{x}{10} \times \frac{60}{1}$
	TOTAL										
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR INTEGRITY											
	Less likely	Summative Response Scales								Most likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Leadership											$\frac{x}{10} \times \frac{100}{1}$
Dedication											$\frac{x}{10} \times \frac{70}{1}$
Honesty											$\frac{x}{10} \times \frac{60}{1}$
Self-discipline											$\frac{x}{10} \times \frac{40}{1}$
Responsibility											$\frac{x}{10} \times \frac{40}{1}$
Reliability											$\frac{x}{10} \times \frac{40}{1}$
Punctuality (earliest arrival time to work											$\frac{x}{10} \times \frac{30}{1}$

often)											
Regularity or Absenteeism											$\frac{x}{10} \times \frac{30}{1}$
	TOTAL										
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR COMPATIBILITY											
	Less likely	Summative Response Scales								Most likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Team work											$\frac{x}{10} \times \frac{80}{1}$
Contributions to the immediate community											$\frac{x}{10} \times \frac{20}{1}$
Hospitality											$\frac{x}{10} \times \frac{20}{1}$
Special contributions to section/branch											$\frac{x}{10} \times \frac{20}{1}$
Relation to customer											$\frac{x}{10} \times \frac{10}{1}$
	TOTAL										
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR USE OF RESOURCES											
	Less Likely	Summative Response Scales								Most likely	
	1	2	3	4	5	6	7	8	9	10	
Use of resources											$\frac{x}{10} \times \frac{400}{1}$
	TOTAL										

d) CONFIDENTIAL SECRETARIES, DATA ENTRY OPERATORS, TELEPHONE OPERATORS, LIFT OPERATORS											
PERFORMANCE RELATED ASPECT											
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR COMPETENCE											
	Less likely	Summative Response Scales								Most likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Hardwork (quantity)											$\frac{x}{10} \times \frac{100}{1}$
Quality of work											$\frac{x}{10} \times \frac{100}{1}$
Initiative											$\frac{x}{10} \times \frac{60}{1}$
Expertise											$\frac{x}{10} \times \frac{30}{1}$
Supervision											$\frac{x}{10} \times \frac{40}{1}$
Reporting											$\frac{x}{10} \times \frac{20}{1}$

Work Planning											$\frac{x}{10} \times \frac{30}{1}$
Creativity											$\frac{x}{10} \times \frac{60}{1}$
	TOTAL										
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR INTEGRITY											
	Less likely	Summative Response Scales								Most likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Leadership											$\frac{x}{10} \times \frac{100}{1}$
Dedication											$\frac{x}{10} \times \frac{70}{1}$
Honesty											$\frac{x}{10} \times \frac{60}{1}$
Self-discipline											$\frac{x}{10} \times \frac{40}{1}$
Responsibility											$\frac{x}{10} \times \frac{40}{1}$
Reliability											$\frac{x}{10} \times \frac{40}{1}$
Punctuality (earliest arrival time to work often)											$\frac{x}{10} \times \frac{30}{1}$
Regularity or Absenteeism											$\frac{x}{10} \times \frac{30}{1}$
	TOTAL										
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR COMPATIBILITY											
	Less likely	Summative Response Scales								Most likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Team work											$\frac{x}{10} \times \frac{80}{1}$
Contributions to the immediate community											$\frac{x}{10} \times \frac{20}{1}$
Hospitality											$\frac{x}{10} \times \frac{20}{1}$
Special contributions to section/branch											$\frac{x}{10} \times \frac{20}{1}$
Relation to customer											$\frac{x}{10} \times \frac{10}{1}$
	TOTAL										
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR USE OF RESOURCES											
	Less Likely	Summative Response Scales								Most likely	
	1	2	3	4	5	6	7	8	9	10	

Use of resources											$\frac{x}{10} \times \frac{400}{1}$
	TOTAL										

e) SECURITY, COACHING, LAUNDRY, BINDERY, HOUSE KEEPING, GUEST HOUSE STAFF											
PERFORMANCE RELATED ASPECT											
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR COMPETENCE											
	Less likely	Summative Response Scales								Most likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Hardwork (quantity)											$\frac{x}{10} \times \frac{100}{1}$
Quality of work											$\frac{x}{10} \times \frac{100}{1}$
Initiative											$\frac{x}{10} \times \frac{60}{1}$
Expertise											$\frac{x}{10} \times \frac{30}{1}$
Supervision											$\frac{x}{10} \times \frac{40}{1}$
Reporting											$\frac{x}{10} \times \frac{20}{1}$
Work Planning											$\frac{x}{10} \times \frac{30}{1}$
Creativity											$\frac{x}{10} \times \frac{60}{1}$
	TOTAL										
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR INTEGRITY											
	Less likely	Summative Response Scales								Most likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Leadership											$\frac{x}{10} \times \frac{100}{1}$
Dedication											$\frac{x}{10} \times \frac{70}{1}$
Honesty											$\frac{x}{10} \times \frac{60}{1}$
Self-discipline											$\frac{x}{10} \times \frac{40}{1}$
Responsibility											$\frac{x}{10} \times \frac{40}{1}$
Reliability											$\frac{x}{10} \times \frac{40}{1}$
Punctuality (earliest arrival time to work often)											$\frac{x}{10} \times \frac{30}{1}$
Regularity or Absenteeism											$\frac{x}{10} \times \frac{30}{1}$

	TOTAL									
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR COMPATIBILITY										
	Less likely	Summative Response Scales							Most likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Team work										$\frac{x}{10} \times \frac{80}{1}$
Contributions to the immediate community										$\frac{x}{10} \times \frac{20}{1}$
Hospitality										$\frac{x}{10} \times \frac{20}{1}$
Special contributions to section/branch										$\frac{x}{10} \times \frac{20}{1}$
Relation to customer										$\frac{x}{10} \times \frac{10}{1}$
	TOTAL									
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR USE OF RESOURCES										
	Less Likely	Summative Response Scales							Most likely	
	1	2	3	4	5	6	7	8	9	10	
Use of resources										$\frac{x}{10} \times \frac{400}{1}$
	TOTAL									

f) LIBRARY, RADIO, PRESS, CREATIVE ARTS STAFF											
PERFORMANCE RELATED ASPECT											
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR COMPETENCE											
	Less likely	Summative Response Scales								Most likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Hardwork (quantity)											$\frac{x}{10} \times \frac{100}{1}$
Quality of work											$\frac{x}{10} \times \frac{100}{1}$
Initiative											$\frac{x}{10} \times \frac{60}{1}$
Expertise											$\frac{x}{10} \times \frac{30}{1}$
Supervision											$\frac{x}{10} \times \frac{40}{1}$
Reporting											$\frac{x}{10} \times \frac{20}{1}$
Work Planning											$\frac{x}{10} \times \frac{30}{1}$
Creativity											$\frac{x}{10} \times \frac{60}{1}$
	TOTAL										

ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR INTEGRITY											
	Less likely	Summative Response Scales								Most likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Leadership											$\frac{x}{10} \times \frac{100}{1}$
Dedication											$\frac{x}{10} \times \frac{70}{1}$
Honesty											$\frac{x}{10} \times \frac{60}{1}$
Self-discipline											$\frac{x}{10} \times \frac{40}{1}$
Responsibility											$\frac{x}{10} \times \frac{40}{1}$
Reliability											$\frac{x}{10} \times \frac{40}{1}$
Punctuality (earliest arrival time to work often)											$\frac{x}{10} \times \frac{30}{1}$
Regularity or Absenteeism											$\frac{x}{10} \times \frac{30}{1}$
	TOTAL										
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR COMPATIBILITY											
	Less likely	Summative Response Scales								Most likely	Formula
	1	2	3	4	5	6	7	8	9	10	
Team work											$\frac{x}{10} \times \frac{80}{1}$
Contributions to the immediate community											$\frac{x}{10} \times \frac{20}{1}$
Hospitality											$\frac{x}{10} \times \frac{20}{1}$
Special contributions to section/branch											$\frac{x}{10} \times \frac{20}{1}$
Relation to customer											$\frac{x}{10} \times \frac{10}{1}$
	TOTAL										
ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR USE OF RESOURCES											
	Less Likely	Summative Response Scales								Most likely	
	1	2	3	4	5	6	7	8	9	10	
Use of resources											$\frac{x}{10} \times \frac{400}{1}$
	TOTAL										

Note: For each achievement criteria performance measurement there should be a Box for Head of Department/Unit Max Score

Maximum Score

Appraisal

Maximum Score

HOD/Unit Max. Score

yes No

Appraisal Acceptance

HOD/UNIT
Justification____

Note: If Appraisal's Score is within plus/minus 10% of HOD/UNIT Max. Score then accept and record the average between both scores. i.e. Appraisal score attained and the HOD/Unit Max. Score

Note: If Appraisal's Score is below (minus 10% of HOD/UNIT Max. Score and the HOD/UNIT has justified his/her score without alteration) then accept and record only the HOD/UNIT Max. Score but do not save the result also flag the score for the attention of the appraisal auditor for resolution.

Note: If Appraisal's Score is above (plus 10% of HOD/UNIT Max. Score and the HOD/UNIT has justified his/her score without alteration) then accept and record the average between both scores. i.e. Appraisal score attained and the HOD/Unit Max. Score

Appraisal Model

$$Effective\ Score = \left[\frac{Achieved\ Total\ Scores\ (X)}{Total\ Available\ Score\ (Y)}\right] \times 100$$

The following should be featured under the Non academic staff appraisal

(a) State below in order of importance the main duties performed in your job during the period of report?

………..
....

(b) Was there any joint discussion between you and your supervisor on how to accomplish the tasks? And when?

………...
....

(c) Were you properly equipped professionally/technically/administratively to perform the jobs allotted to you? YES/NO. If not, what were your difficulties or constraints?

…………………………………………………………………………………………………

……………………

(d) In the light of (c) above, state the various difficulties encountered in carrying out your duties and the efforts you and your Supervisor put in to rectify them?

………...
....
………...
....

(e) What were the methods adopted by your Supervisor to assist you in solving the difficult problems?
………...
....

(f) Was there any periodic (three months, six months) review of your methods/techniques by your Supervisor to achieve the desired goals?. …….…………………………………………………
………...
....
………...
....

(g) After the review, did your performance measure up to the prescribed standards set at the beginning of the year?...
....
………...
....

(h) If the answer to (g) above is NO, state what solution or admonition was given for the shortcomings?

………...
....

(i) How did your performance relate to the total accomplishment of the goals set for your Faculty/College/Department/Unit and the vision of the University?
………...
....

………………………………………………………………………………………………
…………….............

(j) State any ad-hoc duties performed during the period, if any:
………………………………………………………………………………………………
…………….............

………………………………………………………………………………………………
…………….............

(k) How did the performance of ad hoc duties affect your real duties?
Positively ( ) Negatively ( )

………...
....

If negative, did you bring this to the attention of your Supervisor?

………………………………………………………………………………………………
……………….....

(l) State the period that you have been on the schedule of duty referred to in (a) above:
………………………………………………………………………………………………
……………….............
………………………………………………………………………………………………
…………….............

From:…………………………… To:……………
…………………………

STAFF APPRAISED COMMENT AND RECOMMENDATION ON

Job Performance

Comment on duties performed during the period of this report:

(a) Looking back on the past year, which jobs assigned to you do you think you have undertaken satisfactorily?

……
……

(b) i. What were the factors to which you ascribe your success?

………
………

ii. What were the factors to which you ascribe your failure?

………

………

(c) Based on your response to (a) and (b) above, include in <u>not more than two-pages if any</u>, a paper on your observations of current challenges facing the University and your suggestions on the way forward.

………
………

(d) Do you think that you need more training or experiences to enable you do your job better? YES/NO.

If so, what kind?…………………………………………………………………………………………

………

Did go for any training course during this just concluded appraisal session? Yes/No. Was there any examination conducted after the training period(s)? Yes/No.
If Yes, Please upload certificate(s) awarded.

(e) Is the most effective use being made of your capabilities in your present job?
………

………

(f) Do you think that your abilities could be better used in your present job or in another kind of job?

………

……...

(g) During the period of this report did you have job satisfaction? If not, what were the reasons?

……....

……..

(h) Any other comment on issues not mentioned above?

……...

……...

(i) Date Report was submitted to the Reporting Officer ……………………………………………………….

Individual………………………………………….. Position……………………………………….. Qualification…………………………………. Appraiser…………………………………………..	Department/Service………………………………………………………………….. Appraisal Period………………………………………………………………………… Date of Appraisal……………………………………………………………………………..

S/No	Work Parameter	Possible Range of Point Scores for							Performance	
		(1) Very Poor	(2) Poor	(3) Fair	(4) Good	(5) Very Good	(6) Excellent	(8) Very Outstanding	(8) Grade	(9) Point Score
1	Hardwork (Quantity)	0 – 20	21-40	41 - 50	51-61	66 - 80	81 – 90	91 - 100		
2	Quality of Work	0 – 20	21-40	41 - 50	51-61	66 - 80	81 – 90	91 - 100		
3	Initiative	0 – 10	11 – 20	21 – 30	31 – 44	45 – 50	51 – 55	56 - 60		
4	Creativity	0 – 10	11 – 20	21 – 30	31 – 44	45 – 50	51 – 55	56 - 60		
5	Expertise	0 – 4	5 - 8	9 – 14	15 – 21	22 – 24	25 – 27	28 - 30		
6	Supervision	0 – 6	7 – 12	13 – 21	22 – 28	29 – 32	33 – 36	37 - 40		
7	Reporting	0 – 4	5 - 8	9 – 14	15 – 21	22 – 24	25 – 27	28 - 30		
8	Work Planning	0 – 4	5 - 8	9 – 14	15 – 21	22 – 24	25 – 27	28 - 30		
9	Leadership	0 – 20	21-40	41 – 50	51- 61	66 - 80	81 – 90	91 - 100		
10	Dedication	0 – 10	11 – 24	25 – 34	35 – 55	56 – 60	61 – 65	66 – 70		
11	Honesty	0 – 10	11 – 20	21 – 30	31 – 44	45 – 50	51 – 55	56 - 60		
12	Self Discipline	0 – 6	7 – 12	13 – 21	22 – 28	29 – 32	33 – 36	37 - 40		
13	Responsibility	0 – 6	7 – 12	13 – 21	22 – 28	29 – 32	33 – 36	37 - 40		
14	Reliability	0 – 6	7 – 12	13 – 21	22 – 28	29 – 32	33 – 36	37 - 40		
15	Punctuality	0 – 4	5 - 8	9 – 14	15 – 21	22 – 24	25 – 27	28 - 30		
16	Regularity	0 – 4	5 - 8	9 – 14	15 – 21	22 – 24	25 – 27	28 - 30		
17	Team Work	0 – 10	11 – 20	21 – 35	36 – 50	51 – 60	61 – 70	71 - 80		
18	Community	0 – 3	4 – 6	7 –	11 –	15 –	17 – 18	19 – 20		

	Contribution			10	14	16				
19	Hospitality	0 – 3	4 – 6	7 – 10	11 – 14	15 – 16	17 – 18	19 – 20		
20	Special Contribution	0 – 3	4 – 6	7 – 10	11 – 14	15 – 16	17 – 18	19 – 20		
21	Relation to Customer	0 – 1	2	3	4	5 – 6	7 – 8	9 - 10		
22	Use of Resources	0 – 40	41 – 90	91 – 140	141 – 250	251 – 300	301 – 350	351 – 400		

Parameter-Grade Matrix: Performance Appraisal Scheme

$$If\ Actual\ Achievement\ is \begin{cases} -intolerable\ below\ trarget - write\ Very\ poor\ (VP) \\ -tolerable\ but\ well\ below\ target - write\ poor\ (P) \\ -only\ a\ little\ below\ target - write\ fair\ (F) \\ -equal\ to\ Target - write\ Good\ (G) \\ -only\ a\ little\ above\ target - write\ Very\ Good\ (VG) \\ -well\ above\ target - write\ Excellent\ (E) \\ -Far, far\ above\ target - write\ Very\ Outstanding\ (VO) \end{cases}$$

Quantitative Score	Performance Grade	Descriptive Grade (Overall Performance Grade)
91 – 100%	1st Class	Very Outstanding
81 – 90%	2nd Class	Excellent
66 – 80%	3rd Class	Very Good
50 – 65%	4th Class	Good
41 – 49%	5th Class	Fair
21 – 40%	6th Class	Poor
0 – 20%	7th Class	Very Poor

Performance Classification Scheme

There should be a section provided for auditing the appraised staff especially those that do not accept their HOD/Unit heads scores. This section should only be made available only after an appraisal have been done/carried out.

This auditing should/must be done by consulting firms with personnel staff whose discipline should be in Industrial Engineering/Industrial Psychologist.

The features they (Appraisal Auditors) perform should include but not limited to

- Economic function vis-à-vis social responsibility
- Corporate structure
- Health of earnings
- Service to the shareholders
    - i) Minimization of risk to investment
    - ii) Reasonable return on investment
    - iii) Reasonable appreciation of capital/performance over a period of time
- Research and development

- Analysis of board of directors
  i) The quality of each director and his contribution to the board
  ii) The extent to which the directors work as a team
  iii) Whether the directors act as trustees for the organization
- Fiscal and Financial Policies
- Production efficiency
- Sales Vigor
  i) The extent to which the past sales potential has been realized
  ii) The extent of development of sales personnel
  iii) The extent to which the present sales policies of the company enable its management to realize further sales potential
- Executive Evaluation

i.e. they must be able to provide scores for the above indices.

The following indices can only be performed after the auditors (Consultant Appraisal Firm) has resolved all issues especially for staff that do not agree with their HOD's scores and these are

ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR MANAGEMENT										
	Less likely	Summative Response Scales							Most likely	Formula
	1	2	3	4	5	6	7	8	9	10	
hardwork										$\frac{x}{10} \times \frac{100}{1}$
Initiative										$\frac{x}{10} \times \frac{60}{1}$
reporting										$\frac{x}{10} \times \frac{20}{1}$
planning										$\frac{x}{10} \times \frac{30}{1}$
supervision										$\frac{x}{10} \times \frac{40}{1}$
teamwork										$\frac{x}{10} \times \frac{80}{1}$
leadership										$\frac{x}{10} \times \frac{100}{1}$
Use of resources										$\frac{x}{10} \times \frac{400}{1}$
	TOTAL									

ACHIEVEMENT CRITERIA PERFORMANCE MEASUREMENT FOR PRODUCTIVITY											
	Less likely	Summative Response Scales								Most likely	Formula
	1	2	3	4	5	6	7	8	9	10	
hardwork											$\frac{x}{10} \times \frac{100}{1}$
quality											$\frac{x}{10} \times \frac{100}{1}$
initiative											$\frac{x}{10} \times \frac{60}{1}$

creativity											$\frac{x}{10} \times \frac{60}{1}$
Use of resources											$\frac{x}{10} \times \frac{400}{1}$
	TOTAL										

Loyalty includes (Dedication and Self-discipline)

Diligence includes (Reporting, Self-discipline and responsibility)

Natural Motivation includes (Leadership, Reliability and community contribution)

For the above two achievement criteria performance measurement (management and productivity), the scores are pulled from the already analyzed result for each staff.

MOTIVATION OF STAFF (ACADEMIC AND NON ACADEMIC)

The following should be suggested to the management top staff (VC/MD) for he/her to choose what he will like to enforce for the motivation of his staff

The IT Admin in charge of this software should be able to reset this section when there is a change in-top management (VC/MD) and each adopted setting for any administration be save on the records for that tenor. There should be room for him to make adjustment.

Training

- Sponsorship to higher education in specialist areas such as computer literacy courses (Grade level 1-7)
- Specialist short term training programmes such as workshop, conferences and seminars (junior and senior staff)
- Educational scholarships for the staff/workers (junior staff Grade level 1-7 and senior staff/ excluding top management staff)
- Adequate training for new employees
- In-house training for reprimands*
  - i) Specialist training*
  - ii) Selective training*
  - iii) Deficiency training*
  - iv) Special scholarship*
- Special counseling Units staff (for very poor grade appraised staff)*

  Note: * indicates compulsory for selection

Non-Financial Fringe Benefits

- Staff transportation plan
- Staff canteen subsidy
- Uniform for workers (where applicable)
- Staff medical scheme
- Staff housing loan
- Travelling for sports and games
- Insurance schemes

- promotions
- There should be room for personal addition(s)

Financial staff sustenance items (this should be enforced in his/her monthly salary)

- Housing allowance
- Mass increase in salaries (as at when due)
- Transport allowance
- Leave allowance
- Feeding allowance
- Gifts
- Study allowance
- Lunch allowance
- Christmas bonus
- There should be room for personal addition(s)

Trophies Award

- Gold Trophy
- Silver Trophy
- Bronze

Note: The achievement parameter measurements are grouped or categorized into 4 and these are

1. Competence
2. Integrity
3. Compatibility
4. Use of resources

And based on these should award be given together with the overall best three staff for the department/faculty for academic staff and faculty for nonacademic staff in the institutions.

And based on these should award be given together with the overall best three staff for the department staff/worker and for staff of the company/firm.

Note: From the above achievement parameter measurements, the software should be able to determine the

- Which is the biggest strength of the department and faculty in terms of (**competence**, **integrity, compatibility** and (**prudence in spending** (to be determine from the **minimum effective use of resources with maximum satisfaction of the department/faculties needs** which is an indices to be determined by the staff of the departments and the faculties))
- Which is the personnel biggest weakness in relative terms to: (**competence, integrity, compatibility** and (**prudence in spending** (to be determine from the **minimum effective use of resources with maximum satisfaction of the department/faculties needs** which is an indices to be determined by the staff of the departments and the faculties))
- In what specific work parameter are the personnel most highly distinguished? In what are the worst off?

Note: There should be designed barges and certificates (designed in **Gold-for 1st overall performances**, **silver-for 2nd overall performances** and **bronze-for 3rd overall performances** positions for civil service and companys/firms) and also there should be beautifully designed **1st book of records** and **2nd**

book of records and records for those who made the list of the **Member of the Hall of Fame** in the software.

The below table shows what the software should be able to generate for the staff

S/No	Types of Motivators	Name/Description	Wining Criteria
1	Best Workers Identity Display	Best worker of the month picture	Overall best performer of the month
		Best worker of the quarter picture	Overall best performer of the quarter
		Biannual best workers' picture	Overall best performer of the first six month
		Annual best workers' picture	Overall best performer of the year
		2nd list of best performers in assorted work achievement parameters	"Excellent" performance in the selected work parameters
		1st list of best performers in assorted work achievement parameters	"Very Outstanding" performance in the selected work parameters
		2nd Book of Records	"Very Good" 3-year overall performance
		1st Book of Records	"Excellent" 3-year overall performance
		Member of the Hall of Fame	"Very Outstanding" 3-year overall performance
2	Gifts	Communication devices: Radios, Videos, Mobile Phones, Televisions, Computers etc.	Annual and 3-years above "good" performance
		Semi-monumental: Bicycles, Motorcycles, vehicles, etc.	Annual and 3-years above "good" performance
		Monumental: Land, Houses	6-years overall "Very good", "Excellent" and "Very Outstanding"
		Books	Annual performance from "Good", to "Very Outstanding"
		Wares and Households	Ditto
		Custom made clothes	Ditto
		Certificate of competence 3rd class	"Very good" quarterly, biannual and annual performance
		2nd class	"Excellent" quarterly, biannual and annual performance

3	Achievement Certificates	1^{st} class	"Very Outstanding" quarterly, biannual and annual performance
		• Certificate of Integrity 3^{rd} class 2^{nd} class 1^{st} class	The same as the certificate of competence
		• Certificate of Compatibility 3^{rd} class 2^{nd} class 1^{st} class	Ditto
		• Certificate of Prudence/use of resources 3^{rd} class 2^{nd} class 1^{st} class	Ditto
		• Certificate of Management 3^{rd} class 2^{nd} class 1^{st} class	Ditto
		• Certificate of Productivity 3^{rd} class 2^{nd} class 1^{st} class	Ditto
		• Certificate of Loyalty 3^{rd} class 2^{nd} class 1^{st} class	Ditto
		• Certificate of Leadership 3^{rd} class 2^{nd} class	Ditto

		1st class	
		• Certificate of Diligence 3rd class 2nd class 1st class	Ditto
		• Certificate of Natural Motivation 3rd class 2nd class 1st class	Ditto
		• Certificate of Public Relation 3rd class 2nd class 1st class	Ditto
		• Certificate of Punctuality and Regularity 3rd class 2nd class 1st class	Ditto
		• Certificate of Devotion/dedication 3rd class 2nd class 1st class	Ditto
		• Certificate of Honesty 3rd class 2nd class 1st class	Ditto
		• Certificate of Team Cooperation 3rd class 2nd class 1st class	Ditto

		• Certificate of Quality performance 3^{rd} class 2^{nd} class 1^{st} class	Ditto
		• Certificate of Creativity 3^{rd} class 2^{nd} class 1^{st} class	Ditto
		• Certificate of Planning 3^{rd} class 2^{nd} class 1^{st} class	Ditto
4	Financial	Some money to accompany every achievement certificate, rank promotion, best worker identity displays and souvenirs, monumental gifts etc.	Based on criteria for winning other motivators; the establishment to decide the actual amount but must be store/recorded
5	Training and scholarship	• Special Training	Best overall cumulative performance
		• Selective Training	“Very Good” to “Very Outstanding” annual and 3-year performance
		• Skills Improvement Training	“Very Poor” to “Fair” annual 3-year overall performance
		• Leadership Training	The same as in skills improvement Training
		• Special Scholarship Award	The same as in selective training
6.	Promotion	• Rank Promotion	3-year “Excellent” or “Very Outstanding” performance
		• Promotion into existing positions	3-year “Very Good” to “Very Outstanding” performance
		• Step Incremental Promotion	Ditto

Motivation Action Scheme

Period / Performance level	Monthly	Quarterly	Biannually	Annually	3-yearly	6-yearly
Very Poor	Counseling	Written warning	Stern written warning	Stern written warning and deficiency training	Lay-off at the end of the period	
Poor	Counseling	Verbal warning	Written warning	Stern written warning and deficiency training	Last stern warning	Lay-off at the end of the period
Fair	Counseling	Counseling	Counseling	Written advice and deficiency training	Written advice and deficiency training	
Good	Assorted best worker displays if any	Achievement certificate and associated cash and kind if any	Achievement certificate and associated cash and kind if any	Achievement certificate and associated cash and kind if any	Achievement certificate and associated cash and kind if any	
Very Good	Assorted best worker displays if any	Assorted 3rd class achievement certificates, appropriate displays plus associated cash	Assorted 3rd class certificates, displays plus associated cash, gifts	3rd class certificates, displays, gifts, souvenir, cash etc.	Incremental promotion, Estab. 2nd Book of Records	Monumental Gifts plus cash
Excellent	Assorted best worker displays	Assorted 2nd class achievement certificates, displays plus associated gifts and cash	Assorted 2nd class achievement certificates, displays gifts and cash	2nd class certificates, displays, gifts, souvenir, cash etc.	Position promotion if available plus Estab. 1st Book of Record	
Very Outstanding	Assorted best worker displays	Assorted 1st class achievement certificates, displays, gifts, cash, etc	Assorted 1st class achievement certificates, displays gifts and cash	1st class certificates, gifts, souvenir, cash etc.	Rank promotion or position promotion plus Hall of Fame Membership	

The below must be flagged on the notices in the software

- Government establishments should annually publish their work-plan to include the government goals they intend to pursue in the year and the periods they hope to achieve them.
- There should be known days to be celebrated with fanfare. The press should be invited and any interested member of the public allowed to come to verify all achievements
- Schedule verification days are to be known (displayed in the software) to all the staff at the beginning of the year and should not be changed except for completely unavoidable events beyond the control of the government/establishment. This appears a way the chief executive of an establishment can be objectively assessed by the tax payers.
- Performance Award days for monthly, quarterly, biannual or annual awards, if any are to be known (displayed in the software) to all the staff at the beginning of the year and should not be subject to change.
- The appraisal dates should be known well ahead of time.
- The schedule of auditors visiting dates too may be specified in the software and strictly adhered to. And the use of internal and external auditors should be encouraged.
- The motivators to be applied for any appraisal year/season should be judiciously selected strictly applying the “variety” and the “resources availability principles”
- And for each year, every award given and the receivers must be recorded

Pseudo code for software

1) If setting for academic & non academic staff (page 1) has been Performed

2) If academic and non academic staff profiles are captured (page 2-3) from existing database

3) Else determine the number of academic and non-academic staff (academic-staff page 42-40, non academic staff page 49-73)

4) If the period for appraisal (semester or yearly page 3) and computational right has been assign to Estab/personnel and the choice to carry out motivation has been determine by VC and management.

5) Then, capture academic & non-academic staff data and check for data fitness (page 8, 31-39). If data capture is fit for analysis

6) Then perform appraisal (academic page 15-30, non-academic page 73-77).

7) If stress level for (academic and non-academic staff Page 9-15, 38-42),

8) Then, It's time to allow external auditors to verify appraisal result

9) Then, perform motivation if set in item 4 (page 93-101)

End if,

End if,

End if,

End if,

End if,

End if,

End if,

End.

Note: The academic appraisal and non-academic appraisal are two unique entities that must be performed separately.

PREVENTIVE MAINTENANCE (PM) MODEL

Job specification:

Each specification should indicate:

1) The identification number and name of the item of equipment.

2) The location of the item.

3) The maintenance schedule reference number (and task reference) from which the task was extracted.

4) The job specification reference number. (Each job specification should be allocated its own individual reference number).

5) The frequency at which the particular task must be carried out.

6) The trade or trades required to carry out each operation.

7) The specific details of the work to be done – the procedure and method, details of inspections, tests, servicing, etc. allowable limits and tolerances.

8) Components to be replaced.

9) Special tools and equipment to be used.

10) Reference drawings, manuals, etc. applicable.

11) Safety procedures to be followed.

FACULTY IDENTIFICATION No	**MAINTENANCE SCHEDULE REF No………………..** **ITEM No ………………..**	**JOB SPECIFICATION No STANDARD**
FACILITY DESCRIPTION STEAM STOP VALVE	**JOB DESCRIPTION:-** OVERHAUL OF STEM STOP VALVE	
LOCATION:- ALL STEAM LINES ON SITES	**TRADE:-** MECH FITTER	**FREQIENCY:-** ANNUALLY

1. OBTAIN permission-to-Work
2. Sot off the two steam. Stop valves on each side of valve to be overhauled. Secure in the closed position with locks and chain.
3. Allow time for the line steam pressure to drop to zero.
4. Remove valve from line. Fit blanks to open ends of the line. If no replacement valve is available
5. Remove bridge cover nuts and withdraw cover, spindle, and valve lid.
6. Dismantle gland, gland packing, hand-wheel, spindle and valve lid.
7. Examine the following components.

GLAND for bending and cracking VALVE SPINDLE in way of packing and at screw thread VALVE LID FACE for scoring and pitting, machine if necessary STUDS on valve chest for condition and fit, hammer test for brittleness. VALVE SEAT FACE for scoring and pitting, machine if necessary. VALVE CHEST FLANGES for condition and flatness 8. Compare dimensions of valve seat, valve lid, and spindle with manufacturers drawings. All spindles, valves, etc. to be brought back to drawing dimensions on renewal. 9. Water test valve chest to twice working pressure. 10. Assemble valve and water test twice working pressure. 11. Remove blanks and replace valve in line. 12. Remove chains and locks on adjacent stop valves. 13. RETURN permit-to-work.	
DRAWINGS REQUIRED	
SPECIAL TOOLS REQUIRED	
OBSERVED SAFETY REGULATIONS	**CLEAR SITE ON COMPLETION**

A standard job specification (for the annual overhaul of steam stop valves)

Maintenance aids:

Much of the work of preventive maintenance consists primarily of examination to ensure that certain components and assemblies are fit for further operational service. As a result of this, certain and special tools are needed as aids to preventive maintenance process. So of these vital tools include:

1) **Temperature indication stickers**: they are flexible self-adhesive strips which, when the temperature of the surface to which they are adhered to reaches a specified level an indicating panel on the strip turns black. These strips can be affixed easily to potential thermal trouble spots.

Uses:

i) To provide proof of excessive temperature exposure
ii) To detect temperature anomalies
iii) To measure temperature gradients

2) **Electronic thermometers (portable)**: It’s an instrument which consists of a small, pencil-shaped probe carrying a heat sensitive semiconductor element connected by flexible cable to a pocket size indicator unit. They are also used as temperature indicators and are capable of measuring solid, liquid, gas or surface temperatures

Area of application includes:

i) Bearing temperatures; “hot spot” location

ii) Checking air conditioning, refrigeration and heating systems
iii) Pipe and vessel temperatures
iv) Tracing temperature gradients and patterns

3) **Temperature probes**: temperature probes, thermocouples and resistance thermometers comes in a variety of shapes, sizes and mountings, built into the plant at inaccessible, thermally-strategic points and can give warnings of temperature anomalies before they cause trouble.

Uses:

i) Temperatures of bearings, cylinder heads.
ii) Winding temperatures of electric motors, generators and transformers

4) **Pistol thermometers**: Is a self-contained, battery-powered, portable instrument which measures the precise surface temperature of any object, moving or stationary, without surface contact for surfaces such as in any liquid or solid – plastic, chemical, metal, rubber, ceramic, fabric, glass.

Area of application includes:

i) Scanning surfaces for "hot spots" (breakdown of insulation or refactory lining)
ii) Checking furnace and tube temperatures
iii) Facing brick and refactory temperatures
iv) "Trouble shooting" on machinery, duct work, bearings, service mains.
v) Electrical wiring, motors, transformers, etc.
vi) Checking process systems.
vii) Balancing heating elements.
viii) Temperatures of molten metals.
ix) Ascertaining temperature profiles.

5) **Leak detector**: comes in various forms however, there is a simple hand-held, battery-operated instrument designed to detect rapidly minute air or gas leaks in both pressure and vacuum systems.

Uses:

i) Pipes and pipe-lines leakages.
ii) Orifice leaks

6) **Industrial stethoscopes**: A tunable stethoscope is a highly sensitive listening device. Faint noises are highly amplified for easy detection and finds application in both mechanical and electrical equipment.

Area of application includes:

i) Broken ball races and worn bearings

ii) Tracing fluid flow in pipes
iii) Detecting source of noises transmitted through solid objects and structures
iv) Examining live electrical equipment provided they are insulated.

7) **Stroboscopes**: Is a stroboscopic light device with mechanisms which, rotates, reciprocate and vibrate though appears to be stationary or made to go through their operations in slow motion.

Uses:

i) is use to detect faults and defects which might not be visible where the parts are stationary.

8) **Vibration meters**: These are devices that can pick up and measure the various characteristics of vibrations generated from moving or rotating parts.

Examples include:

i) accelerometer
ii) vibration sensors

Uses:

i) Enable overhauls to be planned when they are required
ii) The possibility of unexpected breakdown id minimized.

9) **Dye penetrants (for crack detection):** they are used to provide simple cheap but also a sensitive searching means of detecting the existence of surface cracks and faults in manufactured components.

Examples include:

i) A portable transistor strobe-flash

10) **Ultrasonic flaw detectors**: They are often portable, battery powered ultrasonic flaw detector which transmits ultrasonic pulse when pressed against the face of the material.

Examples include:

i) A portable transistor vibration meter
ii) An ultrasonic flaw detector etc.

11) **Wall thickness gauge**: the D-meter wall Thickness gauge is a small portable digital display ultrasonic unit designed for making rapid non-destructive thickness measurements on steel or aluminum.

Examples include:

i) D-meter wall Thickness gauge

Uses/application:

i) Universal wall thickness gauging
ii) Thickness of ship's plate
iii) Extent of eroded or corroded areas thickness on pipe walls; erosion at bends.
iv) Depth of erosion or pitting
v) Wall thickness of pressure vessels, petroleum and other storage tanks.

12) **Optical inspection devices**: they are simple robust instruments for viewing the inside of small bores, tubes and holes.

Examples include:

i) An Allen Bore Viewer

13) **Tank Viewers**: Is a portable optical instrument which is used to inspect the internal surfaces of medium sized tanks, vessels etc.

14) **Endoscope**: They are slim tubular optical instruments that enable the user to look inside cylinders, tubes and similar hollow parts, particularly when the access to the hole is small.

Uses/application:

i) Inspecting a heat-exchanger.
ii) Inspecting welds in gas tanks
iii) HP gas cylinder inspection
iv) Inspecting inside a large marine diesel engine cylinder without dismantling.

15) **Inspection mirrors**: They are used for the internal inspection of larger hollow components and for viewing normally inaccessible parts of all kinds of plant, machines and equipment.

Examples include and Uses/application:

i) **Angle of mirror adjustable by enclosed mechanism;** used for operating control in handle
ii) **Illuminated inspection mirror;** used for inspecting a fire extinguisher
iii) **Telescopic inspection mirror;** used for viewing inaccessible places

16) **Fibrescopes**: are portable flexible inspection instruments for the internal inspection of curved pipes, shaped hollow components and machinery.

17) Tachometer: Are devices used to measure the speed of any revolving part. They could be mechanical or electrical based.

Examples include:

i) Tacho-H meter

ii) Mechanical Tachometer

	Sheet No ……………..					
Identification Symbol	Description of Facility	Location	Facility Register Id. No:	Type	Priority Rating	Remarks
B-66-41)	Centre lathe: Lang 13” swing Model 3.6. Serial No 62 B/10	Maintenance work-shop		Machine tool	5	
B-66-41)	Electricmotor (driving lathe) Brookhirst Igranic Ltd No C11360/61/2; 5h.p.	-		Electric (minor)	5	Type SC: 400/440 volts 3 phase: 50 cycles
B-66-41)	Shaper “Invicta”. Type 2M B. Elliott (Machinery) Ltd. London Serial No B.E.C. 1901/4	-		Machine tool	5	
B-66-41)	Electric motor (driving shaper) Brook Motors. 3 h.p. No L131 651	-		Electric (minor)	5	A.C class E: INT rating. Frame C182: 1420 r.p.m. 400/440 V. 3ph 50
B-66-41)	Milling M/C (Universal) B. Elliott (Machinery) Ltd. London Serial No B.E.C. 011236/120	-		Machine tool	5	
B-66-41)	Electric motor (driving milling m/c) Newman: 3 h.p. Conn Diag… No C123006 Ed 30 25	-		Electric (minor)	5	Class F: 1425 r.p.m, 400/440 V: 3 ph: 50~ CMR rating: 4.9amps
B-66-41)	Steam boiler: Paxman economic No 22653: W.P. 60 P.S.I. Davy Paxman & Co Ltd. Colchester	Boiler house		Mechanical (major)	3	Space heating and water heating service
B-66-41)	Steam boiler: Paxman economic Clyde Automatic Oil Burners – Glasgow Machine No H2 584: Inst No: A8319	-		Mechanical (major)	3	Electric supply 415 V: 3ph 50~ Fitted to above boiler
B-66-41)	Steam boiler:	Boiler house		Mechanical	1	Process steam

	Cleaver Brooks Packaged boiler: Marshalls of Gainsborough. Model 621-300 h.p. No 96785			(major)		Capacity 10.350 Ibs/hr W.P 150 P.S.I: Hyd T.P. 283 Test date 17-6-68
B-66-41)	Steam boiler: Cleaver Brooks Packaged boiler: Marshalls of Gainsborough. Model 621-300 h.p No 97522	-		-	1	Details as above Hyd test date 12-6-70
B-66-41)	Air compressor: Atlas Copoo Type BT 3E: No F 30 09 30 Max Press 125 P.S.I. 8.8 kg cm3 1460 r.p.m.	Comp room		Mechanical (major)	1	Air supply to control instruments and gears
B-66-41)	Electric motor: Direct coupled to compressor. 30 h.p. Brooks Motors No X367273	-		Electric (major)	1	Frame C180 M: 1460 r.p.m C.M.R. rating: 39 amp 415 volt: 3ph. 50~
B-94-41)	After-cooler (air comp) finned. Tubes: air cooled, fan blown. Type TD3: No 910	-		Mechanical	1	Max W.P 20 Kg cm3 Test press 26… ... Test date: 1969
B-90-341)	Electric motor (fan) for cooler. A.S.E.A. No 110-016-C: 0-33 h.p. Type MT 71A 14-4	-		Electric (minor)	1	415 volts Y: 1400 r.p.m. 3 ph. 50~: 0.25 kw 1 amp
B-66-41)	Air receiver (air comp) Hodson & Brown Ltd: No HB 9775 To B.S.S. 487/60 GR D	-		Pressure vessel	1	W.P. 100-125 P.S.I. HYD Test Press 195 P.S.I Date of test 3-9-69

An inventory sheet

Identification Symbol: Each item must possess a positive means of identification

Description of facility: A brief description of the item concerned; where possible, nameplate details should also be included. This could be or include

i) Major mechanical plant – pumps, compressors, boilers.

ii) Major electrical plant – incoming transformers, breakers and switchgear, rectifiers, main motors and starts

iii) Minor mechanical plant – valves, hydraulic rams.

iv) Minor electrical plant – small motors and starters. Small electrical items transferable between different machines.

v) Instruments and instrumentation systems

vi) Pressure vessels, receivers, gas holders.

vii) Lifting gear, lifting machines, lifting tackle, hoists, lifts, jacks, cranes, slings.

viii) Machine Tools – lathes, shapers, grinders

ix) Fire-fighting services – detection systems, alarm systems, fire pumps, extinguishers, appliances.

x) Factory and office services – heating, ventilation, hot and cold water.

xi) Mobile plant – trucks, dumpers, small generator sets, mobile pumps, compressors.

xii) Major spares – the inventory should also include major spares. i.e. **rolls, gear boxes, heat exchanger tube nests**, etc. which may be taken out of one machine, reconditioned, returned to stores and then used on another machine. These items will probably be in storage, but could also be in the workshops undergoing reconditioning.

Priority Rating: this rating ranges from 1 down to 5 in descending order and indicates the relative importance of that particular item within the production process.

No. 1 Rating – Applied to all items whose efficient operation is vital to the production process. Usually a failure of one of these items would immediately affect or halt production; alternately, it would create a safety hazard.

No. 2 Rating – The failure of one of these items would not immediately affect production but could do so within a very short space of time.

No. 3 and 4 Rating – Similar to rating No. 2 but in descending order of importance.

No. 5 Rating – Plant whose failure would not affect production or create a safety hazard in any way whatsoever. They usually applies to non-productive items of equipment

Location: Often includes

01	Machine shop
02	Welding shop
03	Fitting shop

04 Pattern shop
05 Foundry
06 Press shop
07 Boiler house
08 Compressor room
09 Maintenance workshop etc.

Identification No: Location:

Type of Facility: Installation Date: Start date of Use:

Manufacturer: Date of Manufacture:

Serial No: Specification: Size: Model:

Capacity: Speed:

Total Weight: Power and Service requirements:

Connection Details: Foundation Details: Overall Dimensions:

Headroom, clearance and access dimension:

a) For the withdrawal and maintenance of components.
b) For maneuverability through restricted openings, doorways, passages, gangways, etc.

Reference drawing numbers:

Reference numbers of service manuals:

Interchangeability with other units:

General facility register

Identification No: Location:

Type of Facility: Installation Date: Start date of Use:

Manufacturer: Serial No: Date of Manufacture:

Type: Specification: Rating: Frame Size:

Power/kW: Winding: Speed: Total Weight:

Voltage: Current: Phase: Frequency:

Shaft details: Diameter, length, Keyway, height:

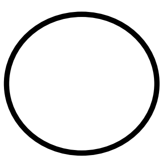
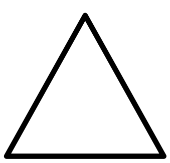
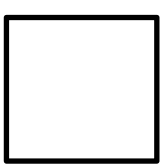
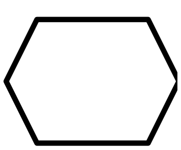
Bearings: D.E., N.D.E., H.D. bolts: Diameter and centres: Reference drawing numbers: Interchangeability with other motors: Identification number of associated starter gear:	Lubrication:

Electric motor facility register

MAINTENANCE SCHEDULE Description of facility "MBM" 15 kW Air Cooled Diesel Engine					Location: Pump House	Identification No of Facility 16-52-3	
Ref Drawing Nos						Schedule Ref: No 52693	
Srvice Manuals – Ref Nos 8234						Date Originated: 25 April 1973	
Associated Equipment: Generator No. 16-60-3						Modifications	Date
Item No.	Description	Availability for maintenance	Job spec. applicable for each item	Trade	Time Required for each item	Remarks	
	Daily – "A" Service						
1	Check level of fuel in service tank	R					
2	Check level of lubrication in oil in sump	R					
3	Clean oil bath air filter and top up with oil to the marked level	S/D					

R ……… Running maintenance
S/D……. Shut down maintenance

Maintenance schedule for diesel engine

<table>
<tr><th colspan="14">STANDARD CODING FOR LUBRICATION SYMBOLS</th></tr>
<tr><th>FREQUENCY CLASSIFICATION</th><th colspan="2">LUBRICANT CLASSIFICATION</th><th colspan="11">LUBRICANT CLASSIFICATION GRADE NUMBERS</th></tr>
<tr><th rowspan="2">SHAPE-FREQUENCY</th><th rowspan="2">COLOUR</th><th rowspan="2">OIL</th><th colspan="11">VISCOSITY RANGE-REDWOOD SECS</th></tr>
<tr><th>35-40</th><th>41-45</th><th>46-55</th><th>56-75</th><th>76-100</th><th>101-130</th><th>131-170</th><th>171-300</th><th>301-450</th><th>451-600</th><th>601-1000</th></tr>
<tr><td rowspan="13"> Daily
 Weekly
 Monthly
 Scheduled for frequencies other than those above</td><td>RED</td><td>Lubricating-Light</td><td>1</td><td>2</td><td>3</td><td>4</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>Black</td><td>Lubricating-Medium</td><td></td><td></td><td></td><td></td><td>5</td><td>6</td><td>7</td><td></td><td></td><td></td><td></td></tr>
<tr><td>Orange</td><td>Lubricating-Heavy</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>8</td><td>9</td><td>10</td><td>11</td></tr>
<tr><td>Blue</td><td>Hydraulic</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>White</td><td>Slideway</td><td></td><td></td><td></td><td colspan="2">4/5</td><td colspan="2">6/7</td><td colspan="2">8/9</td><td></td><td></td></tr>
<tr><td>Red stripes on white ground</td><td>Special Purpose. This applies to any oil outside the above classification</td><td colspan="11">Grade numbers do not apply.
Sequence numbers are applied to differentiate between types of special purpose grease</td></tr>
<tr><td rowspan="2">COLOUR</td><td rowspan="2">GREASE</td><td colspan="11">N.L.G.I. No.</td></tr>
<tr><td colspan="3">1</td><td colspan="3">2</td><td colspan="3">3</td><td colspan="2">4</td></tr>
<tr><td>Yellow</td><td>Lithium</td><td colspan="3">1</td><td colspan="3">2</td><td colspan="3">3</td><td colspan="2">4</td></tr>
<tr><td>Pink</td><td>Soda</td><td colspan="3">1</td><td colspan="3">2</td><td colspan="3">3</td><td colspan="2">4</td></tr>
<tr><td>Green</td><td>Lime</td><td colspan="3">1</td><td colspan="3">2</td><td colspan="3">3</td><td colspan="2">4</td></tr>
<tr><td>Yellow strip on black ground</td><td>Special Purpose. This applies to any grease outside the above classification</td><td colspan="11">Grade numbers do not apply.
Sequence numbers are applied to differentiate between types of special purpose grease</td></tr>
</table>

Standard coding for lubrication symbols

After identifying units with identical components that needs maintenance, the below maintenance model is then applied to these components to determine the most appropriate Preventive Maintenance (PM) interval of length of time needed before each components should be subjected to a maintenance activity. All identical components present in a unit needing maintenance should be subjected to the model at a given time. Hence, different components should be grouped differently for the model to be applied.

The model can also be applied to just an individual component if it does not have other identical components.

The generalized **stationary model** for scheduling **preventive maintenance** is given below

$$A(L) = \frac{L - dCo.H(L) - dPM}{L} \qquad \text{eqn. 1}$$

where A(L): Component availability given a PM interval of length L.

L: PM cycle length, L>0, L>>dCo, L>>dPm

dCo: Average downtime due to Co

dPM: Average downtime due to PM; where

Co: Corrective maintenance

H(L): The expected number of failures in a PM cycle length L.

When L is small, the equation below is to be used instead

$$A(L) = \frac{L - dCo.H(L)}{L + dPM} \qquad \text{eqn. 2}$$

The values of dCo and dPM can be readily estimated from the component failure data. For the model formulation of **multi PM scheduling** (which was applied to an electrical department) the following assumption is made.

1. Consider a single component system

2. Define a multi-PM (L) which contains the shortest sequence of PMs which repeats itself

3. The intervals between successive PMs are equal and of length L

4. Each PM has its own frequency ratio (R_i) in L. The frequency ratio R_i is the proportion of PM of type i in a multi PM interval L. This ratio is obtained from the user's multi PM maintenance policy. Alternatively they can be calculated from the historical maintenance data by the formula

$$R_i = \frac{No\ of\ type\ i\ PM}{Total\ no\ of\ PM}$$

5. The component is repaired immediately upon failure and its average repair time is dCo time units.

6. PM_i requires dPM_i time units on average

7. Each PM renews the component to a specific level and influence of each PM is not accumulated.

8. The failures are assumed independent, that is; a failure will not affect another.

9. The lives span of components following Co or PMi are independently distributed with identical distribution (**Weibull distribution**)

10. The lives span of components following Co or PMi are multiple censored

Let:

t: the intervals between PM;

L: the multi-PM interval length

K: the number of PM types

H(L): the expected number of failures in the interval (O,L)

$H_i(t)$: the expected number of failures

dPM_i: the average down time due to PM_i

dCo: the average down time due to Co

DPM: the total down time due to PMs in the interval (O,L)

DCo: the total down time due to Cos in the interval (O,L)

R_i: the frequency ratio of PM_i in L

R_{min}: the minimum value of R_i for i = 1, …………K

n_i: the number of PM_i in L

Now our objective is to **estimate the optimal multi-PM** interval Lopt to maximize availability, A(L), where

$$A(L) = \frac{L - DCo - DPM}{L} \qquad \text{eqn. 3}$$

The application of equation 3 can be problematic if L is small. In general one can use instead the following availability equation to overcome such problem

$$A(L) = \frac{L-DCo}{L+DPM} \qquad \text{eqn. 4}$$

It follows that

$$T_{opt} = \frac{L_{opt}}{\sum_{i-1}^{k} n_i} \qquad \text{eqn. 5}$$

The total down time due to PMi or Co can be obtained in the following manner:

$$DCo = H(L).dCo = \{\sum_{i=1}^{K} n_i.H_i(t)\}.dCo \qquad \text{eqn. 6}$$

and

$$DPM = \sum_{i=1}^{K} n_i.dPM_i \qquad \text{eqn. 7}$$

where

$$n_i = int\left[\frac{R_i}{R_{min}}\right] \qquad \text{eqn. 8}$$

To calculate Hi(t), the discretization and simulation methods stated below are used and applied both for the single and multi PM optimization models

$$H(L) = \int_{t=0}^{L} M(t)dt \qquad L \geq 0 \qquad \text{eqn. 9}$$

$$M(t) = f(t) + \int_{r=0}^{t} M(r).g(t-r)dr \qquad \text{eqn. 10}$$

The equation 10 is a second kind of **Volterra equation** and can also be solved using **Simpson rules**.

If x is even (x = 2, 4, 6, ...) then

$$Mx = fx + \frac{v}{3}\sum_{j=0}^{x} W_{xj}K[xv, jv, mj] \qquad \text{eqn. 11}$$

If x is odd (x = 3, 5, 7,.........) then

$$Mx = fx + \frac{v}{3}\sum_{j=0}^{x-3} W_{xj}K[xv, jv, mj] + \frac{3}{8}v\{K[xv, x-3v, M_{x-3}] + 3K[xv, x-2v, M_{x-2}] + 3K[xv, x-v, M_{x-1}] + K[xv, xv, M_x]\}$$

A shortcoming of the method above is the need for having starting values. In some cases, it is impossible to obtain the starting values. To calculate the integral equation of H(L) in equation 9, the simulation method is used as follows

$$\int_a^b f(x)dx = \frac{\Delta}{2}\sum_{i=0}^{n-1}\{f(i\Delta + a) + f((i+1)\Delta + a)\} \qquad \text{eqn. 12}$$

where Δ equal to 7 days and $n = \frac{(b-a)}{\Delta}$

Example

Base on the company's policy of replacing only when an item fails, the following data was obtained for each of the four tubes used.

where T_1 = Tube 1

T_2 = Tube 2

T_3 = Tube 3

T_4 = Tube 4

Readings obtained

Clock	Time to failure			
	T1	**T2**	**T3**	**T4**
0	1,230	1,050	1,750	1,430
1,050	180*	1,340	700	380
1,230	1,900	1,160	520	200*
1,430	1,700	960	320*	1,720
1,750	1,380	640*	1,460	1,400
2,390	740*	1,880	820	760
3,130	1.660	1,140	80	20*
3,150	1,640	1,120	60*	1.090
3,210	1,580	1,020*	1,140	1,030
4,230	560	1,590	120	10*
4,240	550	1,580	110*	1,410
4,350	440*	1,470	1,050	1,300
4,790	1,400	1,030	610*	860
5,400	790	420	1,960	250*
5,650	540	170*	1,170	1,120
5,820	370*	1,830	1,540	950
6,190	1,910	1,460	1,170	580*
6,770	1,330	880	590	1,460

$$dCo = \frac{Total\ downtime\ due\ to\ corrective\ maintenance}{Total\ number\ of\ replacement}$$

therefore; dCo = 1 hour

with their preventive maintenance policy of replacing all four tubes when one tube fails or after a PM interval of L before failure is reached

Table 2

Clock	Time to failure			
	T1	**T2**	**T3**	**T4**
0	1,230	1,050*	1,750	1,430
1,050	1,900	1,340*	1,460	1,720
2,390	1,660	1,880	1,140	1,090*
3,480	1,400	1,590	1,050*	1,410
4530	1,910	1,830	1,960	1,120

How the clock new sequence data is arrived at for table 2

From the first row entry (row 1) in Table 1, the minimum time to failure on that row is 1,050. So enter all entry on the first line into the new table 2. The minimum time to failure on the first line is added to the first line clock data.

0 + 1050 = 1050

The answer now becomes the second data entry to the clock

We will find that for T1, we have four stared entry. Then which entry between the **first star** and the **second star entry** on T1 that meets the highest data. This will be 1,900. This will then be the next entry for T1. For T2 we also check which entry between the **first star** and the **second star entry** on T2 meets the highest data, this will be 1,340. For T3 also check which entry between the **first star** and the **second star entry** on T3 that meets the first highest data and this will be 1,460. For T4 also check which entry between the **first star** and the **second star entry** on T4 that meets the first highest data this will be 1,720.

From the second row entry (row 2) in Table 2, the minimum time to failure on that row is 1,050. And to now generate the third row clock data, we add the minimum time to failure (which will now be 1,340) on the new generated second row data to the second line clock data. This will be:

1,050 + 1,340 = 2,390

Then to populate the new third row entry on Table 2 we fall back on Table 1. Then, we check which entry between the **second star** and the **third star entry** on T1 that meets the highest data, this will be 1,660. this will then be the next entry for T1. For T2 we also check between the **second star** and the **third star entry** on T2 that meets the highest data, this will be 1,880. For T3 also check between the **second star** and the **third star entry** on T3 that meets the highest data this will be 1,140. For T4 also check between the **second star** and the **third star entry** on T3 that meets the highest data this will be 1,090.

From the third row entry (row 3) in Table 2, the clock data on that row is 2,390. And to now generate the fourth row clock data, we add the minimum time to failure (which will now be 1,090) on the new generated third row data to the third line clock data. This will be:

2,390 + 1,090 = 3,480

Then to populate the new fourth row entry on Table 2 we fall back on Table. Then, which entry between the **third star** and the **fourth star entry** on T1 that meets the highest data, this will be 1,400. This will then be the next entry for T1. For T2 we also check between the **third star** and the **fourth star entry** on T2 that meets the highest data, this will be 1,590. For T3 also check between the **third star** and the **fourth star entry** on T3 that meets the highest data, this will be 1,050. For T4 also check between the **third star** and the **fourth star entry** on T4 that meets the highest data, this will be 1,410.

From the fourth row entry (row 4) in Table 2, the clock data on that row is 3,480. And to now generate the fifth row clock data, we add the minimum time to failure (which will now be 1,050) on the new generated third row data to the third line clock data. This will be:

3,480 + 1,050 = 4,530

Then to populate the new fifth row entry on Table 2 we fall back on Table 1. Then, which entry between the **fourth star** and the **last star entry (if any)** on T1 that meets the highest data, this will be 1,910. This will then be the next entry for T1. For T2 we also check between the **forth star** and the **last star entry (if any)** on T2 that meets the highest data, this will be 1,830. For T3 also check between the **forth star** and the **last star entry** (if any) on T3 that meets the highest data, this will be 1,9650. For T4 also check between the **fourth star** and the **last star entry (if any)** on T4 that meets the highest data, this will be 1,120.

Note: we stop the iteration process when we get to where there is no upper bound star data entry to any of the column.

Therefore, the total number of PM carried out will be 4 from the table above

Since it takes 2 hours to replace all the four tubes, then the hours spent in the PM will then be

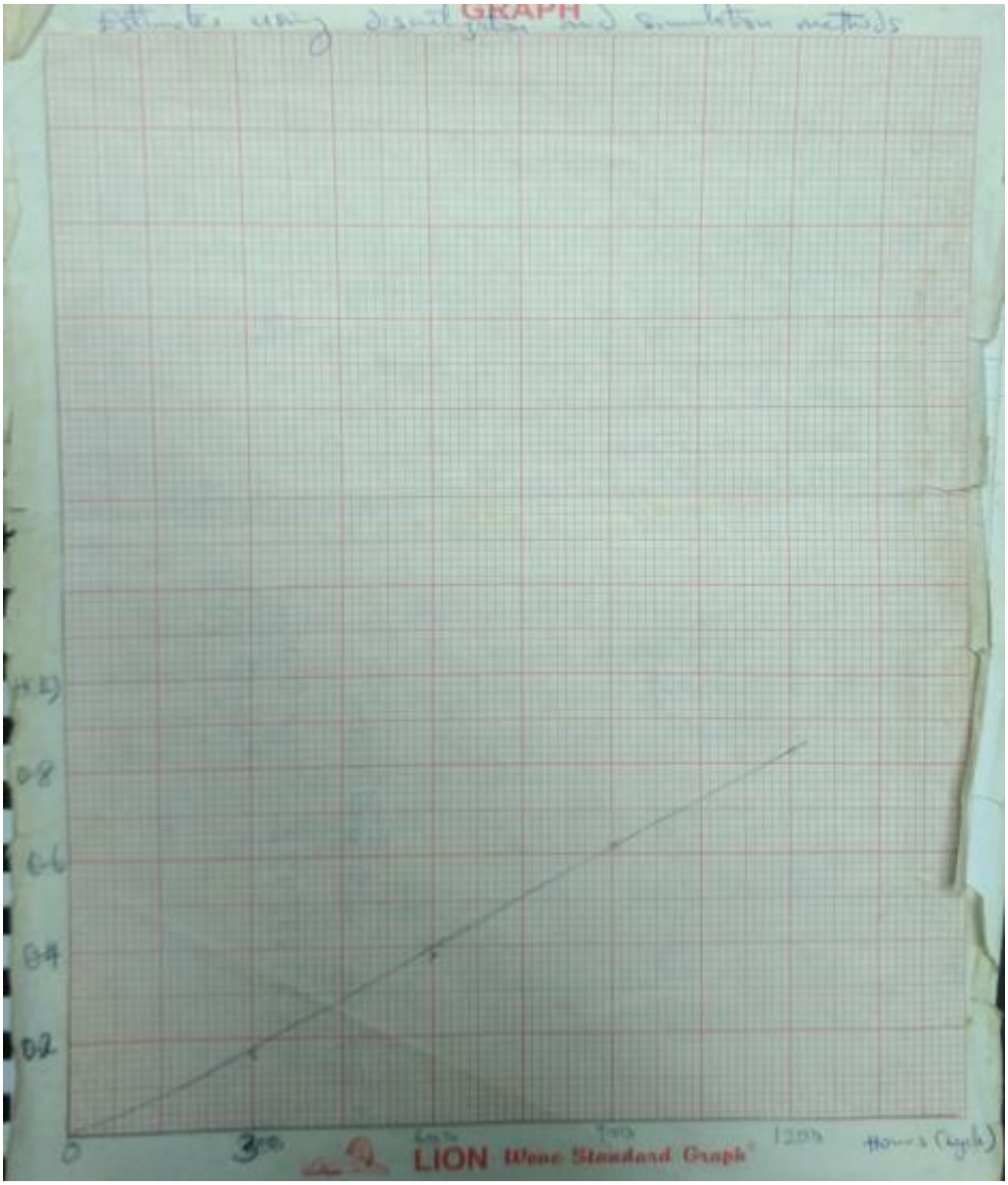
2 x 4 = 8 hours

Therefore,

$$dPM = \frac{Total\ downtime\ due\ to\ preventive\ maintenance}{Total\ number\ of\ replacement\ activity\ done}$$

dPM = 8/4 = 2 hours

A discretization and simulation process is here needed which sample can be found in the attached graph below.



From the graph above, it is found that for an interval of 1200 PM interval L, H(L) is 0.81.

Therefore,

$DCo = H(L).dCo = 0.81$

$$DP_m = \sum_{i=1}^{K} n_i.dPM_i$$

where $n_i = 4$

L = 4,530 cycles (hour)

therefore:

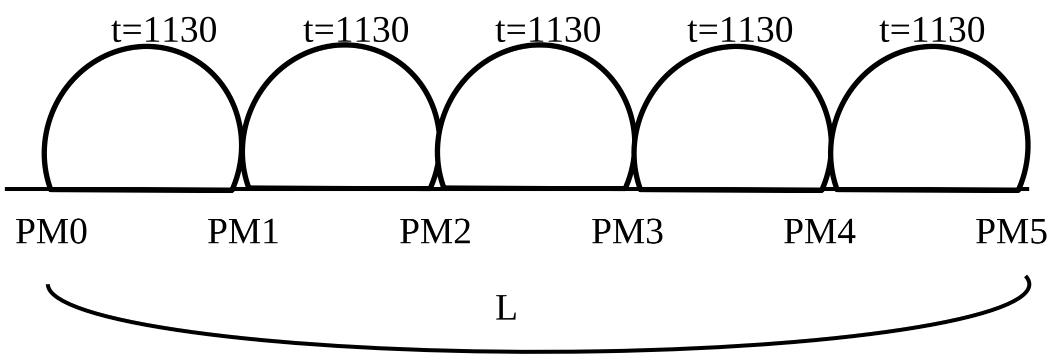
DPm = 4 x 2 = 8

$A(L) = \frac{L-DCo-DPM}{L} = \frac{4530-0.81-8}{4530} = 0.9982$

A(L) from equation will be 0.9982

Lopt = A(L) x 4530 = 0.9982 x 4530

$$T_{opt} = \frac{L_{opt}}{\sum_{i-1}^{K} n_i} = \frac{0.9982\times4530}{4} = 1130.5 \approx 1131$$



COST ESTIMATION SAVE BY PREVENTIVE MAINTENANCE

If the shutdown and replacement of the tubes cost N2000.00 per hour while each tube cost N1000.00 then:

Under Breakdown or corrective maintenance, we observed that the number of corrective maintenance carried out is 17

Cost of tubes will be 17 x N1000 = N17,000.00

Shut down cost will be 17 x N2000 = N34,000.00

Total cost will be N17,000.00 + N34,000.00 = N51,000.00

Under Preventive maintenance, we observed that the number of preventive maintenance carried out is 4 where all four tubes will be replace at once in each operation making 4 x 4 = 16

Cost of tubes will be 16 x N1000 = N16,000.00

Shut down cost for the four operation will be 4 x N2000 = N8,000.00

Total cost will be N16,000.00 + N8,000.00 = N24,000.00

The total amount saved over Break down operation will be N51,000.00 – N24,000.00 = N27,000

Randomized critical examination sheet for method study (***).

	Present Facts		Alternatives	Action
Purpose A	What is done now	Why is it done	What else could be done	What should be done
Purpose B	How is it done now	Why is it done that way	How else could it be done	How should it be done
Purpose C	When is it done now	Why is it done then	When else could it be done	When should it be done
Purpose D	Where is it done now	Why is it done there	Where else could it be done	Where should it be done
Purpose E	Who does it now	Why that person	Who else could do it	Who should do it
			Possible alternatives ↑	Selected alternatives ↑

(to be answered by the Engineer/technician involve for the actual maintenance carried out)

Job Report	Date	Report No
Name of Tradesman	Clock No	Trade
Details of Report Item: Defect/condition: Result: Corrective action: (tied to *** if lubrication is selected etc.) Spares/materials used: Measurements/observations: Remarks: Time taken:		
Facility	Location	Identification No: (note: tied to machine registered card)

JOB REPORT CARD

MACHINE:________________________ LOCATION: ___________________

NUMBER:__________________________ ___________________

Mobil LUBRICATION PROGRAMME

machine register

	ITEM	NO. OF POINTS	LUBRICANT	APPLICATION METHOD	INSPECTION/ APPLICATION FREQUENCY	CHANGE/ REPLACING FREQUENCY	CAPACITY IN GALS/LBS	
A								
B								
C								
D								
E								
F								
G								
H								
J								
K								
L								
M								

Machine register card

Week No…… W/E ……… Department		Mechanical ☐ Red Electrical ☐ Blue Instrumentation ☐ Yellow Lubrication ☐ Green						
Iden. No	Facility	Sun	Mon	Tu	Wed	Thu	Fri	Sat

A layout for a weekly maintenance plan

5000 MILE INSPECTURE

WORKS/DEPORT____________MAKE______________ FLEET No.____________ DATE______________

SPEEDO READING__________LAST INSPECTION DATE__________MILEAGE SINCE LAST INSPECTION____________

CODING: IN ORDER ✓ REPAIRS REQUIRED O IMMEDIATE REPAIRS x

STEERING

Check
- ☐ play at steering wheel
- ☐ Security of column and box
- ☐ Clearance of steering arms throughout travel
- ☐ Security of drop arm and method of locking
- ☐ Wear of steering joints
- ☐ Security of steering arms and method of locking
- ☐ Wear in swivel pins and thrust races
- ☐ Adjustment of lock stops
- ☐ Wheel alignment

TRANSMISSION

Check
- ☐ Gearbox mounting
- ☐ Clutch adjustment, clutch cross-shaft, clevis pins
- ☐ Gearbox coupling flange and flange bolts.
- ☐ Speedometer drive and cable
- ☐ Gear change mechanism and linkage
- ☐ Propeller shaft, universal joints
- ☐ Propeller shaft intermediate bearings and mountings
- ☐ Differential housing, cover plate and oil filter bolts for security and oil seals.
- ☐ Differential drive oil seals
- ☐ Drive shaft studs.
- ☐ Drive shaft oil seals
- ☐ Differential lock operating pipes

BRAKES

Check
- ☐ If brakes operate correctly
- ☐ Braking system, for leaks with footbrake applied
- ☐ Build up of vacuum or air pressure satisfactory.
- ☐ Brake warning gauge and/or device working
- ☐ Security of brake pipes, servos or reservoir.
- ☐ Wear in connections
- ☐ Security of clevis pin
- ☐ Wear in pedal pivot bush
- ☐ Wear in handbrake pivot bush, ratchet or weeping

ELECTRICAL

Check
- ☐ Battery secure and terminals clean
- ☐ Security of terminals and wiring clips
- ☐ Headlamps, sidelamps, rear lights, stoplights, spotlights, indicator lights, horn and wipers.
- ☐ Charging rate, lamp focus switches.

FRAME

Check
- ☐ Freedom from cracks, visibly true
- ☐ Security of rivets and bolts
- ☐ Security of all frame brackets.

- Wear in cross-shafts.
- Wear of brake liners
- Flexible pipes – wear, chafing or weeping.
- Handbrake cable fraying
- Hydraulic reservoir fluid level
- Correct adjustment foot and handbrake

TYRES AND WHEELS

Check
- Tyres for cults and irregular Wear
- Wheel alignment
- Wheel adjustment

EXHAUST

Check
- Security and leaks
- Density of exhaust smoke

ENGINE

Check
- Cooling system, hose pipes, clips and fan.
- Belt adjustment, radiator cap, overflow, pipe water circulation, water pump, radiator mountings.
- Engine mountings, inlet and exhaust manifolds, exhaust pipe security.
- Fuel pump mounting, fuel pump drive
- Fuel pipes, injector pipes, return pipes, filters
- Accelerator controls, venture pipes, stop/start control.
- Starter and dynamo mounting
- Tappets
- Brake compressor or exhauster
- Power steering pump, oil reservoir, steering oil pipes
- Engine performance for excess fuel consumption, pump timing, defective injectors.
- Engine oil, oil filters and air filters

- Serviceability of registration plates and reflectors

OTHER AREAS

Check
- Mounting brackets and pads
- Soundness of cab structure
- Security of doors, locks, windscreens, windows and light
- Security of driver's seat
- Soundness of cab floor boards
- Security of body mounting
- Tipper ram mounting, hinge bar and brackets, tip gear
- Oil reservoir
- Ram seals
- Neuter control valve
- Pump engagement
- Power take-off
- Drive Shaft
- Outrigger bearing
- Drive belts and pulleys
- Air filter
- Pipework and valves

SPRINGS

Check
- Security of axle bolts, lock-nuts and spring clips.
- Broken or displaced leaves
- Wear in shackles, pins and bushes.
- Security of hangers to chassis frame.
- Balance beam trunnions for wears.

M/M M/M M/M M/M

M/M M/M

TYRE TREAD DEPTH M/M

M/M M/M

M/M M/M M/M M/M

REPORT DEFECTS OVERLEAF WORK CHECKED & CLEARED

Job specification sheet (for motor vehicle)

<table>
<tr><td colspan="15">HISTORY RECORD CARD Date from …………….. Date to …………….</td><td colspan="2">Sheet No</td></tr>
<tr><td rowspan="2">Date</td><td rowspan="2">Job Report No</td><td colspan="13" rowspan="2">SUMMARY OF:
Item – Defect – Cause – Corrective – Spare/Materials
Action Used</td><td colspan="2">Cost/Time</td></tr>
<tr><td>Planned Maintenance</td><td>Unplanned Maintenance</td></tr>
<tr><td></td><td></td><td colspan="13"></td><td></td><td></td></tr>
<tr><td></td><td></td><td>J</td><td>F</td><td>M</td><td>A</td><td>M</td><td>JU</td><td>JY</td><td>A</td><td>S</td><td>O</td><td>N</td><td>D</td><td>Facility</td><td>Location</td><td>Identification No:</td></tr>
</table>

A HISTORY RECORD CARD

3 - Pseudo code for Companies/Firms plugin software

1) If setting for staff (page 1) has been Performed.

2) If staff profiles are captured (page 2-3) from existing database

3) Else, determine the number of staff (supervisory-staff & management-staff page 34-58)

4) If Maintenance model is used by the firm/company (page 120-128)

5) If the period for appraisal (monthly, quarterly, half-yearly or yearly page 3) and computational right has been assign to the personnel dept. and the choice to carry out motivation has been determine by Director/CEO and management.

6) Then, capture staff data and check for data fitness (page 5-9, 23-30). If data capture is fit for analysis.

7) Then perform appraisal (staff page 15-23, page 59-74).

8) If stress level for staff is determined (staff Page 9-16, 30-34),

9) Then, It's time to allow external auditors to verify appraisal result

10) Then, perform motivation if set in item 4 (page 75-82)

End if,

End if,

End if,

End if,

End if,

End if,

End if,

End if,

End.